

Chapter 5: Harmonic Patterns

5.1. ABC Patterns

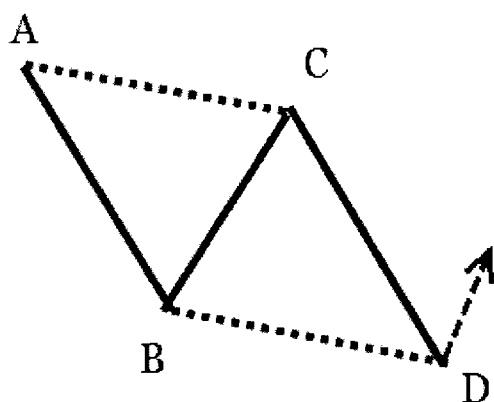
ABC Patterns

The basic “ABC” pattern is first described in H.M. Gartley's book, *Profits in the Stock Market* (1935). This pattern is shaped like a lightning bolt and signals a trend, a retracement and the resumption of the trend. This pattern is also called the “ABC Wave” or 1-2-3 pattern by technical analysts.

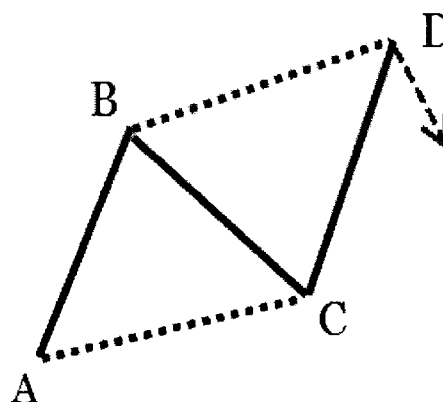
The “ABC” patterns forecast key market turning points and profit targets for traders. “ABC” patterns pinpoint important pivot levels with high and low prices and identify key trading zones.

The key point in identifying a “ABC” is correctly finding the A, B, and C pivot points in a chart. These key pivots are found using for various “pivot strength” levels, and for its correction waves. Once A, B, and C pivots are identified, an Auto-levels algorithm is applied to determine the confluence level “D”. This area is called “Potential Reversal Zone” (PRZ).

The “C” pivot in “ABC” patterns are determined by the Fibonacci retracement of (38.2 to 61.8 percent) of AB swing. The projection from “C” level is measured using fib-ratios of AB and BC swings. Some traders use the confluence of these ratio levels as areas for profit taking.

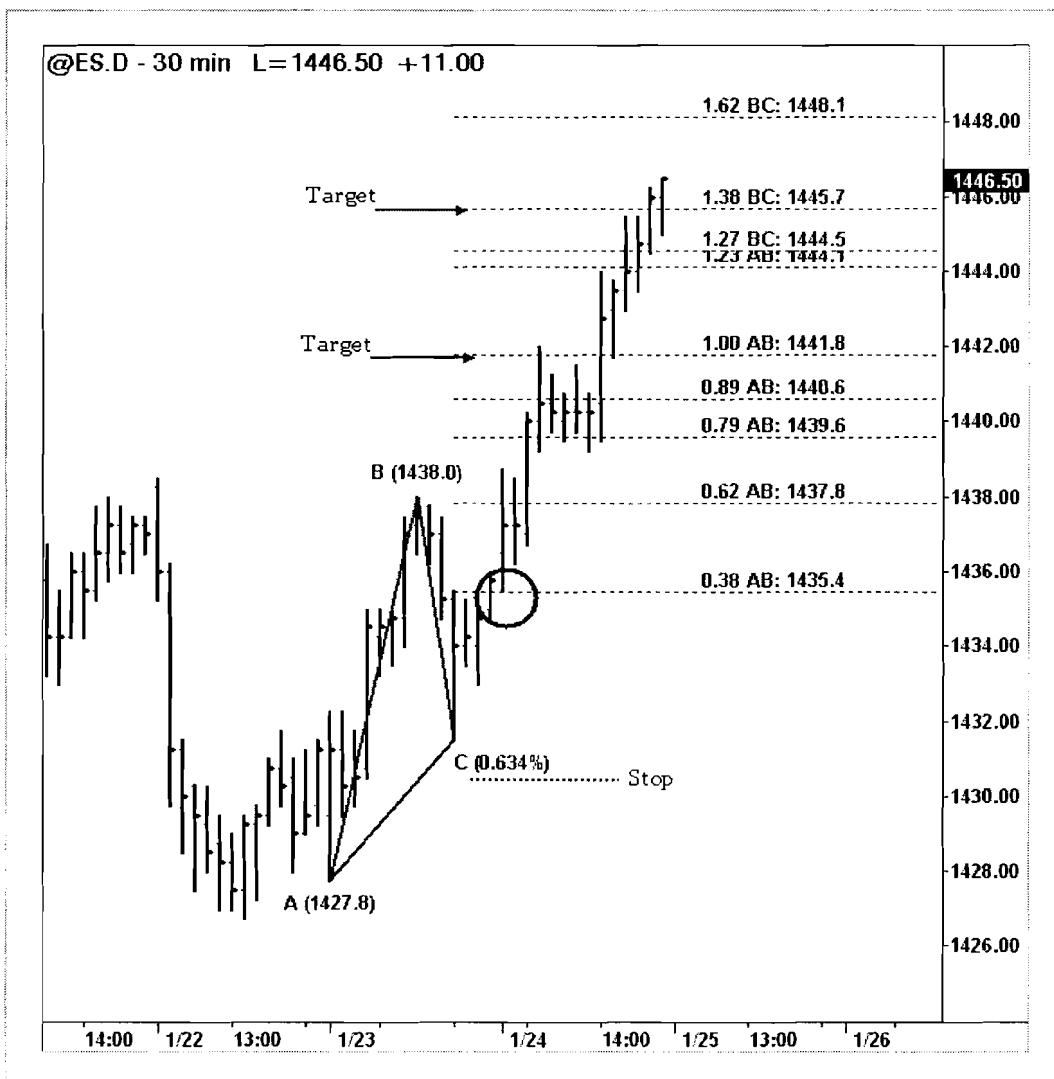


AC = 0.382 to 0.886 AB
BD = 1.232 to 2.618 BC



AC = 0.382 to 0.886 AB
BD = 1.232 to 2.618 BC

ABC Bullish Pattern



Trading an ABC Bullish Pattern

The example above shows an ABC pattern from the S&P Emini futures 30 minute chart. After formation of the AB swing, the "C" pivot point was expected around 62% of AB range. Price trading above the previous bar's high signaled a potential "long" trade.

1. Enter a "long" trade above the previous bar's "high."
2. Place a "stop" order below level "C."
3. Set "targets" at 100% of AB range and at 127% of BC range.

ABC Bearish Pattern



Trading an ABC Bearish Pattern

The example above illustrates an ABC Bearish reversal pattern in daily Gold futures chart. Gold made a 20-day swing high of 658 in December. During mid December, Gold reached a swing low of 621 (at B). At the beginning of January 2007, Gold retraced 78.6% of the AB level to another “swing high” at “C” to complete the ABC “bearish” pattern. A wide-range bar signaled a potential short trade.

1. Enter a “short” trade below the low of previous bar 631 (38% of the AB range).
2. Place a “stop” order above level C at 650.
3. Set “targets” at 100% of AB range (at 605) and 127 to 138% of BC range.

5.2. Gartley Pattern

Gartley Pattern

In 1932, H.M. Gartley described a 5-point “Gartley” trading pattern in his book, *Profits in the Stock Market*. Larry Pesavento has improved this pattern with Fibonacci ratios and established rules on how to trade the “Gartley” pattern in his book, *Fibonacci Ratios with Pattern Recognition*. There are many other authors who have worked on this pattern, but the best work to my knowledge is done by Scott Carney in his books of “Harmonic Trading”.

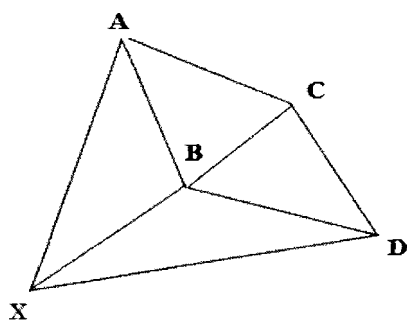
“Gartley” patterns have five points starting at point X, which is the lowest of all points in a bullish setup and the highest of all points in a bearish setup. For bullish Gartley, from X, prices rise to form a higher swing-high at “A”. From A, a retracement swing low “B” is formed within 0.382 to 0.618 of XA range. Another swing high “C” is formed at 0.618 of AB. Point “D” is formed in Potential Reversal Zone (PRZ) within 0.618 to 0.786 of XA swing, or 1.27 to 1.62 of BC range. D is the decision or buy trade point in bullish “Gartley” setup. Point D is also a sell trade point in a bearish “Gartley” setup.

Trade: A confluence of Fib ratio levels is calculated to find a Potential Reversal Zone (PRZ). This is the area where Gartley pattern formations are anticipated for reversals and for potential trade entries. Trades are only placed after “D” formation and if the market makes a reversal bar (wide-range or higher-high) from the PRZ.

Target: The first set of targets are the price levels of C and A. The second target is set at extensions of 1.27 to 1.62 of AD range.

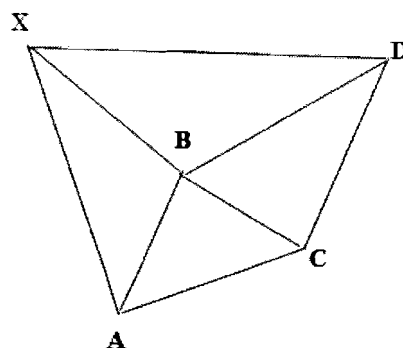
Stop: Once a trade is triggered, a “stop” is placed below D or below PRZ for “long” and above D for “short” trades.

Bullish Gartley



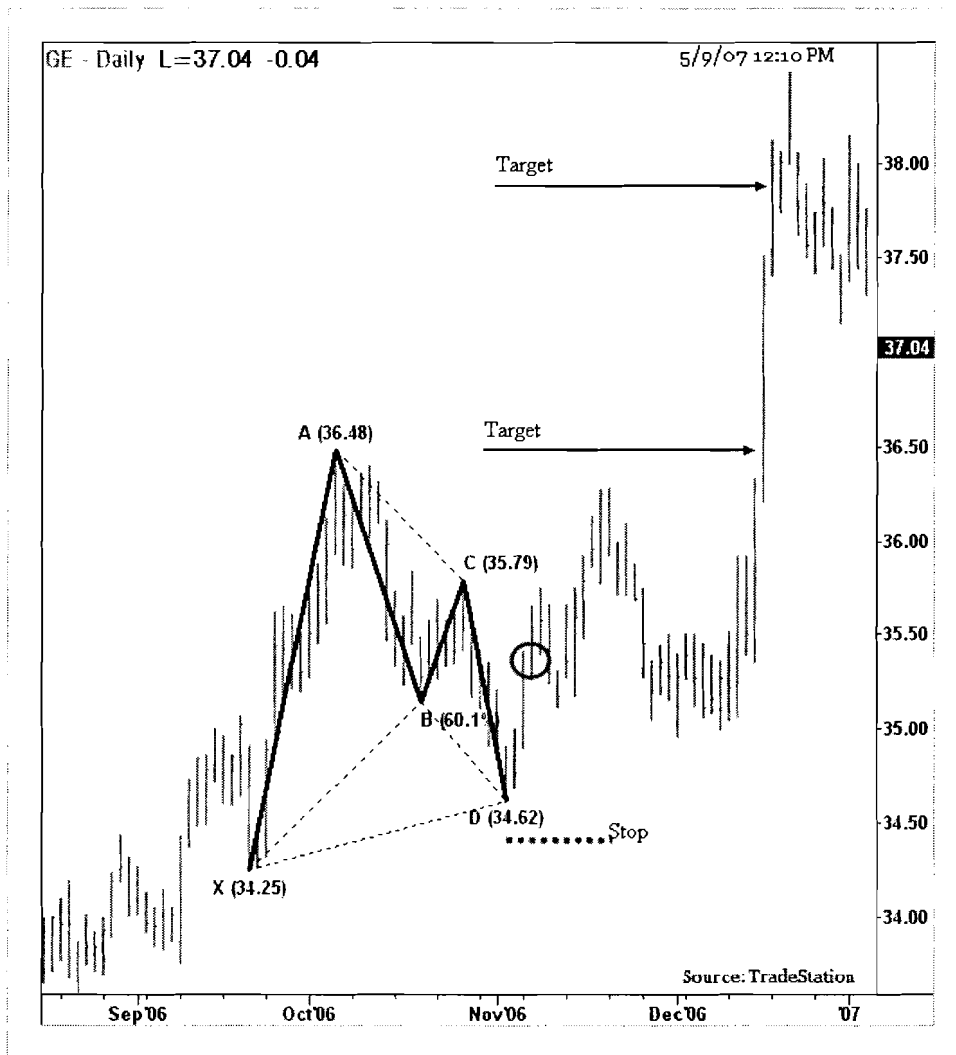
$$\begin{aligned} AB &= 0.618 \text{ XA} \\ AB &= CD \\ AB &= 0.382 - 0.886 \text{ XA} \\ AD &= 0.786 \text{ XA} \end{aligned}$$

Bearish Gartley



$$\begin{aligned} AB &= 0.618 \text{ XA} \\ AB &= CD \\ AB &= 0.382 - 0.886 \text{ XA} \\ AD &= 0.786 \text{ XA} \end{aligned}$$

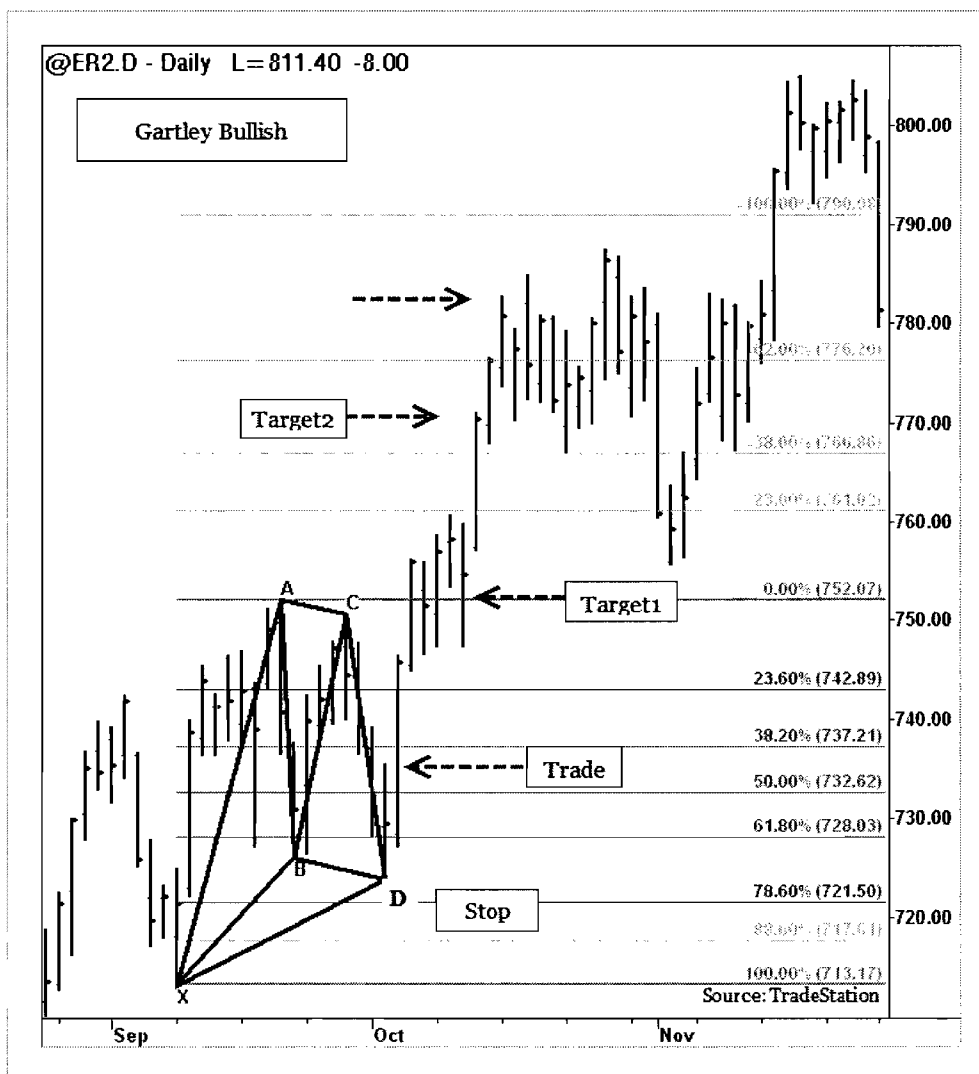
Trading Gartley Bullish Pattern



Trading Gartley Bullish Pattern

The example above shows a bullish “Gartley” formation from the daily GE stock chart. From mid September 2006 to November 2006, GE formed a “Gartley” formation. The B level retracement was 60.1% near the minimum of Gartley’s requirement. The PRZ level was formed at 88.6% area at D. After D level, the price action is closely watched for a “long” trade entry. GE made higher-highs from D level suggesting a completion of the “Gartley” formation. A “long” entry is triggered at \$35.35 area with a “stop” below D level at \$34.30. Targets were placed first at the A level (at \$36.48) and from 138% to 162% of XA range at \$37.50.

Trading Gartley Bullish Pattern

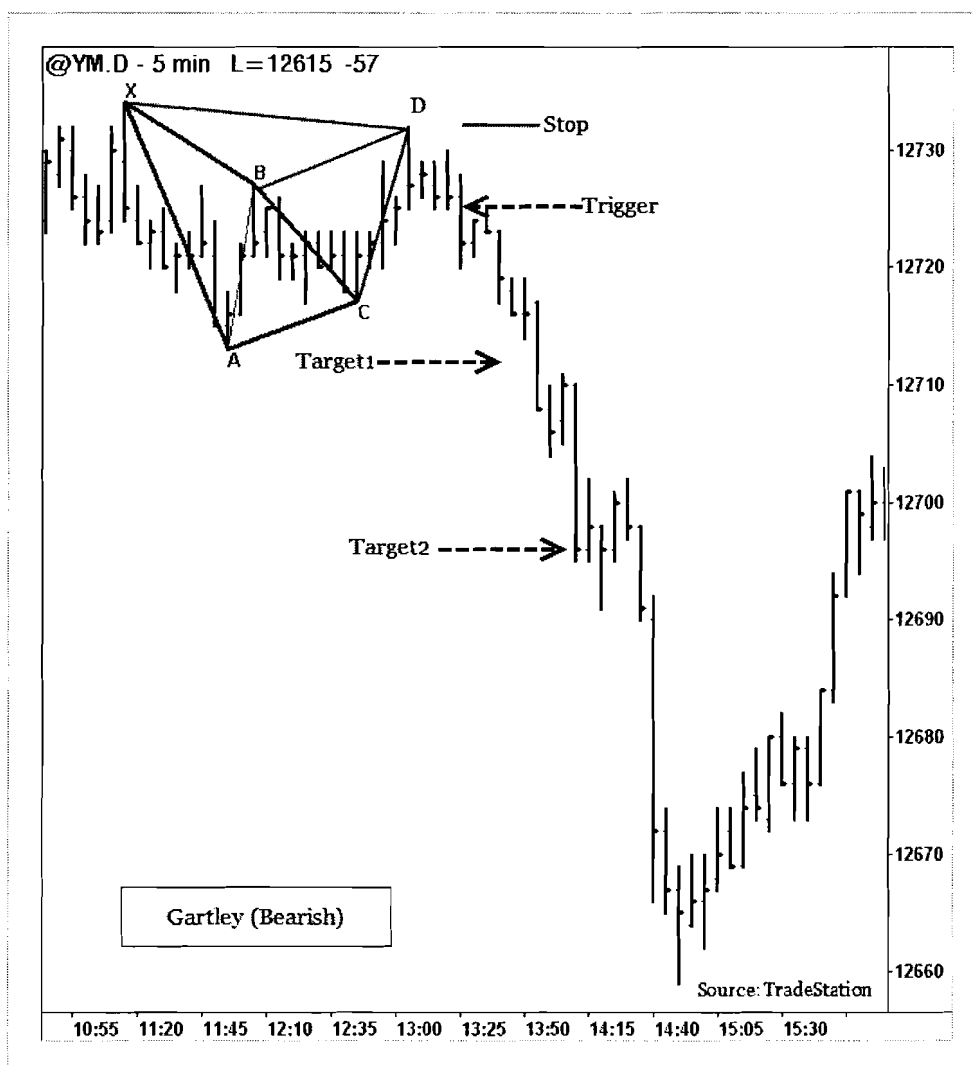


Trading Gartley Bullish

The example above shows a bullish “Gartley” pattern from the Russell Emini (ER2) daily chart. After XA “swing,” the B level was formed near 62% of XA. C swing was formed at 0.886 of AB range. The “Gartley” pattern is completed when AB=CD is formed at D.

1. After completion of D, a “long” trade is entered one-tick above previous bars’ high.
2. A “stop” order was placed one tick below D level (PRZ).
3. Place a target at 127% of XA level around 770 level.

Trading Gartley Bearish Pattern



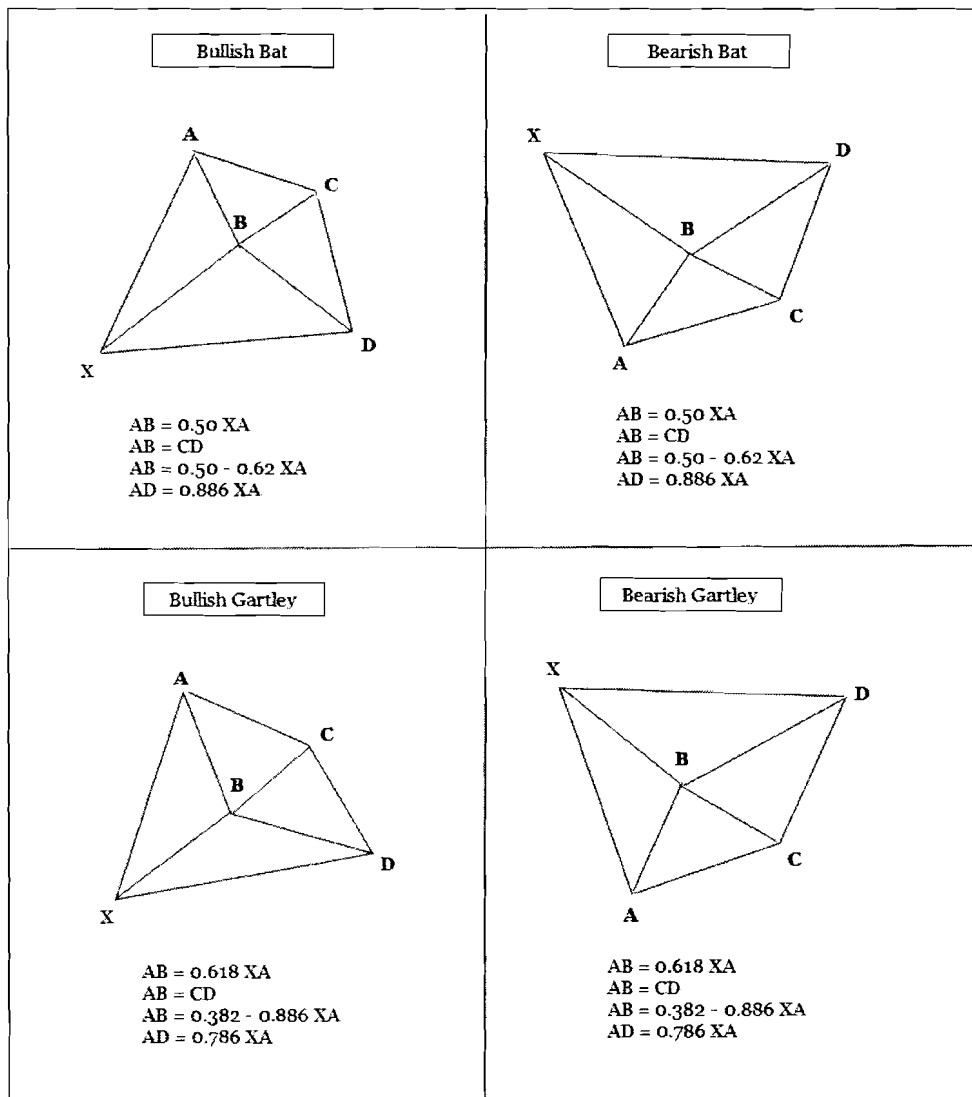
Trading Gartley Bearish Pattern

The example above shows a bearish “Gartley” pattern from the Dow Emini futures (YM) 5m chart. The B retracement was formed at 62% of XA swing. C swing point was formed with 0.786 of AB swing. The D level was formed at 0.886 of XA range with AB=CD confluence.

1. Enter a “short” trade one-tick below the previous bar’s low at ‘D’.
2. Place a “stop” order one tick above the “D” level.
3. Place “targets” at “A” level and at 1.62 of AD range.

Bat and Gartley Comparison

Bat and Gartley



5.3. Bat Pattern

Bat Pattern

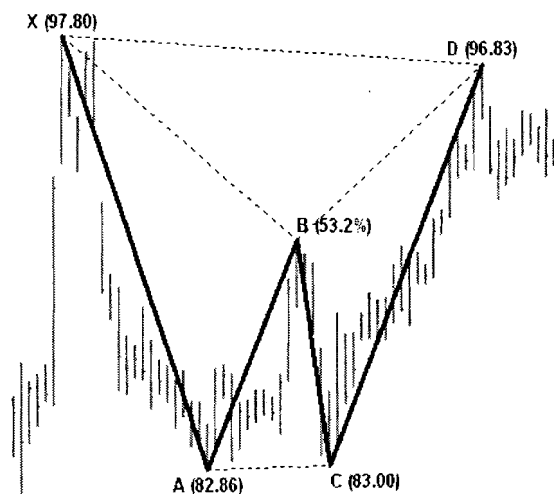
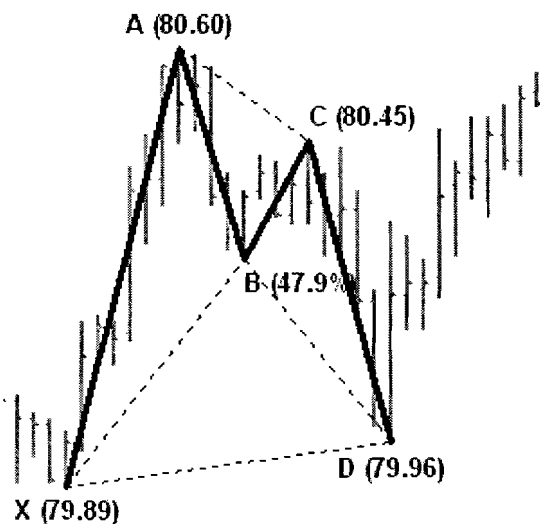
The Bat pattern was discovered by Scott Carney of Harmonic Trading in 2001. The Bat pattern is in the same family of Gartley's 5-point corrective patterns, but has distinct harmonic ratios. Bat pattern incorporates a precise harmonic ratio (0.886 of XA swing). It also demands that the B (center) retracement should be less than 0.618 of XA swing. The B retracement differentiates between the Gartley and Bat patterns. The Gartley pattern must have a 61.8% retracement of XA swing, and in Bat patterns have the same XA swing retracement below 61.8% at "B".

The Bat pattern's Potential Reversal Zone (PRZ) is defined by $1.27AB = CD$ pattern, $1.62BC$, 0.886 XA retracement levels. In bullish or bearish Bat patterns, a reversal from PRZ is anticipated for a potential long and short entry trades.

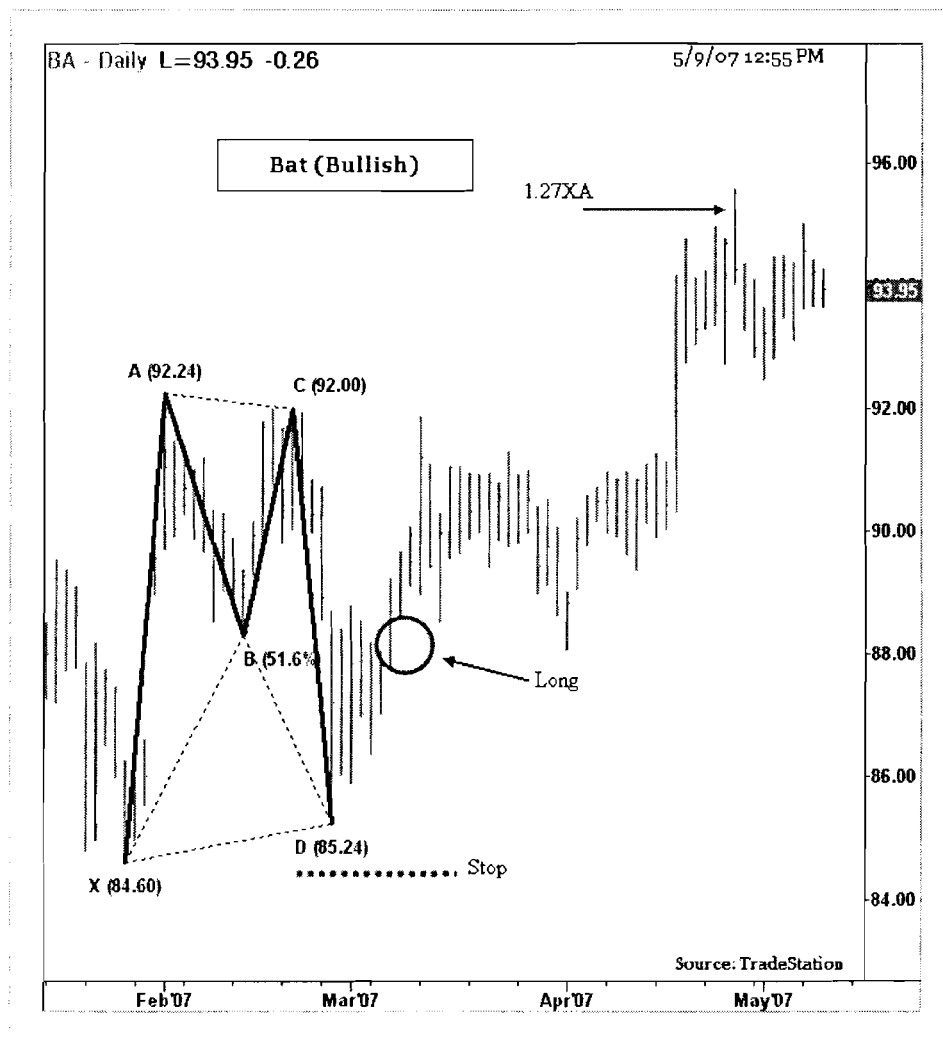
Trade: Once the Bat pattern is completed, wait for a higher-high bar or wide range bar to give a signal to enter a long trade. Enter a "long" trade one tick above the high of the confirmation (higher-high or wide range) bar. For bearish patterns, enter a "short" trade one tick below the low of the lower-low or wide-range bar.

Target: The target for the Bat patterns is similar to the Gartley patterns. First targets may be set at A level or 1.27 of XA swing. The secondary targets could be 1.62 to 2.0 of XA swing level.

Stop: The Bat pattern fails if price trades below the X level. Place a stop order one tick below X level.



Trading Bullish Bat

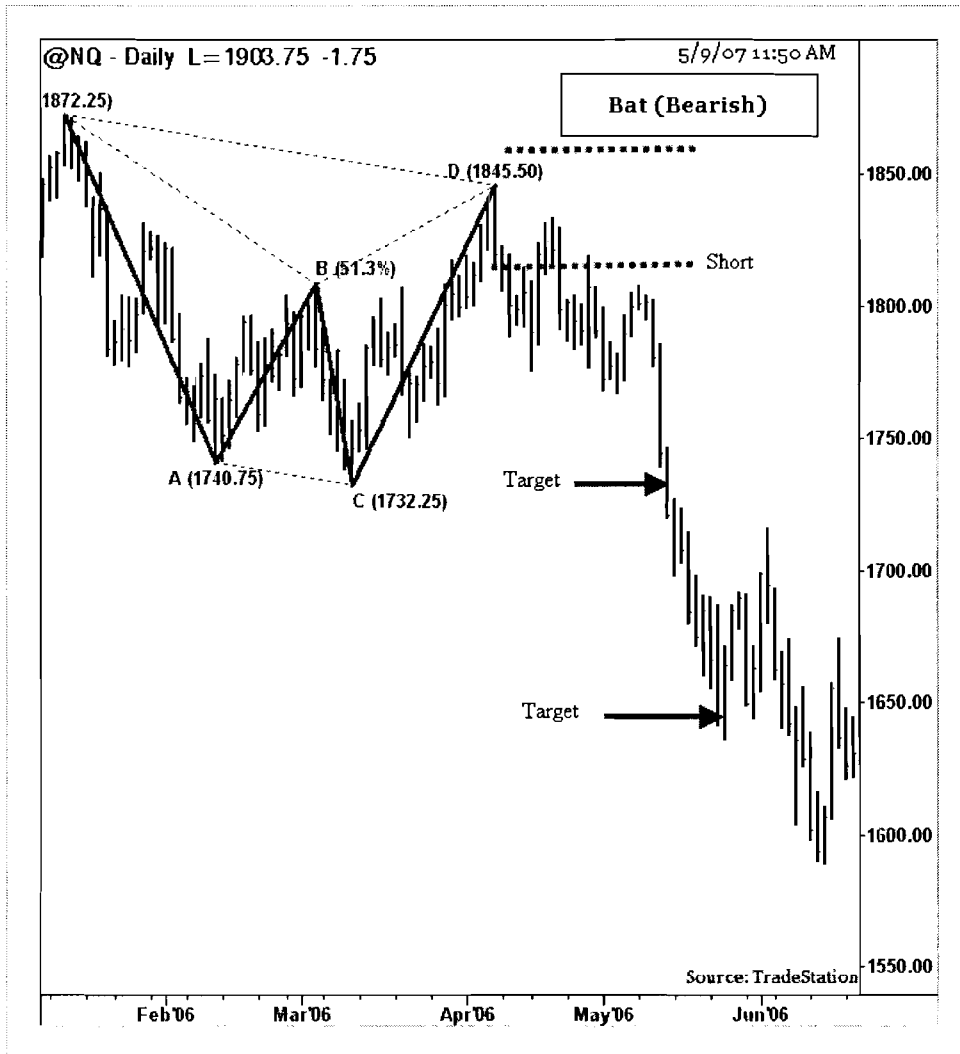


Trading Bullish Bat

The example above shows a bullish Bat formation from Boeing's daily chart. Boeing formed a bullish Bat formation from mid January 2007 to March 2007. After XA swing, a 51.6% retracement is marked for "B" level. A Potential Reversal Zone (PRZ) is computed at 0.886 of XA, 1.62 of BC and 1.27 of BC range. After a series of "higher high" bars, Boeing stock reversed its prices from PRZ (at D). A "long" trade is triggered above the B level.

1. Enter a "long" trade above B level at \$88.
2. Place a "stop" order below the low of D at \$84.60.
3. Place a "Target" 1.27 of XA level at \$94.

Trading Bearish Bat



Trading Bearish Bat

The example above shows a bearish Bat formation from the NASDAQ futures (NQ) chart. NASDAQ futures formed a bearish Bat from January to April in 2006 from the range of 1872 to 1845. The B retracement level was at 51.3% confirming the Bat pattern. The D level was formed at 0.886 of XA range. A PRZ was computed from 1845 to 1853 levels for a reversal at D. Once prices traded inside the PRZ, a reversal bar is anticipated to signal a “short” trade.

1. Enter a “short” trade below the low of the reversal bar at D level (1820).
2. Place a “stop” order above the D level 1847.
3. Place “targets at “A” level and at 1.62 of XA swing at 1660.

5.4. Butterfly Pattern

Butterfly Pattern

The “Butterfly” pattern was discovered by Bryce Gilmore and Larry Pesavento. It is one of the powerful patterns like the Gartley pattern. The “Butterfly” pattern has a distinct retracement level (0.786) of XA swing. In bullish and bearish 5-point swings, the pattern must have 0.786 to 0.886 of XA swing to be valid. In perfect “Butterfly” patterns, the AB swing will be equal to CD (AB=CD).

“Butterfly” patterns usually occurs at market tops and market bottoms. The pattern's success rate is much higher when the retracement and time ratios are harmonically aligned.

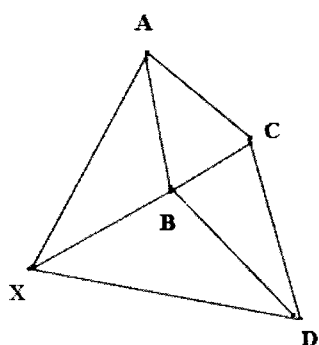
Two primary differences between the “Butterfly” and “Crab” patterns are: 1). The AB retracement must be 0.786 in the “Butterfly” pattern; where as in the “Crab” pattern, it lies between 0.382 and 0.618.2 levels 2). In both patterns the D point extends beyond the X and the C level can be inside or outside of the XA range. The retracement of AB defines level D. In Butterfly patterns, if B is formed at 0.786, the usual retracement of D could be near 1.27 of XA range.

Trade: Once the Butterfly pattern is completed in PRZ level, wait for a confirmation bar, wide range bar or “higher high” close bar to suggest a potential reversal from “D” level. Enter one tick above the high of the confirmation bar.

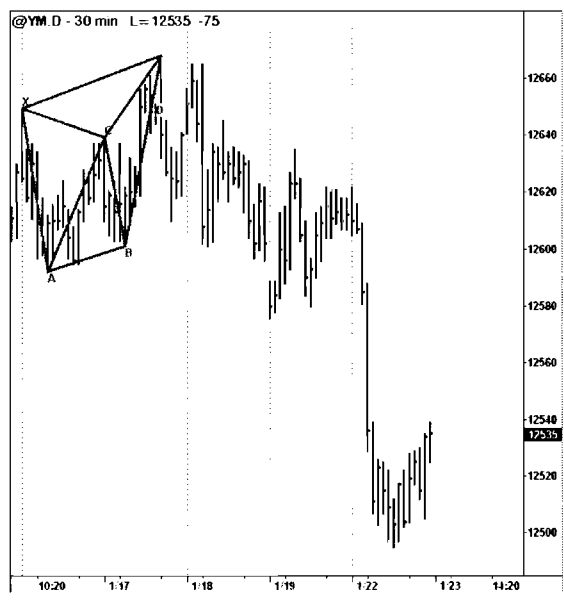
Stop: Place a “stop” order below (bullish) the low of the “Butterfly” pattern. For bearish Butterfly patterns, place a “stop” order above the high of the “Butterfly” pattern.

Target: Targets are set at 100% of AD and 162% of XA from D levels. In bullish butterfly patterns, beyond the A level, targets need to be protected with trailing stops.

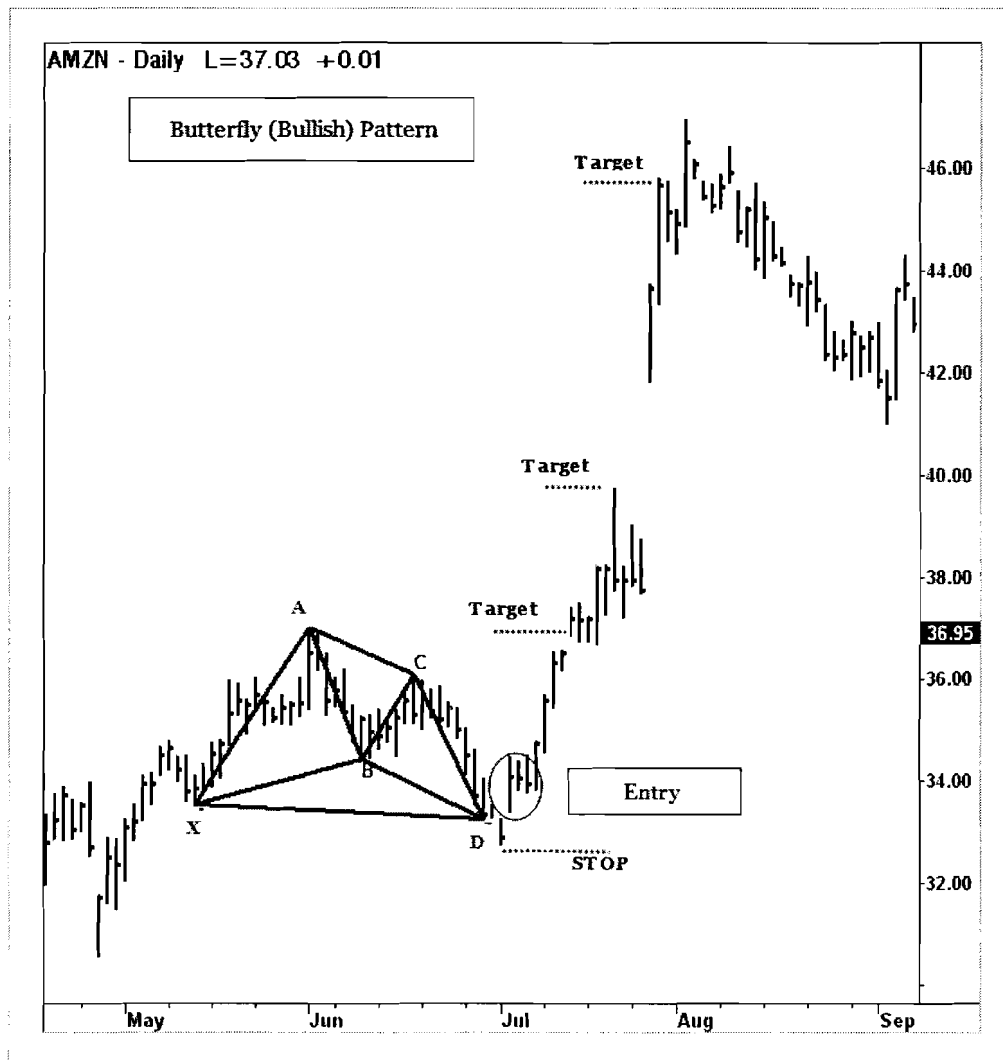
Bullish Butterfly



$$\begin{aligned} AB &= 0.786 \text{ XA} \\ BC &= 0.382 - 0.886 \text{ AB} \\ XD &= 1.27 \text{ XA} \end{aligned}$$



Trading Butterfly Bullish

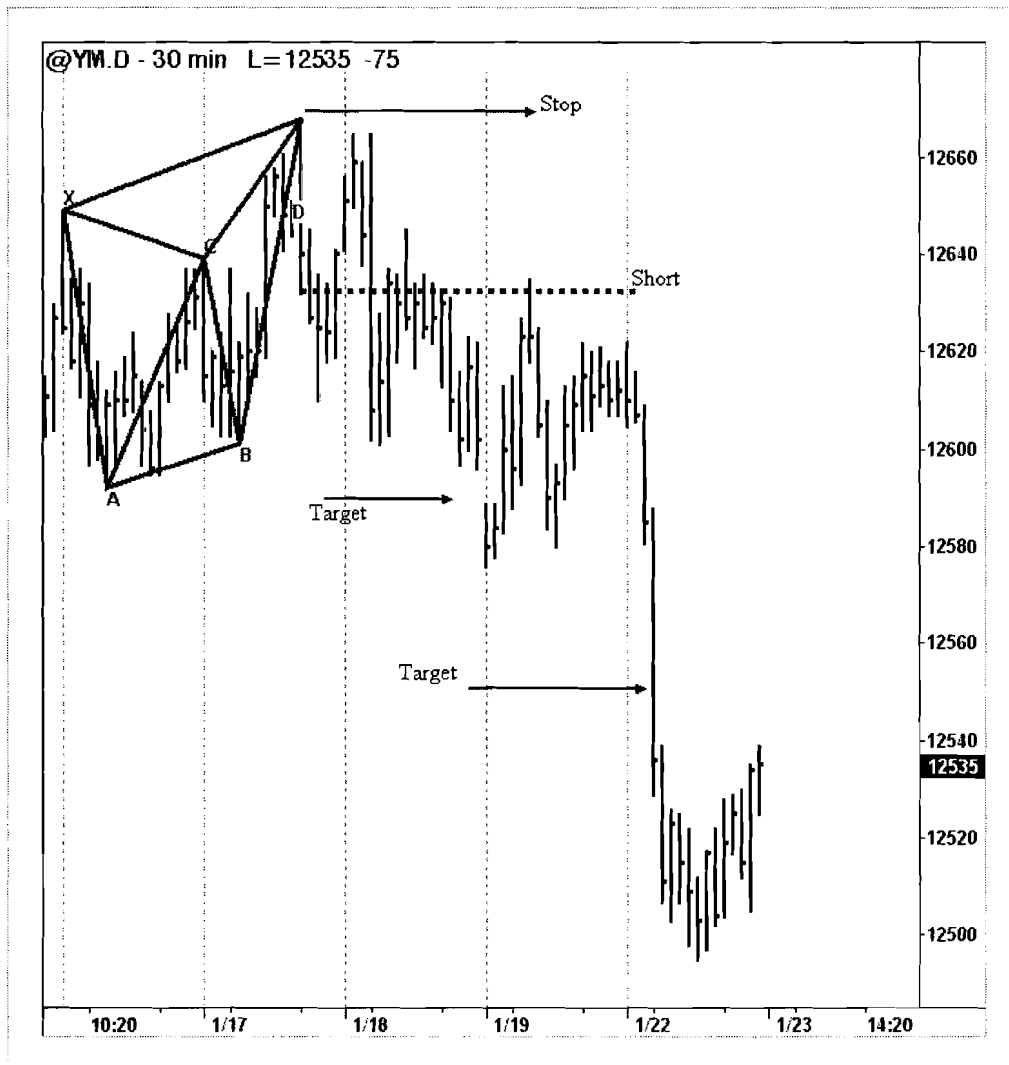


Trading Bullish Butterfly Pattern

The example above illustrates a bullish “Butterfly” pattern from Amazon’s daily chart. From the “swing low” of X to the “swing high” of A, the center of Butterfly level B is formed when prices retrace to 0.786 of XA was made. A retracement of 0.618 of AB retracement is made at C. D is formed with in the Potential Reversal Zone at 1.27 of AB. A “long” trade is triggered from D level as a wide-range bar traded above the previous bar’s high.

1. Enter a “long” trade one tick above the previous bar’s high.
2. Place a “stop” order below the low of D level.
3. The first target is set at “A” level and second target is set at 162% of XA swing.

Trading Butterfly Bearish



Trading Bearish Butterfly Pattern

The example above illustrates a bearish “Butterfly” pattern from the Dow Emini futures (YM) 30 minute chart. YM formed a bearish “Butterfly” pattern from January 16, 2007 to January 17, 2007 between the 12500 to 12660 levels. After completion of D level at 12260, a reversal bar (wide range bar or lower-low bar) is anticipated to signal a “short” trade.

1. Enter a “short” trade below the low of the reversal bar at 12630.
2. Place a “stop” order above the high of the D level at 12665.
3. Place “targets” at “A” level (12590) and another at 1.62 of XA swing (at 12550).

5.5. Crab Pattern

Crab Pattern

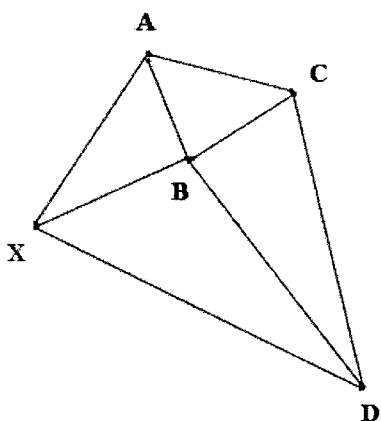
The “Crab” pattern was invented by Scott Carney of Harmonic Trading in 2000. The Crab pattern is another form of the 5-point Gartley extension pattern. The Crab pattern has a distinct extension: 1.62 of XA swing. Crab patterns also have a 0.618 XA retracement to form the center retracement “B” level. The Potential Reversal Zone (PRZ) is formed at 1.27 of AB swings, 1.62 of XA and a distinct 2.62 to 3.62 of BC. These extension patterns form when prices trade outside of XA swing. When the price closes below X, the pattern may be signaling a further correction to 1.62 of XA level to form “Crab” pattern.

Trade: Once the Crab pattern is completed at the PRZ level, look for price-action to confirm the reversal. For bullish Crab patterns, look for a “wide range bar” or “higher highs” from the PRZ level to confirm the Crab pattern. Enter a “long” trade above the confirmation bar. For bearish Crab patterns, enter a “short” trade below the low of the confirmation (lower low) bar.

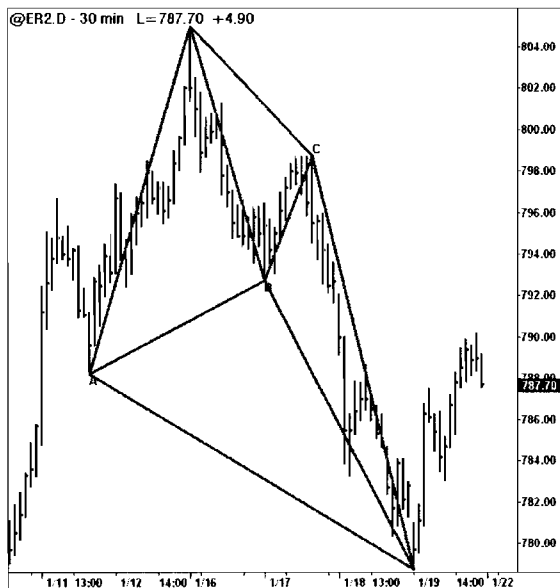
Stop: The bullish Crab pattern fails if prices close below the PRZ levels. Place a “stop” order below the low of the PRZ level. For bearish Crab patterns, place a “stop” order above the high of the PRZ levels.

Target: The bullish Crab patterns result in excellent profits. Set targets at “B”, “C” and “A” levels. Similarly, for the bearish Crab patterns set targets at “B”, “C” and “A” levels.

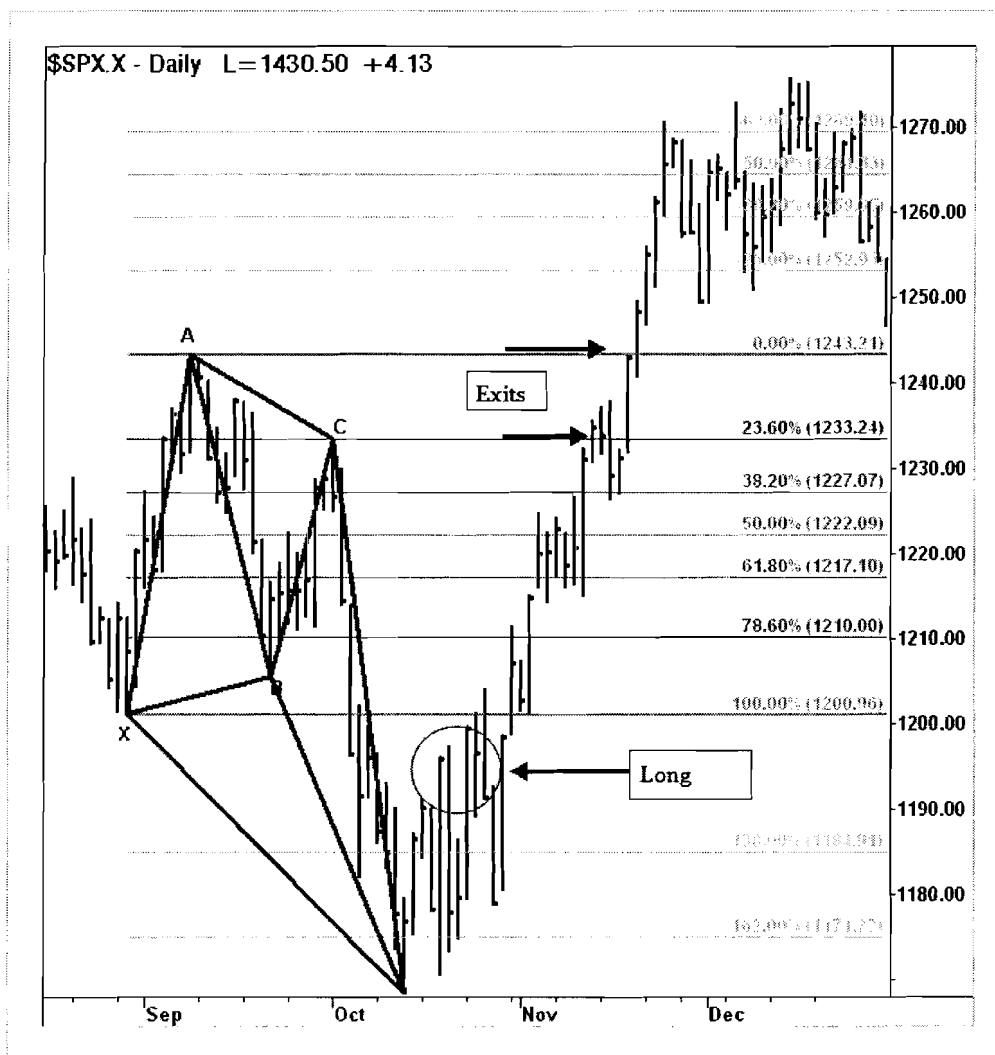
Bullish Crab



$$\begin{aligned} AB &= 0.618 \text{ XA} \\ XD &= 1.618 \text{ XA} \\ BD &= 2.28-3.14 \text{ XA} \end{aligned}$$



Trading Crab Bullish Pattern

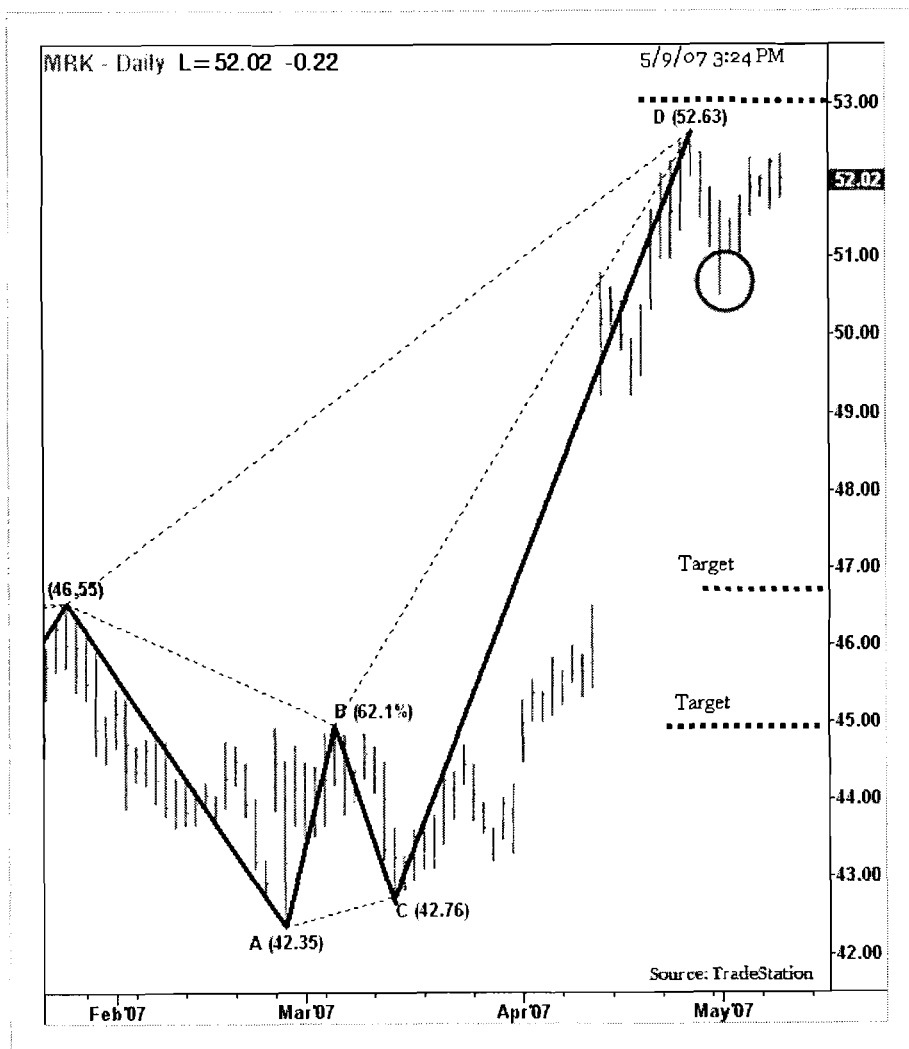


Trading Crab Bullish Pattern

The above example illustrates a bullish Crab pattern from the S & P 500 index (SPX) daily chart. From September 2005 to October 2005, SPX formed a bullish Crab pattern. Prices traded from a high of 1243 to 1172. In October 2005, SPX completed the Crab pattern as prices reversed to confirm the pattern. A wide-range bar at 1195 level confirmed the price reversals.

1. Enter a "long" trade above the confirmation bars' high at 1196.
2. Place a "stop" order below the low of the Crab pattern at 1171.
3. Targets are set at "C" level at 1233 and at "A" level at 1243.

Trading Crab Bearish Pattern



Trading Bearish Crab Pattern

The example above illustrates a bearish Crab pattern from Merck's (MRK) daily chart. MRK formed a bearish Crab pattern as it rallied from \$43 to \$53 from February 2007 to May 2007. After completion of the PRZ at "D" level, a reversal bar from the PRZ confirmed the completion of the Crab pattern. A "short" trade is triggered at the \$51 level.

1. Enter a "short" trade below the low of the confirmation bar at \$51.
2. Place a "stop" order above the high of the "D" level at \$52.75.
3. Targets are set at "X" level at \$47. The second target is set at "B" level at \$45.

Chapter 6: Geometric Patterns

6.1. Triangles

Symmetric Triangle

“Symmetric Triangles” form when the markets are in indecision mode. The “Symmetric triangles” can be easily detected when prices make alternate “lower highs” and “higher lows” in upside and downside slopes defining a symmetry. “Symmetric” triangles form when supply and demand are near equal resulting in market indecision. Most triangles result in a clear breakout and breakdown in the direction of the prior trend.

Trade:

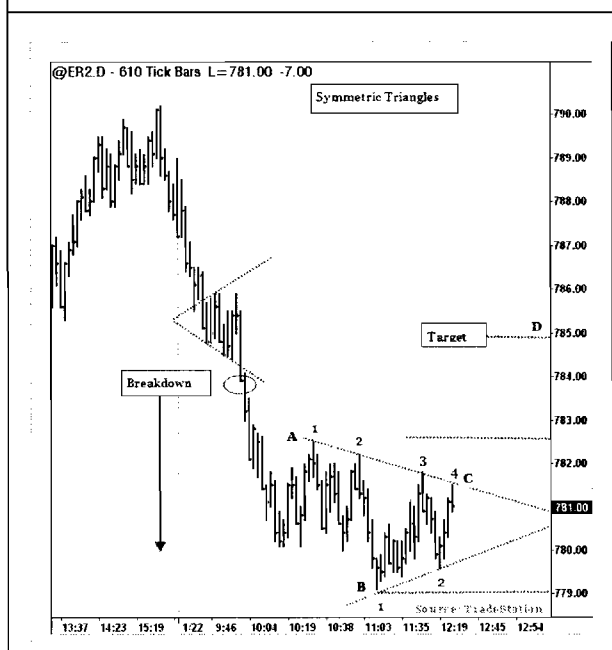
Trades are only initiated at the trend line breakouts of the “Triangle.” Enter trades only when “Symmetric triangle” breakouts/breakdowns are confirmed by price trading one or two ticks above/below the breakout/breakdown bar’s high/low.

Target:

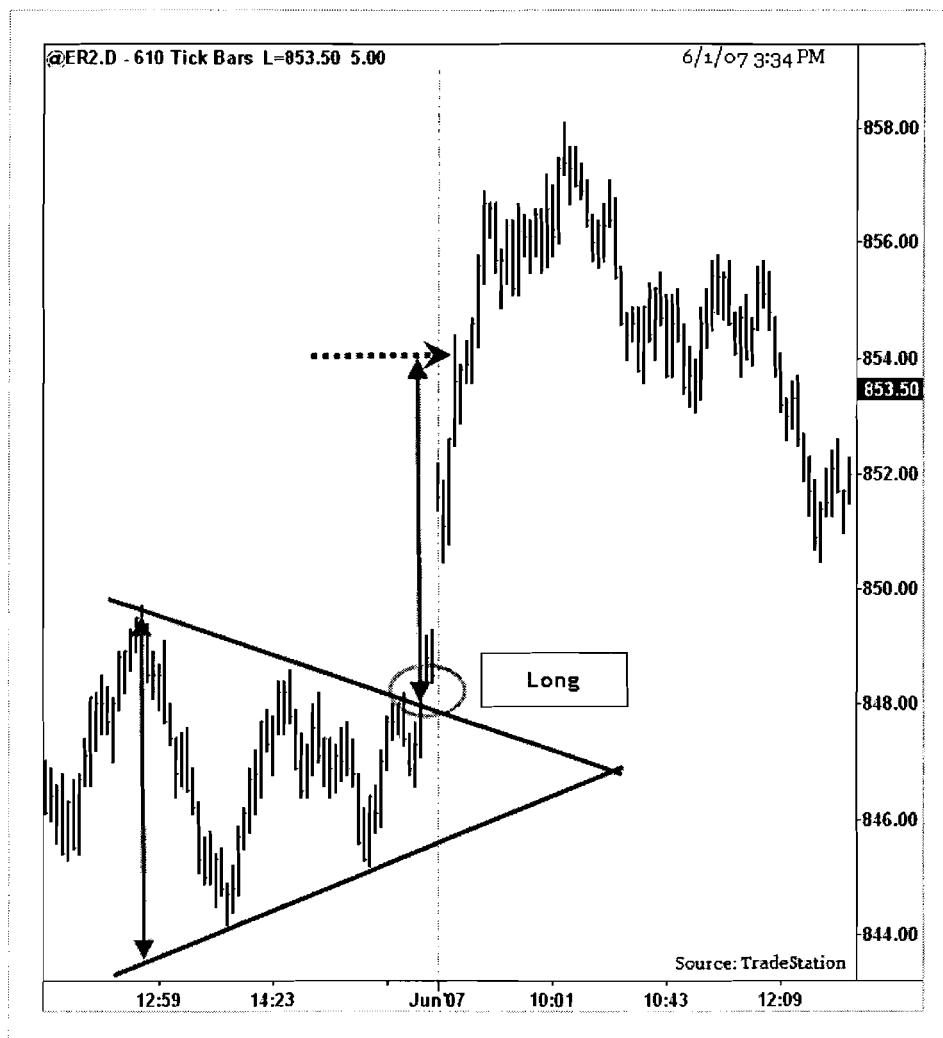
Most “Symmetric triangles” result in 100% of the depth rise/fall of the entire triangle range in the direction of the breakout. Partial trade exit targets are set at 50% of the depth from the trade entry. Protect targets by using trail stops.

Stop:

“Symmetric triangle” failures occur when price results in false breakouts. Stop below the first major “swing low” below the trend line for a long setup. Place a “stop” order above the first major swing high from the trend line for a short-setup.



Trading Symmetric Triangle



Trading Symmetric Triangle

The example above shows a “Symmetric triangle” formation from the Russell Emini (ER2) 610 tick chart. On May 31, 2007, ER2 made lower highs and higher lows to form a “Symmetric triangle.” Late afternoon, ER2 prices rallied and closed above the top trendline to confirm a “Symmetric triangle”. On the following day, ER2 traded higher from the breakout to reach the target levels.

1. Enter a “long” trade above the high of the breakout bar at 848.5
2. Place a “stop” order below the low of the previous swing low at 845.
3. Target the depth of the triangle from the breakout level to 853.

Ascending Triangle

“Ascending Triangles” form when prices attempt to make “higher highs” and “lower lows” suggesting a bullish price trend. The “Ascending triangle” is bound by two trendlines: a horizontal line at the top and an upward slope trend line connecting the lower lows.

“Ascending triangles” form in any market and are quite reliable. The “Triangle” prices must intersect the trend lines at least twice (each) before the pattern is complete. Usually at the third or fourth attempt to trade outside the top trend line results in a breakout. Breakouts occur near the apex of the triangle. This pattern has a high success rate as it meets its target about 75% of the time.

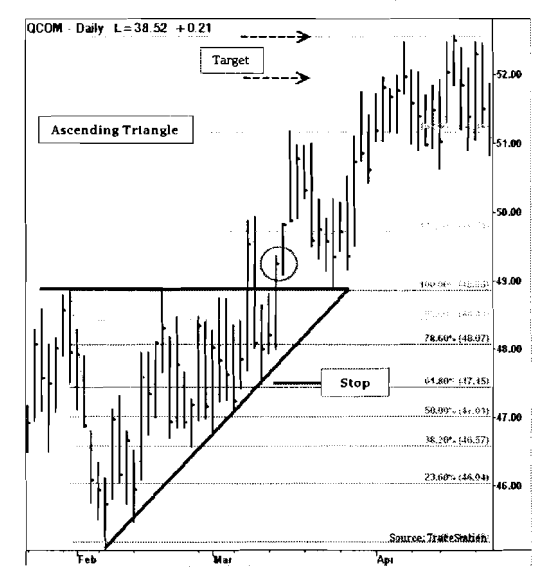
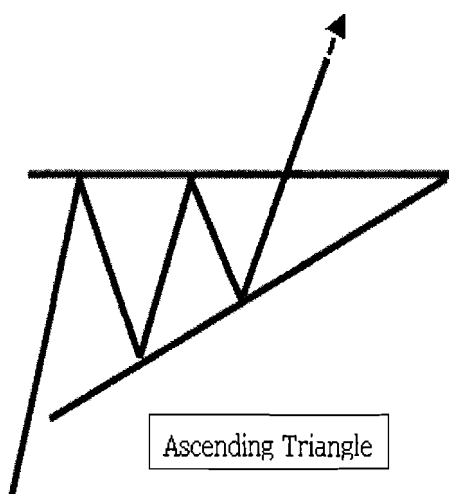
Trade:

Trade a clear breakout of the top trend line. Enter a “long” trade one tick above the high of the breakout bar. Confirm the breakout by volume or other indicators.

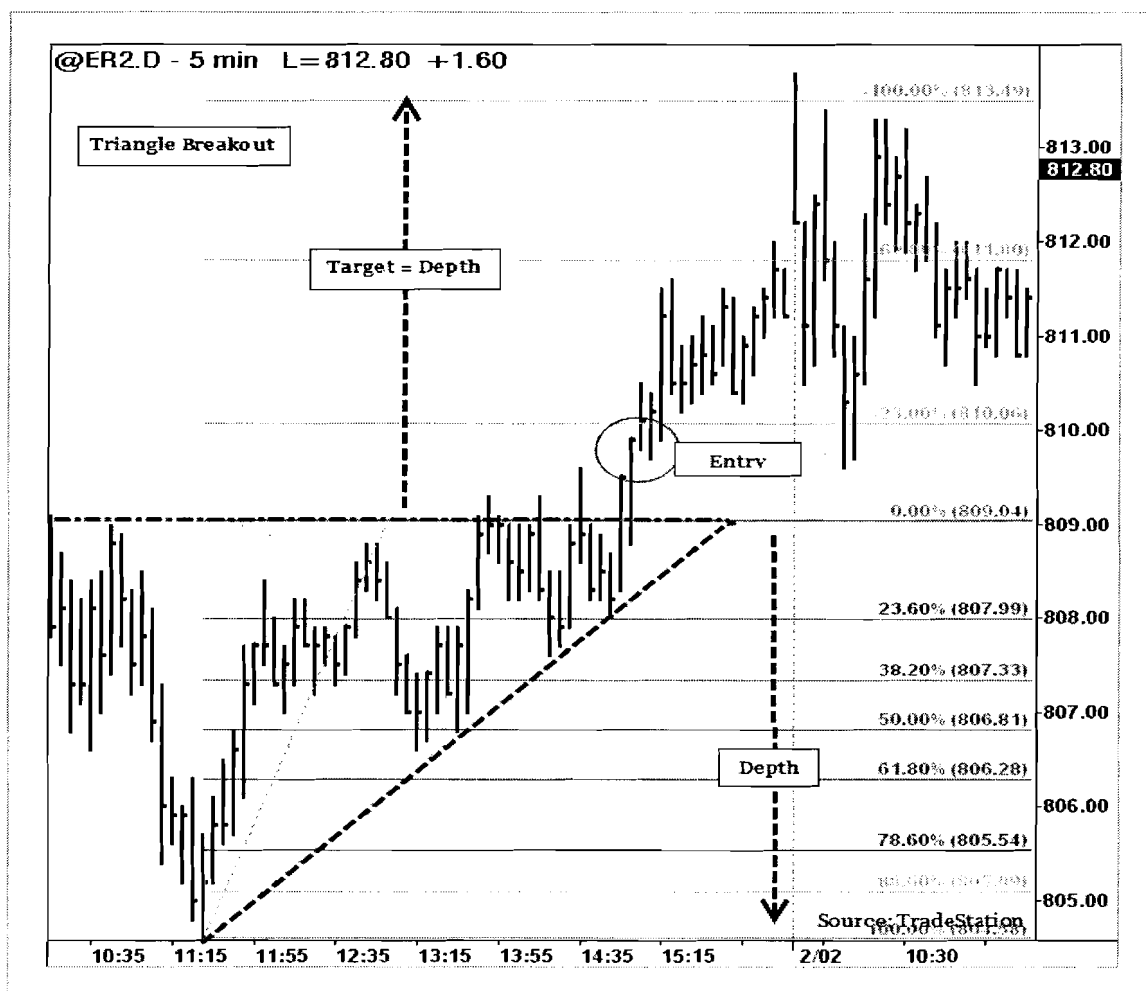
Target:

“Ascending triangles” have excellent success in reaching target areas. The usual target would be the depth of the “Triangle”. Measure the distance (depth) between the top trend line and lowest of the upward slope trend line. Add this depth to the breakout point from the top of the trend line. Targets are also set at 50% of depth level for partial exits.

Stop: Place a “stop” order when the price closes below the low of the lower trend line or a major swing low.



Trading Ascending Triangle



Trading Ascending Triangle

The above example shows an “Ascending triangle” from the Russell Emini (ER2) 5 minute chart. On February 01, 2007, at around 2.30 pm, after few failed breakout attempts, the price closed outside the upper trend line. A trade is triggered above the high of the breakout bar at 809.6. A “stop” order is placed below the low of the last “swing low” at 808. The depth of the triangle is 4 points. Targets are set for 50% of depth (2 points) at 812 and 100% of the depth at 813.

Descending Triangle

“Descending Triangles” are similar to “Ascending Triangles” formation rules except they are bearish. “Descending triangles” form in bear markets and favor breakdowns. A “descending” triangle is bound by two trend lines connecting a downward slope trend line and a flat trend line connecting the lows of the pattern. Trades usually occur near the apex as the price closes outside the bottom trend line suggesting a breakdown. The price must intersect trend lines at least twice before the pattern emerges. Like the Ascending triangles, “Descending Triangles” also have a high success rate.

Trade:

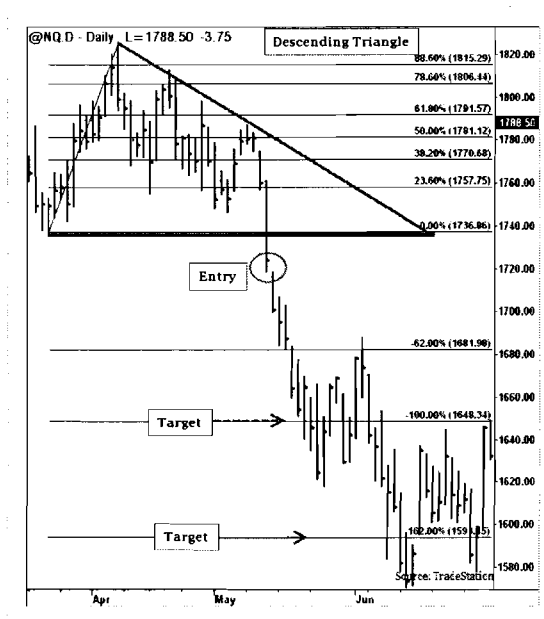
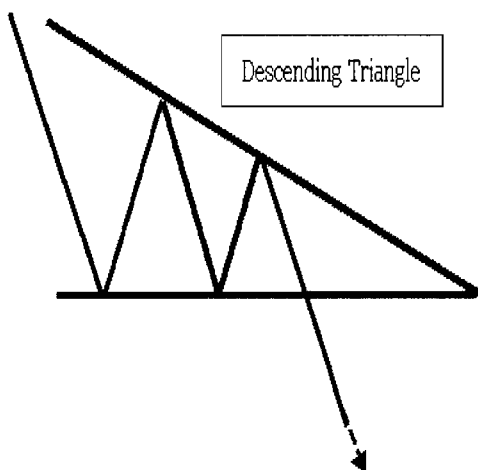
Trade one tick below the low of the breakdown bar (outside of the triangle). Confirm the breakdown with increased volume.

Target:

“Descending” triangles have similar targets like “Ascending” triangles. Measure “Triangle” depth at the lowest and highest points and set targets at 50% and 100% range from the breakdown point.

Stop:

Place a “stop” order outside the downward slope trend line. If price closes above the top trend line, exit the trade.



Trading Descending Triangle



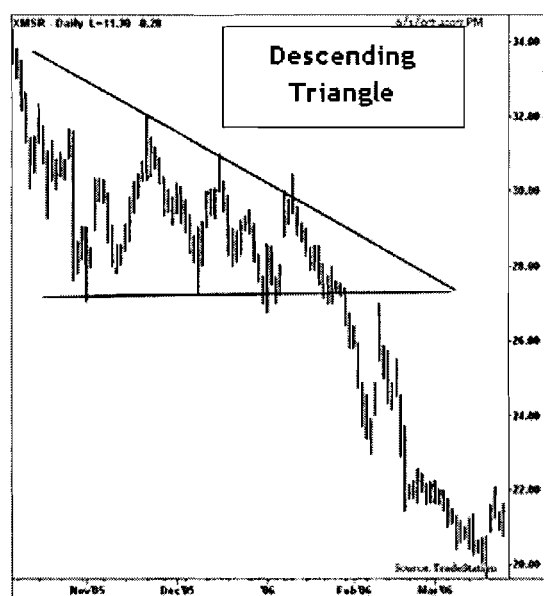
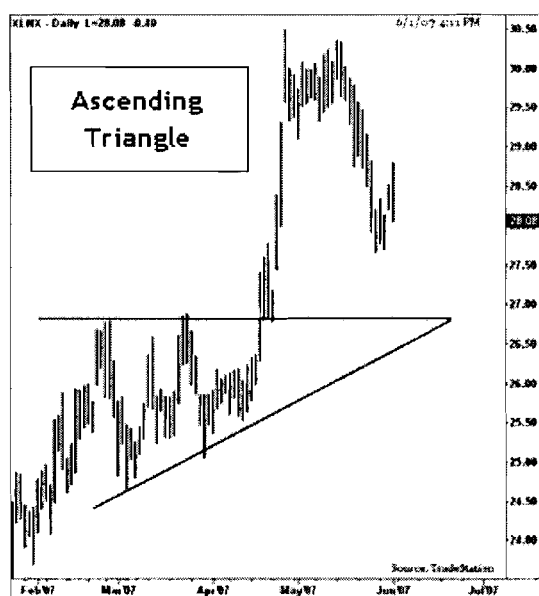
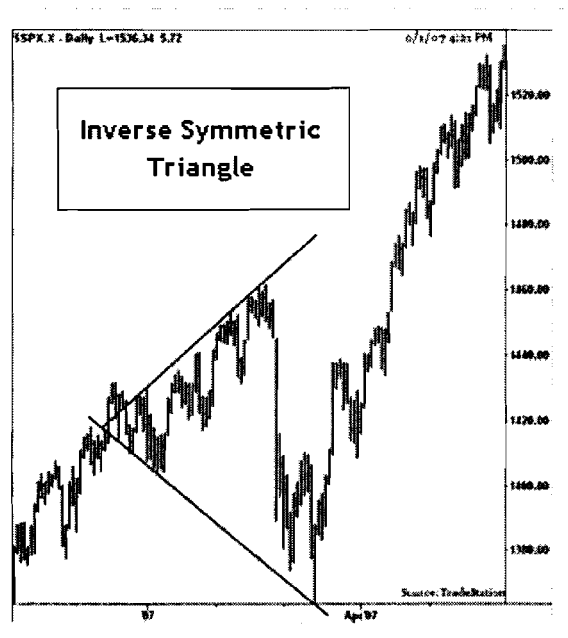
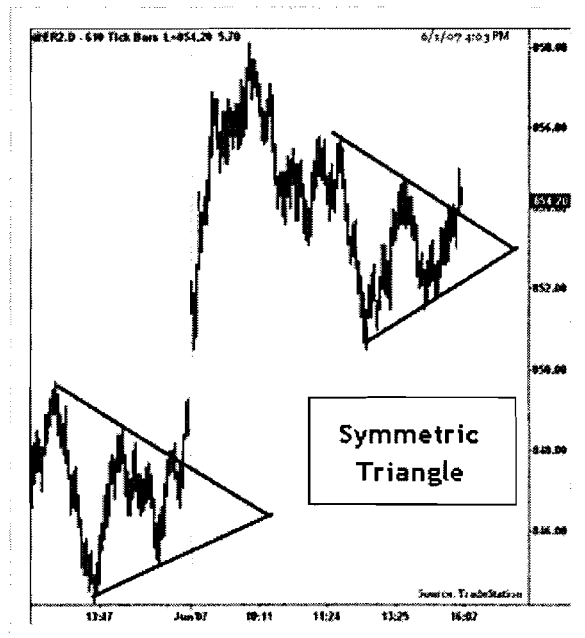
Trading Descending Triangle

The above example illustrates a “Descending triangle” from the daily Gold futures chart. In July 2006, Gold reached 700 and retraced to 620 by the mid of July. Gold made a series of lower highs and a flat bottom near the 620 level to form a “Descending Triangle”. During the first week of September, Gold closed below 620 to trigger a sell-off. The depth of the “Triangle” was 80 (700 to 620) points.

1. A “short” trade was entered at 618 with target of 570.
2. A “stop” order was placed above the trend line at the 640 level.
3. A 50% depth target area can be seen from the trade entry.

1.

Triangle Types



6.2. Rectangle Pattern

Rectangle Pattern

“Rectangle” patterns are continuation patterns showing indecision in trader’s sentiment about bullish and bearish conditions. “Rectangle” patterns are reliable patterns and the direction of the breakout/breakdown is known prior to the rectangle formation. These patterns are continuous and follow in the same prior direction (Up or Down) after the pattern formation.

“Rectangle” formations are bound by two horizontal trend lines, where prices oscillate between the “high” and “low.” The prices must intersect these trend lines at least twice before a breakout or breakdown can result. The volume within the pattern is usually quiet and increases during the breakout/breakdown stages.

Trade:

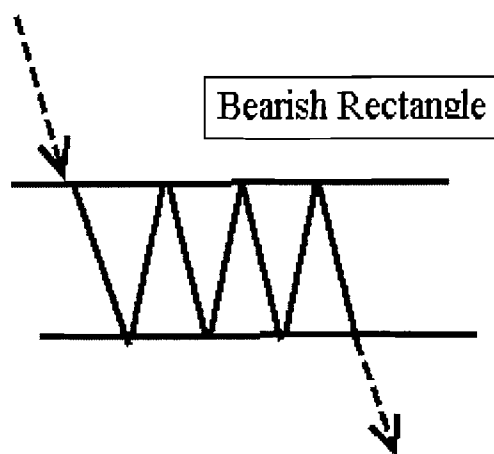
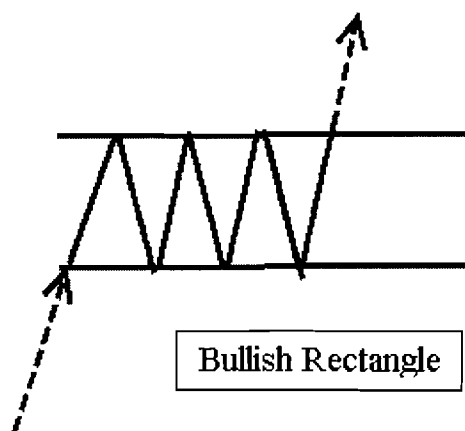
A trade setup occurs when a price closes outside the trend line after at least two penetrations on each side of the boundaries. Trades are entered on a follow-up bar at “high” above the breakout bar or “low” below the breakdown bar.

Target:

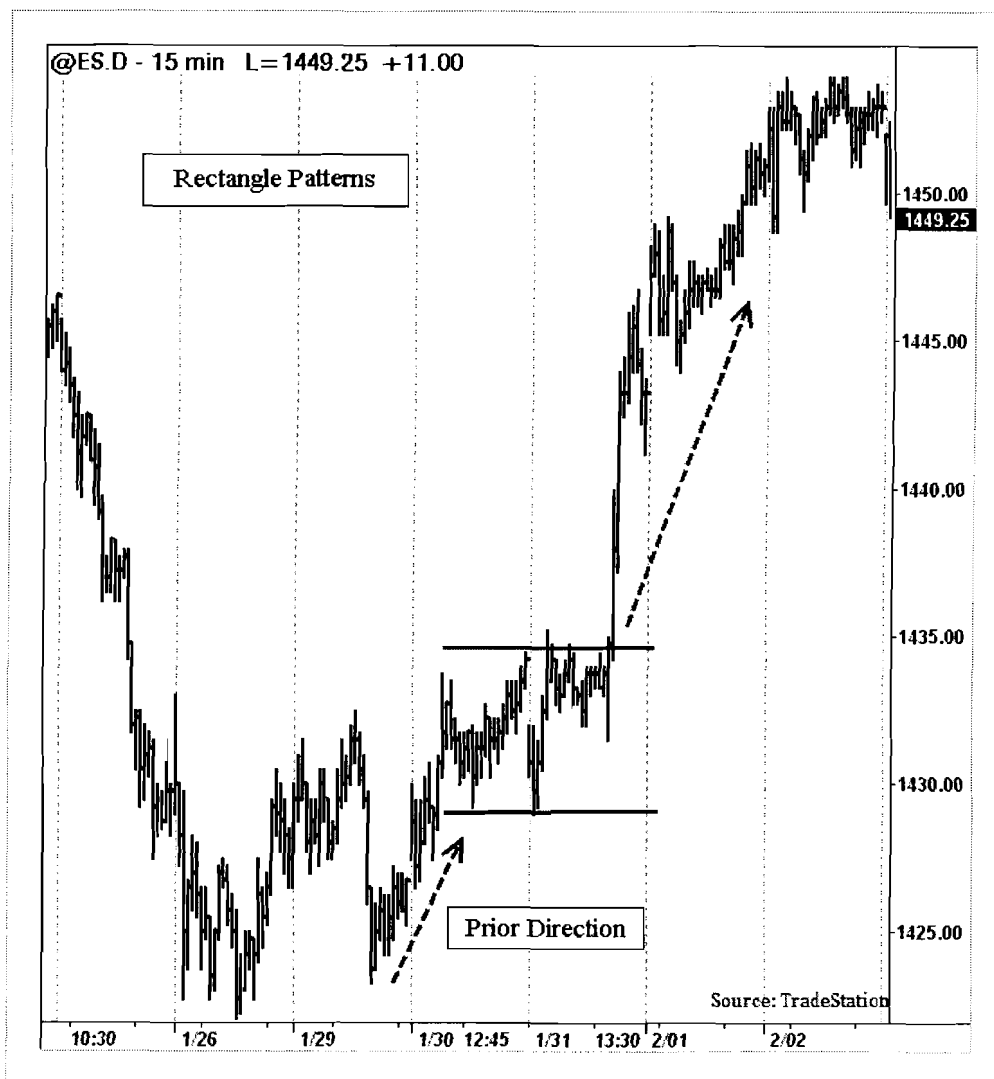
Targets in “Rectangle” formations are based on the depth of the rectangle pattern. The pattern is reliable, and targets are usually set at 70 to 100% of the depth of rectangle from the trade entry.

Stop:

Rectangle patterns fail when prices retrace in to the middle of the rectangle channel. Place a “stop” order just below/above the middle of the channel.



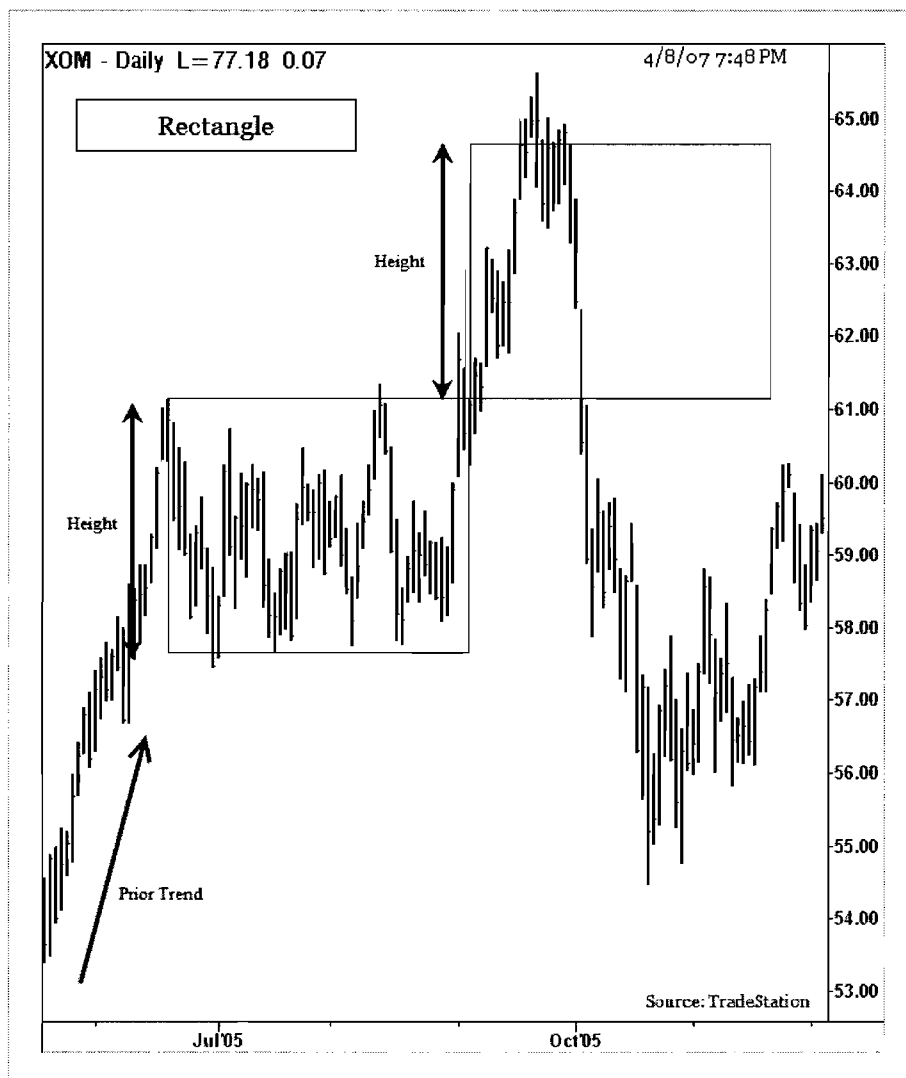
Trading Rectangle Patterns



Trading Rectangle Pattern

The example above illustrates a “Rectangle” pattern from the S&P futures (ES) 15 minute chart. On January 30, ES traded in a tight rectangle channel with highs and lows bound by two parallel trend lines. On the following day, prices traded outside the trend channel suggesting a breakout at 1437. The prior direction before the “Rectangle” formation was upside. A “long” trade is entered above the breakout bar. Targets are placed at the depth of the rectangle from the trade entry at 1440.5. A “stop” order was placed in the middle of the channel at 1433.

Trading Rectangle Patterns



Trading Rectangle Pattern

The example above shows a “Rectangle” formation from the Exxon Mobile (XOM) daily chart. From late June 2005 to September 2005, XOM traded in a narrow range from \$57.5 to \$61 forming a “Rectangle” pattern formation. Late September 2005, XOM broke out of the upper trading channel and closed above \$61. The trades are only entered in prior trend direction.

1. Enter a “long” trade above the breakout bar at \$61.
2. The height of the “Rectangle” pattern is \$3.5.
3. Place a “stop” order below the midline of the “Rectangle” channel at \$59.
4. Target the height of the “Rectangle” pattern above the breakout at \$64.

6.3. Flags

Bull Flag

“Flags” are continuation patterns representing a small pause in the market trend. They can be easily spotted as they appear right after a sudden and quick burst from a trading range. In dynamic and quick markets, Flags form as prices pause and move in the same direction as the prior trend after a clear breakout. Flags are known to be very reliable patterns.

“Bull Flag” patterns can be spotted when the market breaks out from a range and makes “lower highs” and “lower lows” in a tight formation. The trend lines connecting these highs and lows are near parallel. Also, tight and well defined “flags” perform better than short and zigzag “flags.”

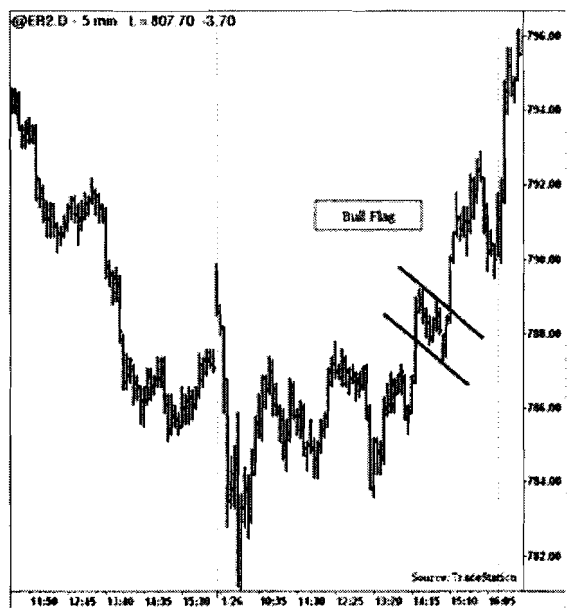
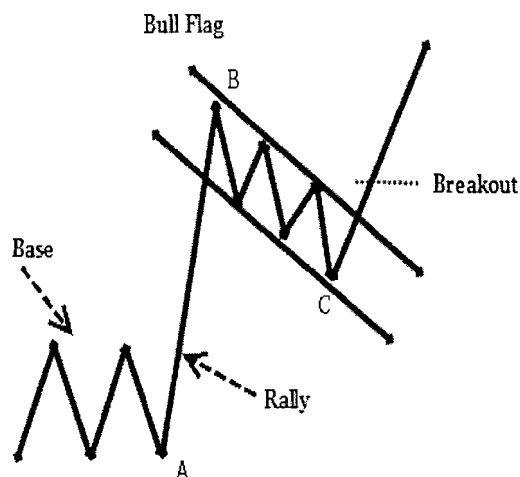
Trade:

After a series of “lower highs” and “lower lows,” connect prices with two parallel trend lines. Wait for a clear breakout to the upside. Price closing outside the upper trend line is the first sign of a breakout. Enter a “long” trade one tick above the high of the breakout bar. Another clear signal of a “Bull flag” breakout occurs when prices trade above the recent “swing high”.

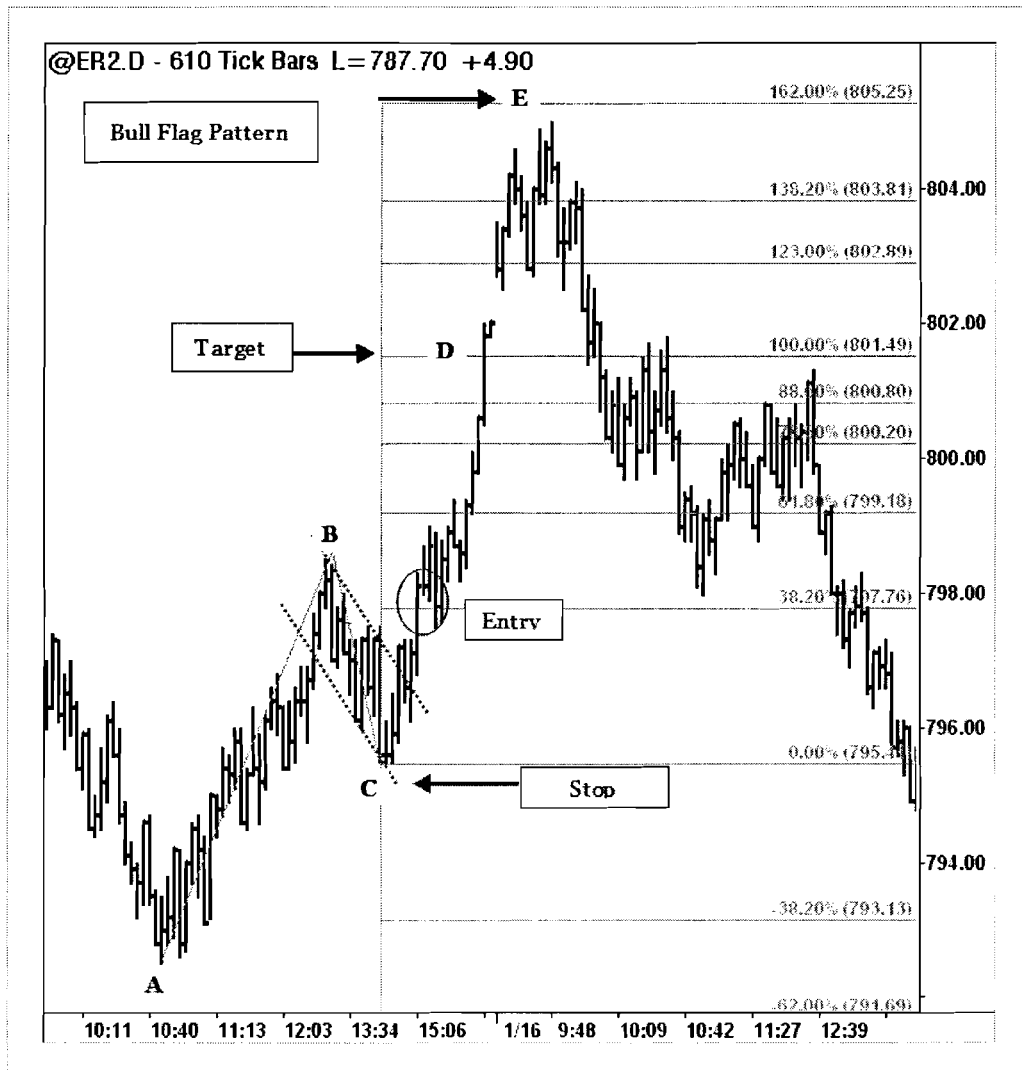
Target:

Measure the prior distance from the “swing low” at point A to the “flag” formation at point B. Target 70% to 100% of this range from C. Secondary targets in bull markets are 138% to 162% of AB from C.

Stop: Place a “stop” order below the “low” of the “flag.”



Trading a Bull Flag



Trading A Bull Flag

The chart above illustrates a "Bull flag" trade from the Russell Emini futures (ER2) 610 tick chart. After a rally from "A" on January 15, 2007, the ER2 made a "swing high" at "B" and formed a "Bull flag" with a series of "higher highs" and "lower lows". A late day rally triggered a breakout from the top trend line as prices "closed" above the previous "swing high" at 797. A long trade is entered with a "stop" order at 795 (below level C). Targets are set at 70% to 100% of the AB range from C and 138% to 162% of the AB range from "C".

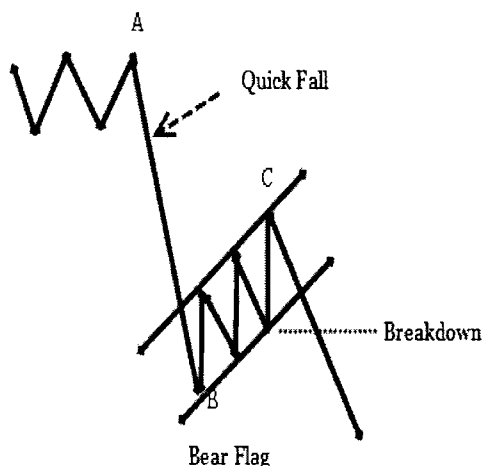
Bear Flag

“Bear Flags” usually occur as markets fall from a base and pause in a downtrend. They are almost identical to “Bull flags,” but in the opposite direction. “Bear flags” can be easily spotted as they make “higher highs” and “higher lows” within the “flag” area. The trend lines connecting “highs” and “low”s are almost parallel. A clear breakdown confirmation is needed to trade these patterns as the price continues in the same direction prior to the “flag” formation. Like “Bull flags,” “Bear flags” are also very reliable.

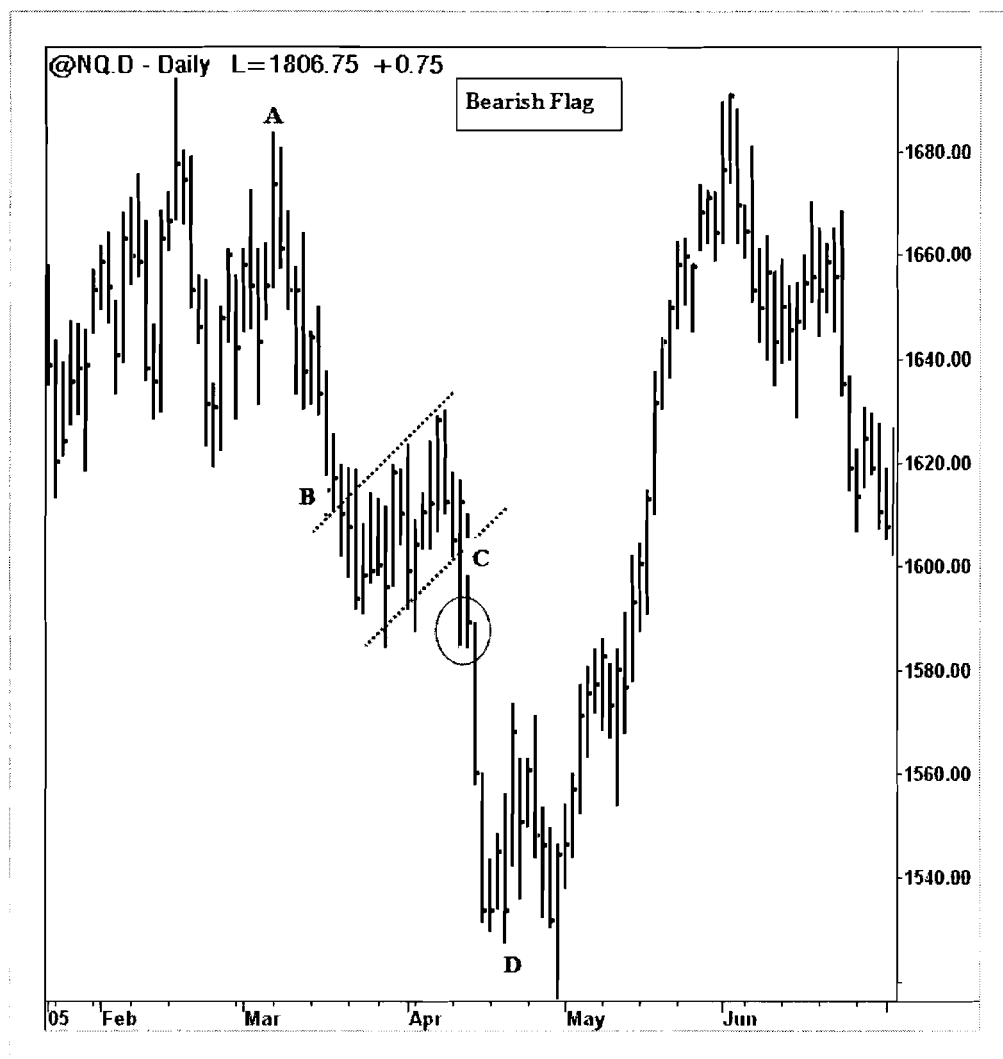
Trade: After a series of “higher high” tops and “higher low” bottoms, prices will breakout of the lower-trend line. Wait for confirmation of breakdown with a long range bar. One of the best confirmations occur when prices “close” below a previous “swing low” (of bear flag). Enter a “short” trade one tick below the “swing low” or previous bars’ low.

Target: A typical target in “Bear flags” is from 76% to 100% of the AB range prior to the “Bear flag”. The secondary targets are from 138% to 162% of the range AB.

Stop: Place a “stop” order above C to protect the “short” trade.



Trading a Bear Flag

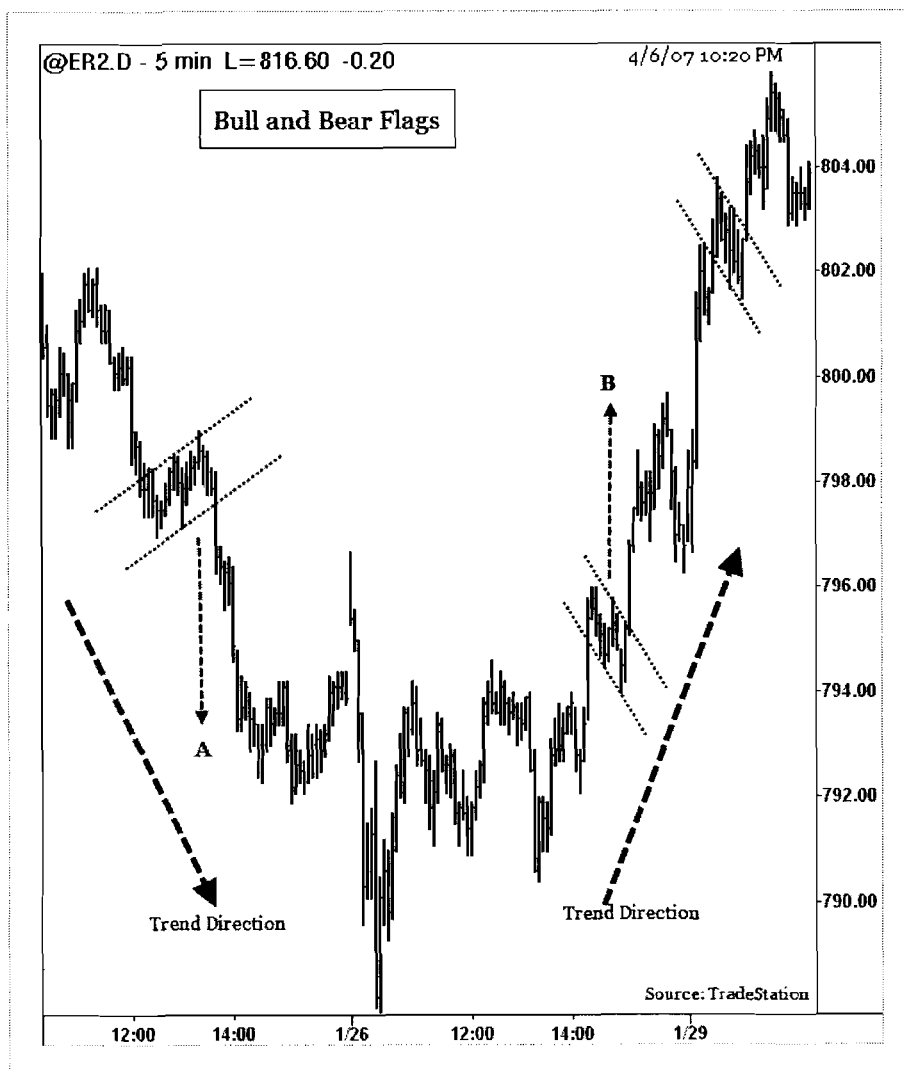


Trading A Bear Flag Pattern

The example above illustrates a “Bear flag” formation from the Nasdaq Emini futures (NQ) daily chart. In March 2005, NQ made a swing high (A) and by late March 2005, NQ prices sold-off to 1580 level. A retracement of 38% into the AB swing (at C) formed a “Bear flag”. A breakdown bar (in the prior down trend direction) below the trendline (at C) gave a short trading opportunity.

1. Enter a “short” trade below the low of the breakdown bar (at 1580).
2. A “stop” order was placed above the “high” of the top trend line (swing high).
3. Targets were set at 70% to 100% of the AB range prior the “Bear Flag” from C.

Trading Bull and Bear Flags



Trading Bull and Bear Flags

The example above illustrates “Bull” and “Bear” flags from the Russell 2000 chart. Intraday charts produce more opportunities to trade “Flags” as the results will be known quicker than day or longer term charts. “Flag” patterns are continuous patterns and trades are only taken in the direction of the current major trend. In the chart above, “trade” A is taken in the direction of a major trend after a breakdown bar from “bearish flag.” A “short” trade is triggered below the low of the breakdown bar. A “stop” order is placed above the high of the “flag. Targets are set at 70 to 100% range of the “Flag.” Similarly, a “long” trade B is triggered in the uptrend. A “stop” order is placed below the low of the “flag” pattern and a “target” is set at 70 to 100% range of the “Flag.”

6.4. Wedge Patterns

Rising Wedge Pattern

“Rising Wedge” patterns are similar to “Symmetric Triangles” but “Rising Wedge” patterns form in an angle where as “Symmetric Triangles” are mostly horizontally formed. “Rising wedge” patterns have higher highs and higher lows and are connected with two angled (slanted) trend lines. These trend lines converge at the top. The price must intersect each trend line at least twice before the pattern fully emerges.

“Rising wedges” are usually bearish in both uptrend and downtrend markets. In addition, they have a high failure rate and are relatively difficult to spot them. They seem to work well in bullish markets.

Trade:

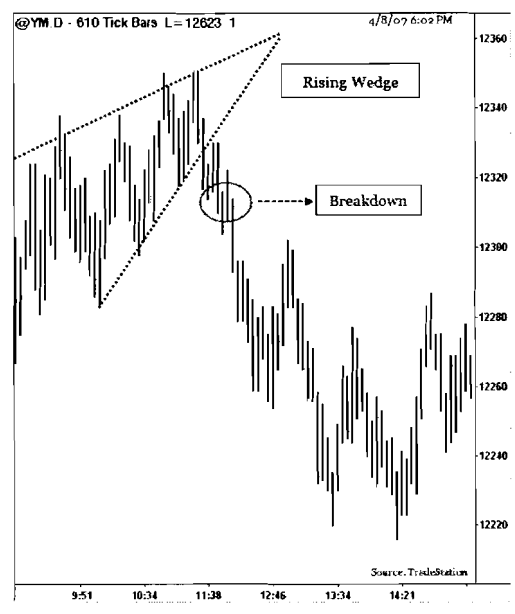
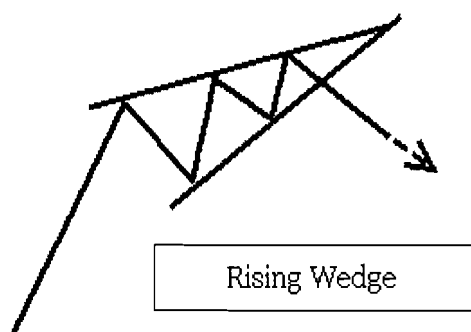
“Rising wedges” are defined by the trend lines connecting the highs and lows of the pattern. The price trading outside the lower trend line signals a potential short trade. A “short” trade is entered when the prices close below the breakdown’s bars low (must be below the trendline).

Target:

After trade entry, a target is set at the lowest point in the wedge formation. Another target measure would be the length of “wedge” pattern from the breakdown level.

Stop:

Place a “stop” order above the last “swing high” of the “wedge” pattern.



Trading Rising Wedge Pattern



Trading Rising Wedge Pattern

The example above illustrates a “Rising wedge” pattern from the Russell Emini futures (ER2) 610 tick chart. ER2 made a “Rising wedge” pattern in a downtrend. The pattern suggests a pullback rally in downtrend. ER2 made higher highs and higher lows with trend lines connecting in an angle suggesting a potential opportunity for a “short” trade when prices close below the trend line.

1. Enter a “short” trade below the low of the breakdown bar (at C).
2. Place a “stop” order few ticks above previous swing high at B.
3. The first target is placed at 100% of the AB range from C.

Falling Wedge Pattern

“Falling Wedge” patterns are similar to “Symmetric Triangles” as they form in an angle; where as the “Symmetrical Triangles” form horizontally. “Falling wedge” patterns have lower highs and lower lows and are connected with two angled, slanted trend lines. These trend lines diverge at the bottom. Another type of “wedge” (inverse) pattern has trend lines converging at the bottom. The trend direction on the breakout from the “Falling Wedge” pattern would be upside.

“Falling wedges” are usually bullish in uptrend and downtrend markets. Similarly, “Falling wedge” patterns have a high failure rate. They are relatively difficult to spot them, and tend to work well in bearish markets.

Trade:

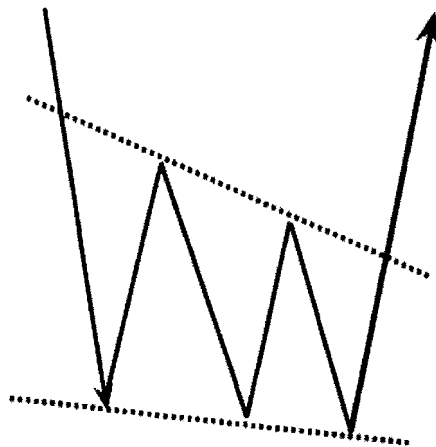
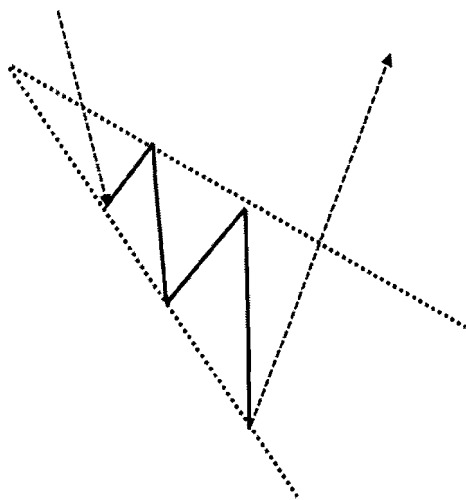
The “wedge” patterns are defined by trend lines connecting the “higher-highs” and “lower-lows.” A trend line breakout suggests a “long” trade. Trades are entered after a clear breakout from the trend line. Enter a “long” trade, one tick above the high of the breakout bar from the trend line.

Target:

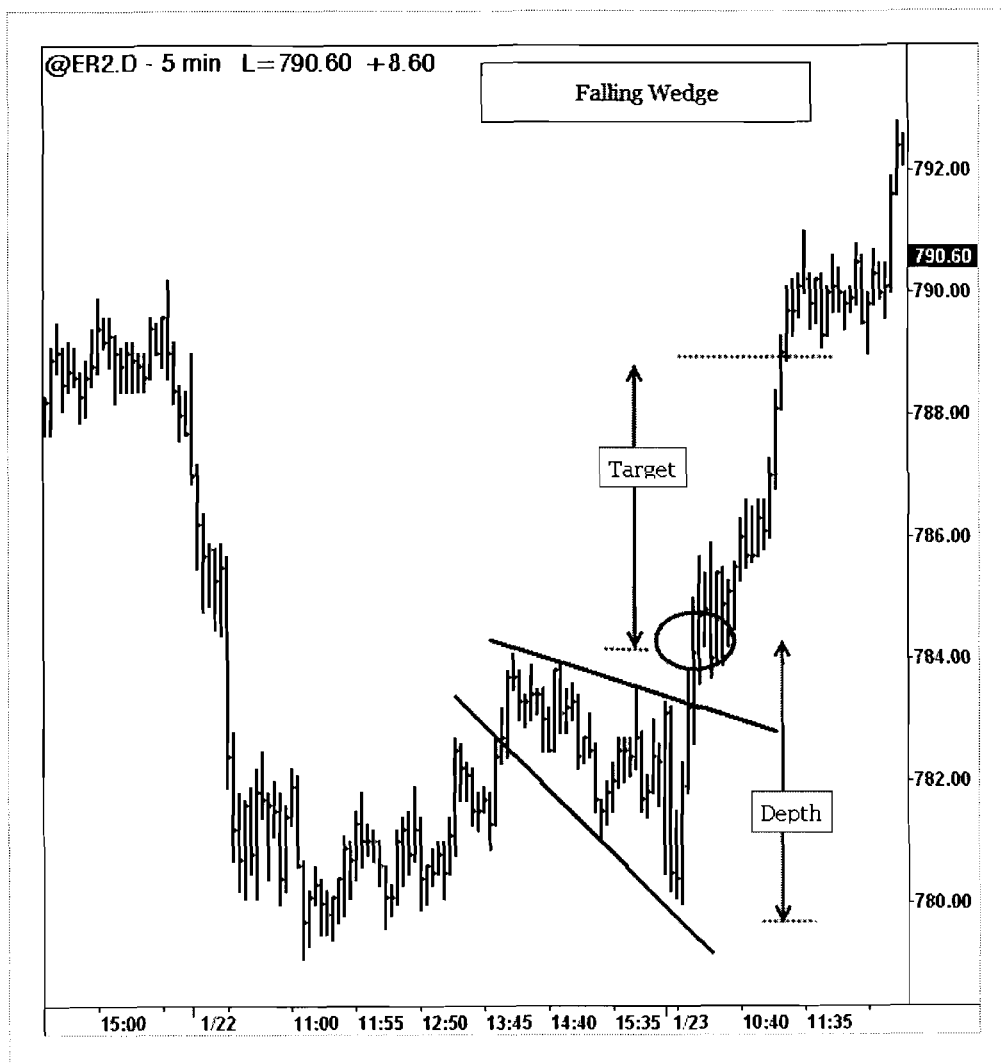
Place a target at the higher “swing high” level of the “wedge” pattern. A secondary target is set at the depth of the wedge pattern from the breakout level.

Stop:

Place a “stop” order below the lowest level of the “wedge” pattern.



Trading Falling Wedge Pattern

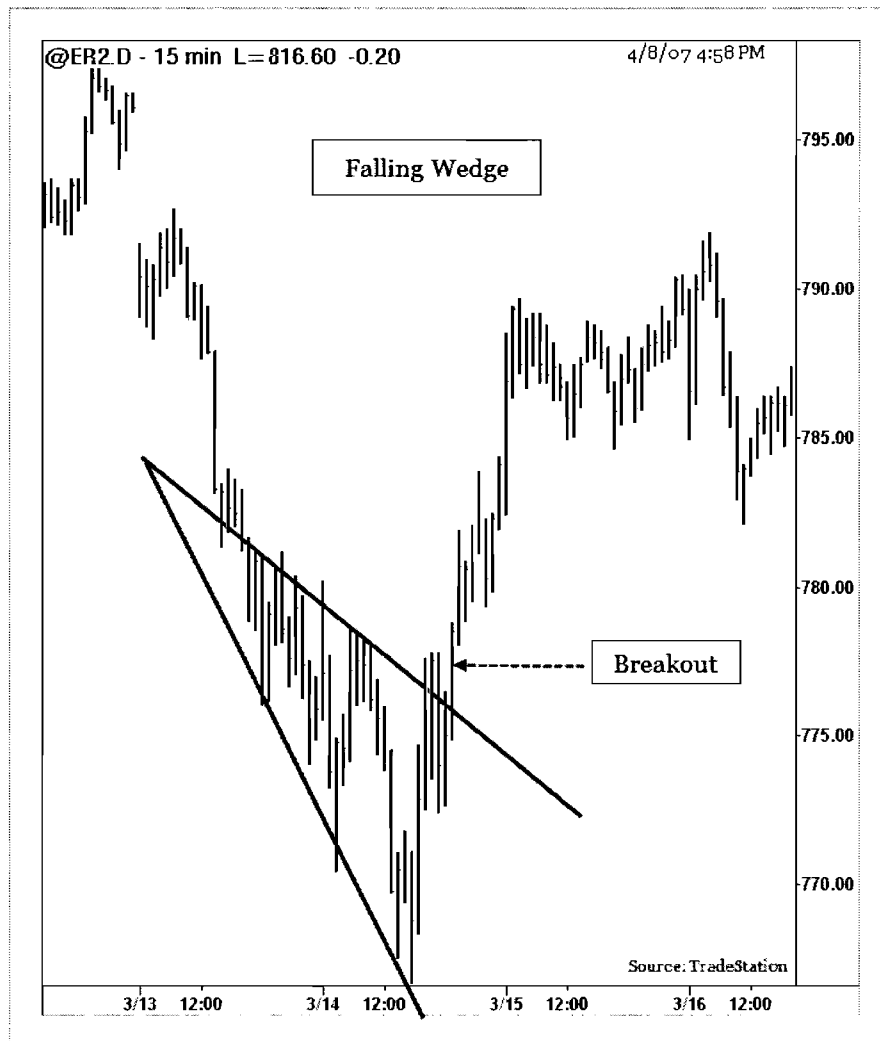


Trading Falling Wedge Pattern

The example above illustrates a “Falling wedge” pattern from the Russell Emini futures (ER2) 5m chart. “Falling wedges” are bullish patterns. On January 22, 2007, during the afternoon’s trading, ER2 made lower highs and lower lows to form a “wedge” pattern. On January 23, ER2 traded higher and closed above the trend line.

1. A “long” trade was entered above the breakout bars high.
2. A “stop” order was placed below the low of the “wedge” at the 780 level.
3. A target is set at the depth of the “wedge” pattern from the trade entry.

Trading Falling Wedge Pattern



Trading Falling Wedge Pattern

The example above shows a “Falling wedge” pattern from the Russell 2000 15 minute chart. A “wedge” pattern developed from March 13, 2007 to March 14, 2007 (until 12pm) as ER2 made lower lows and lower highs. Two trend lines are drawn in the chart above connecting these “lower highs” and “lower lows.” A “close” above the upper trend line signals a “long” trade above the 775 level.

1. Enter a “long” trade above the high of the trend line breakout bar.
2. Place a “stop” order below the low of the “wedge” pattern.
3. Target the depth of the “wedge” pattern from the trade entry level.

6.5. Diamond Pattern

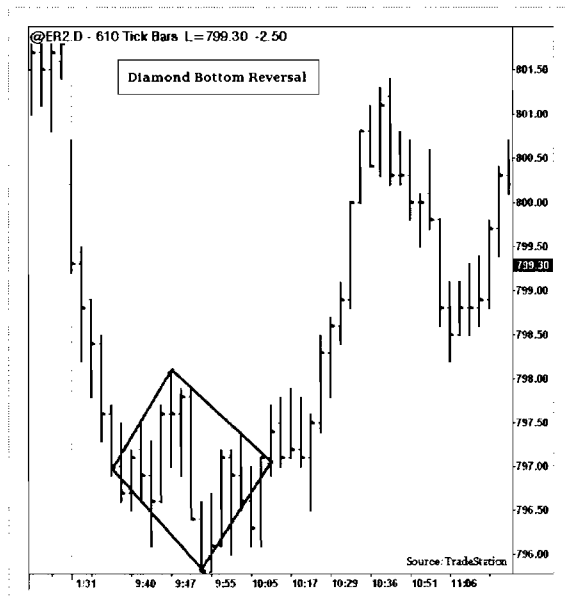
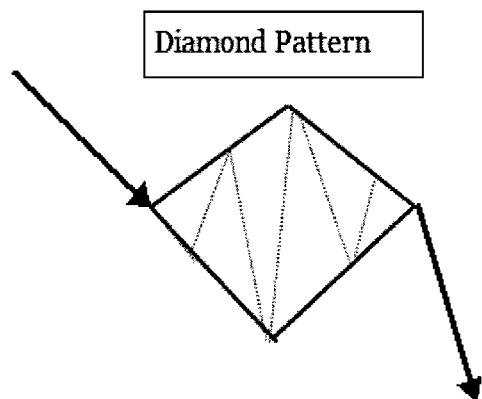
Diamond Pattern

The “Diamond” pattern is very reliable and consists of four-sided price action. It is also similar to adding two “triangle” patterns, Inverted and Symmetrical, together. “Diamond” patterns are both continuation and reversal patterns. If the pattern occurs in the midst of a trend, it will present a continuation of the trend in the same direction prior to the formation. If the pattern occurs at the tops or bottoms, it will be a reversal pattern. Diamond “tops” form more often than Diamond “bottoms.”

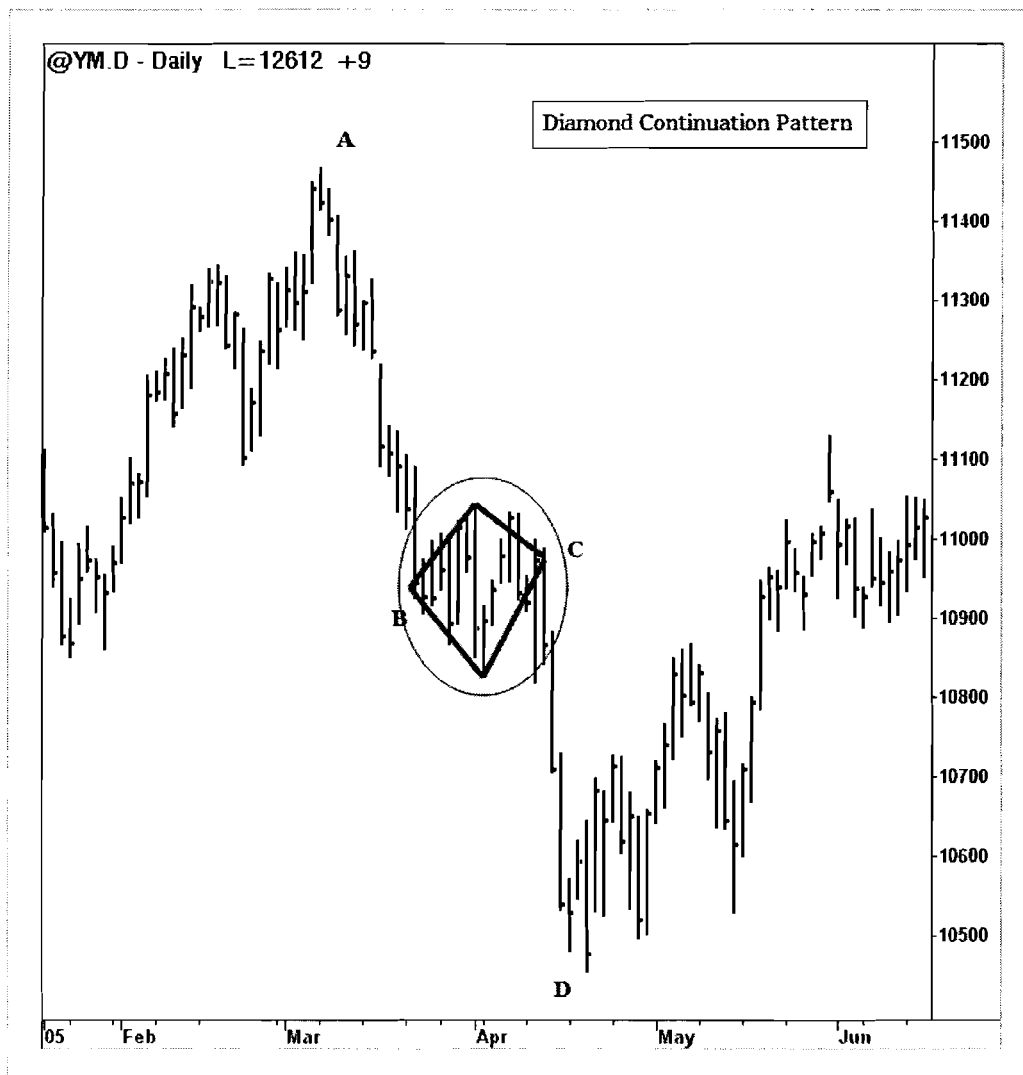
Trade: In “Diamond” patterns, continuous patterns form more often than reversal formations. In a continuation pattern, wait for a breakout of the “Diamond” pattern and trade in the direction of the trend prior to the pattern formation. In reversal formations, trade in the opposite direction of the prior trend. Confirm breakouts and breakdowns by price-action.

Target: Continuation “Diamond” patterns provide excellent target criteria. The prior range before “Diamond” formation is the target from the breakout or breakdown level. In reversal formations, prior major swing highs/lows are set as the targets.

Stop: Place a “stop” order above the high of the “Diamond” pattern for a short-trade and below the low of the “Diamond” for a long-trade.



Trading Diamond Pattern

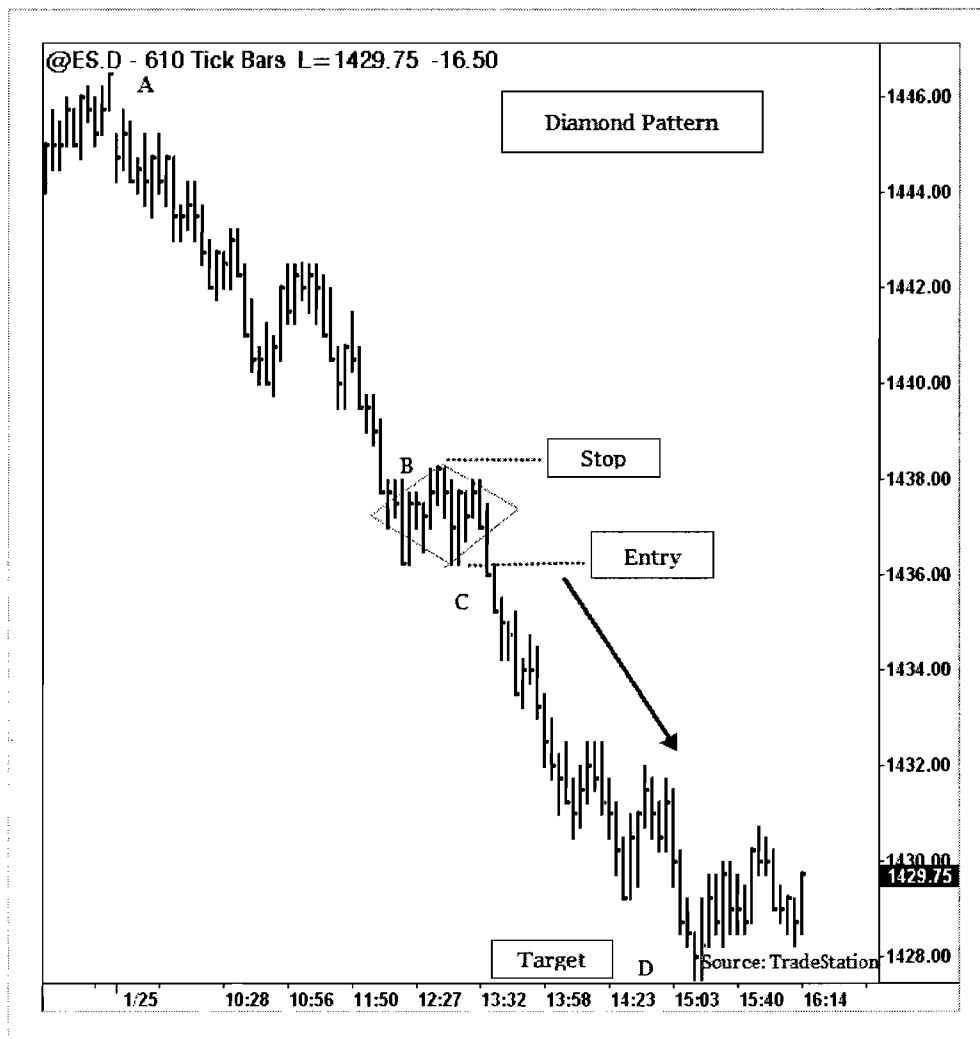


Trading Diamond Pattern

The example above shows a continuation “Diamond” pattern from the daily Dow Emini futures (YM) chart. In March 2005, after making a “swing high” of 11460 (A), YM corrected and declined to 10500 (B). While correcting, YM paused and formed a “Diamond” pattern from mid March to April 2005.

1. Wait for a clear breakdown from the “Diamond” pattern.
2. Enter a “short” in the direction of the prior trend before the “Diamond” pattern at 10850.
3. Place a “stop” order above the high of the “Diamond” at 11050.
4. Target the range of AB (500 pts) from C to 10500.

Trading Diamond Pattern



Trading Diamond Pattern

The example above shows a “Diamond” pattern continuation from the S&P Futures (ES) 610 tick chart. On January 25, 2007, after making a “swing high” of 1446 (at A), the ES corrected and traded lower to 1438 (at B). The ES formed a “Diamond” pattern from 1436 and 1438.

1. Wait for a clear breakdown from the “Diamond” pattern.
2. Enter a “short” trade in the direction of the prior trend (before the “Diamond” pattern) at 1435.
3. Place a “stop” order above the high of the “Diamond” at 1438.
4. Target the range of AB (8 pts) from C to 1428.

Chapter 7: Channels

7.1. Rectangle Channels

Rectangle Channels

“Rectangle Channels” are continuation patterns. Rectangle channel breakouts in the trend direction are more reliable than breakdowns against the trend. Rectangle channels form more in the beginning stages of longer-term trends. A base building before a clear trend run is a very reliable pattern.

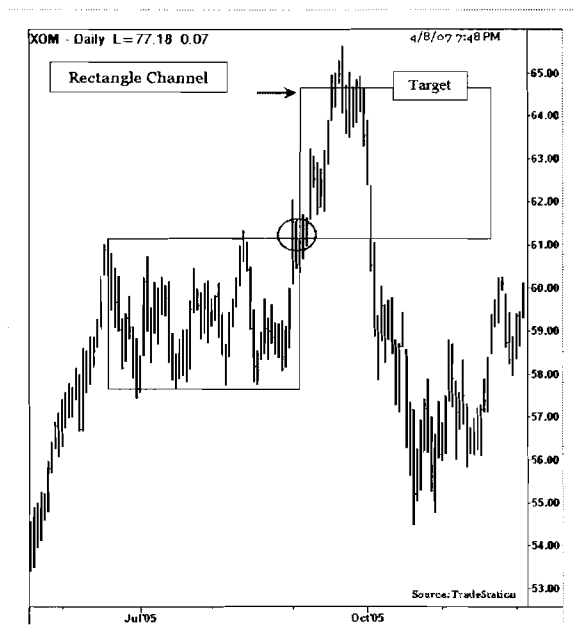
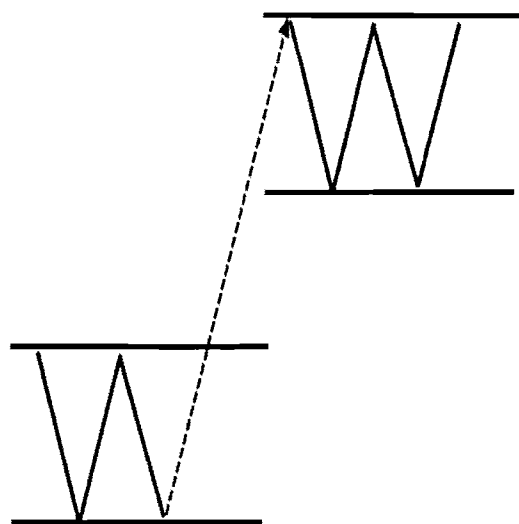
“Rectangle” channel tops are rarely formed compared to the channel “bottoms.” Intermediate trend “rectangle” formations are reliable continuation patterns and they continue in the prior trend directions.

Trade:

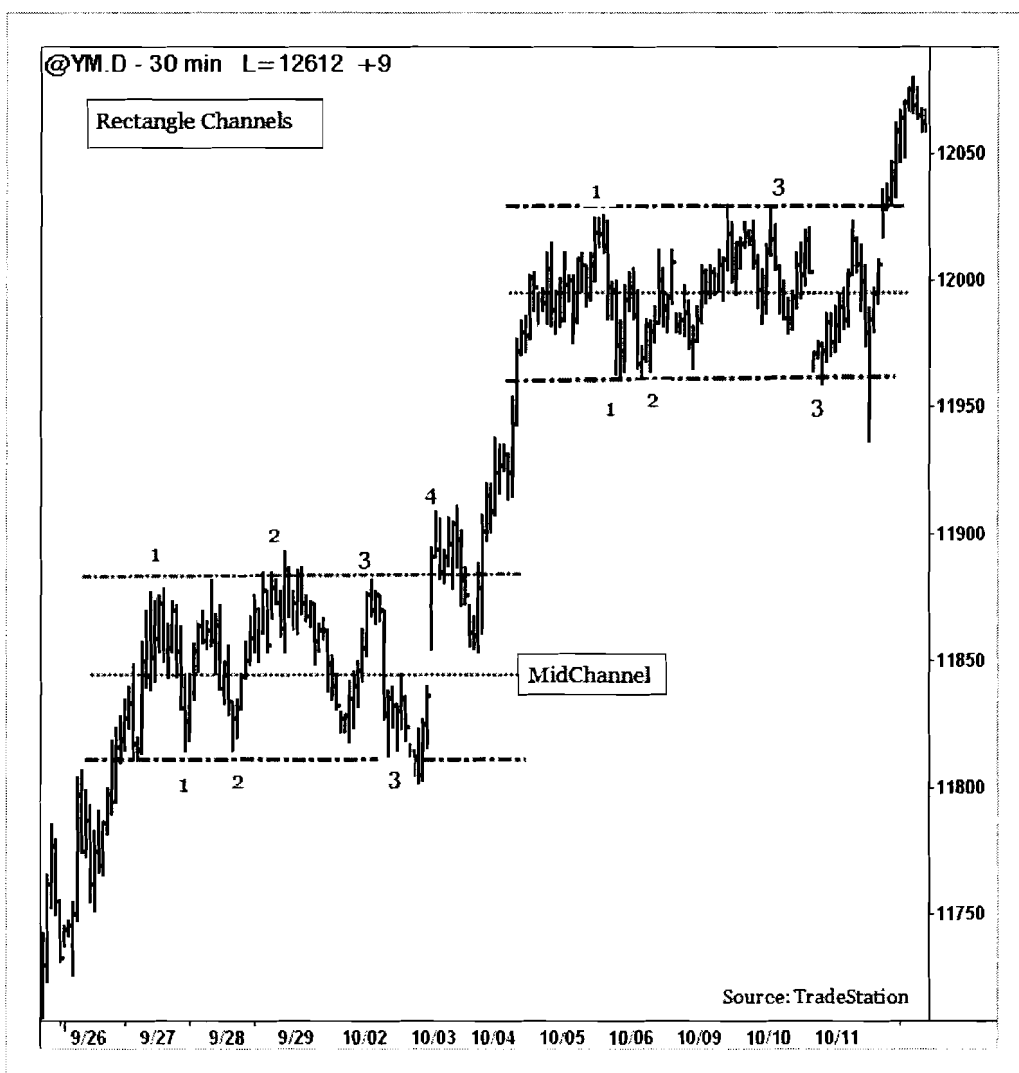
A trade in “Rectangle channels” is triggered when a top (resistance) or bottom trend line (support) is broken and confirmed by price-action. A “long” trade is entered when price closes above the high of the breakout bar. A “short” trade is entered when price closes below the low of the breakdown bar.

Target: The width of the “Rectangle” channel usually defines the resulting target move. The wider the rectangle (like base), the bigger the move. The height of the rectangle, at the top to bottom trend line, is the first target in “Rectangle” channel trades.

Stop: “Rectangle” channel failures can happen if prices trade against the prior trend before the pattern formation. A “stop” order is placed at the middle of the rectangle channel to protect the trade.



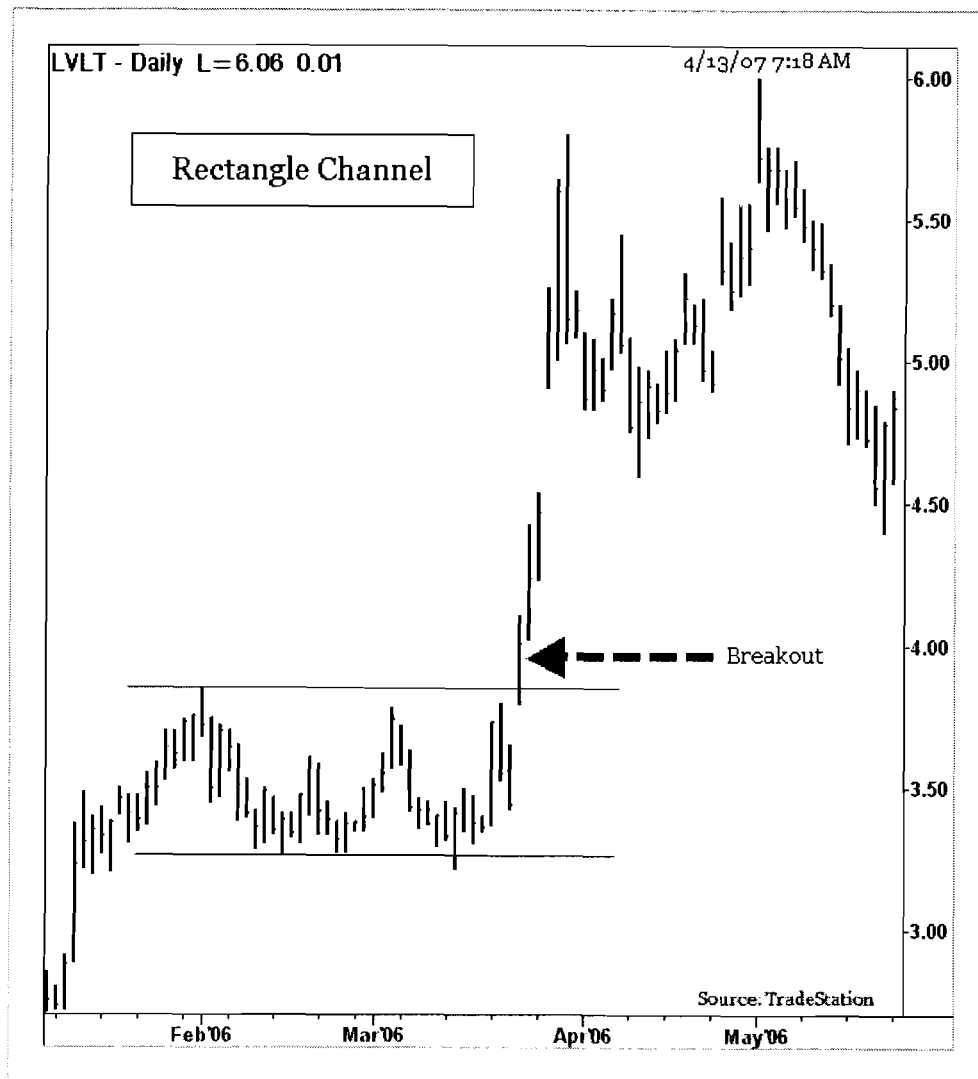
Trading Rectangle Channels



Trading Rectangle Channels

The example above shows A “Rectangle” channel formation from the Dow Emini futures (YM) 30 minute chart. The first “Rectangle channel” pattern was confirmed in early October as it closed below the lower trendline. But there was no trade triggered as prices never traded below the breakdown bar’s low. On October 3, YM traded higher and closed above the upper trend line to confirm a “Rectangle” channel breakout in prior trend. A “long” trade is entered in the direction of the prior trend before the “Rectangle” formation (up side). A “stop” order was placed at the center of the channel. A target is set at the height of the “Rectangle” channel from the breakout level. The following week, a second “Rectangle” channel emerged with a similar long trade setup.

Trading Rectangle Channels



Trading Rectangle Channel

The example above shows a "Rectangle" channel formation from the Level Three Communications (LVL) daily chart. From February 2006 to April 2006, LVL traded in a narrow range "Rectangle" channel from \$3.25 to \$3.75. In late March 2006, LVL traded outside the "Rectangle" channel suggesting a potential upside in the stock. Most "Rectangle" channels are continuation patterns and trades are entered in the prior direction of the trend before the channel formation.

1. Enter a "long" trade above the high of the breakout bar.
2. Place a "stop" order below the low of the "rectangle" channel.
3. Targets are set at rectangle's height from the breakout level.

7.2. Donchian Channel

Donchian Channel

Richard Donchian created one of the best trading systems based on 4-week price channel breakouts and breakdowns. This system is widely used by traders on a daily, weekly and monthly basis. A breakout is described as the price exceeding the high or low of n-Number of days/weeks/months. Donchian method uses 20-period channels. This method works well in both up and down trends, but it evaluates poorly in side-ways markets as most breakout systems do. There are many other variation theories which are created using the Donchian Channel breakout method.

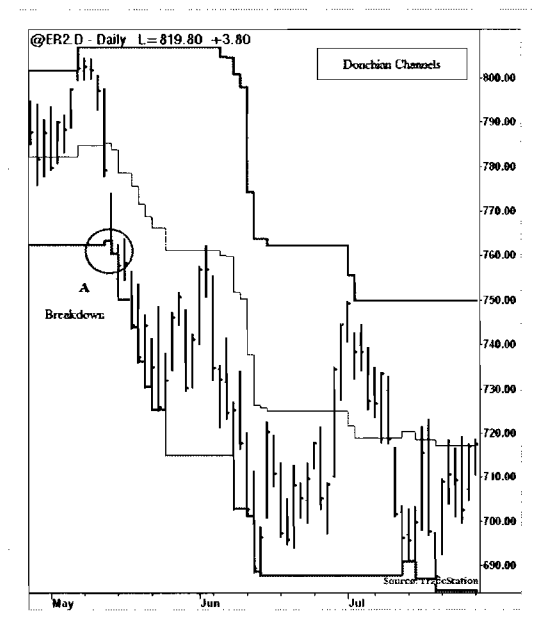
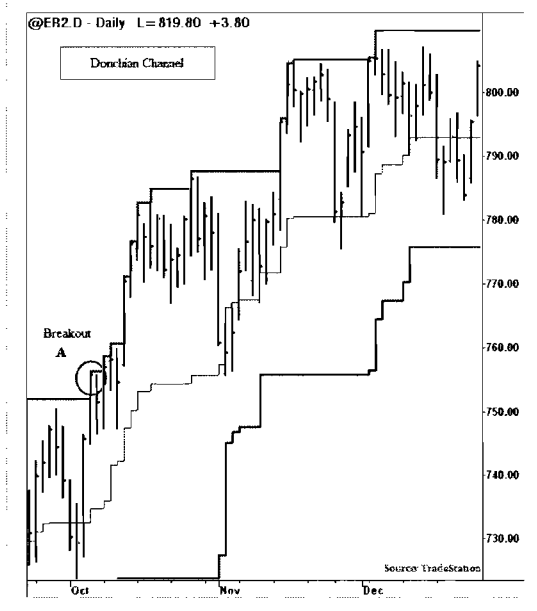
Donchain trading methods suggest entering new trades at retracement in the direction of the channel. Donchain also used a mid-channel between the “highest high” and “lowest low,” and closed positions at mid-channels.

Trade:

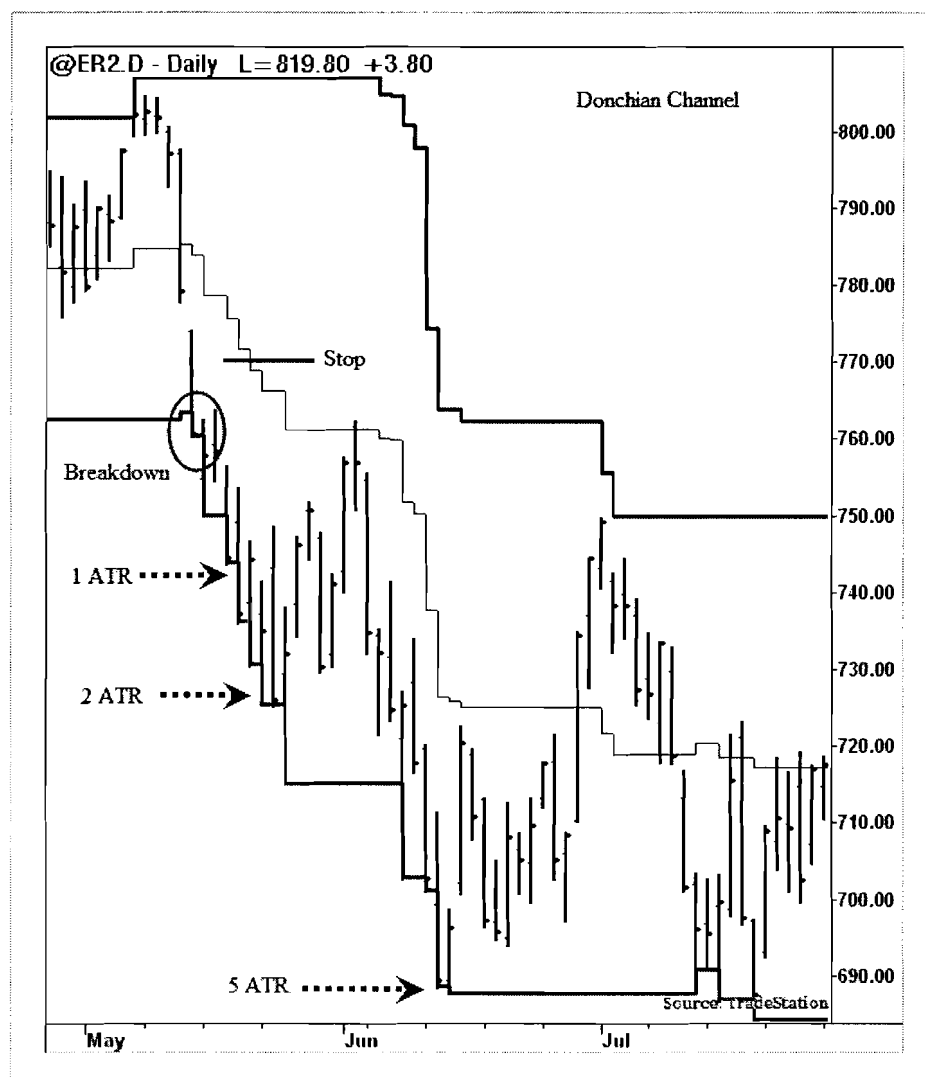
1. Enter “long” or “cover short” when price exceeds the highest high of a 4-week range.
2. Enter a “short” when price falls below the “lowest-low” of a 4-week range.

Target: Most breakouts do not result in trends. However, protection of Donchian channels at the price range or fixed profit range is necessary. Exit trades when prices reach 1.5 to 2 Average True Ranges from the entry.

Stop: Protect trades at mid-channel level or when price trades at a 10-day “low” in a “long” trade or at 10-day “high” on “short” trades.



Trading Donchian Channel

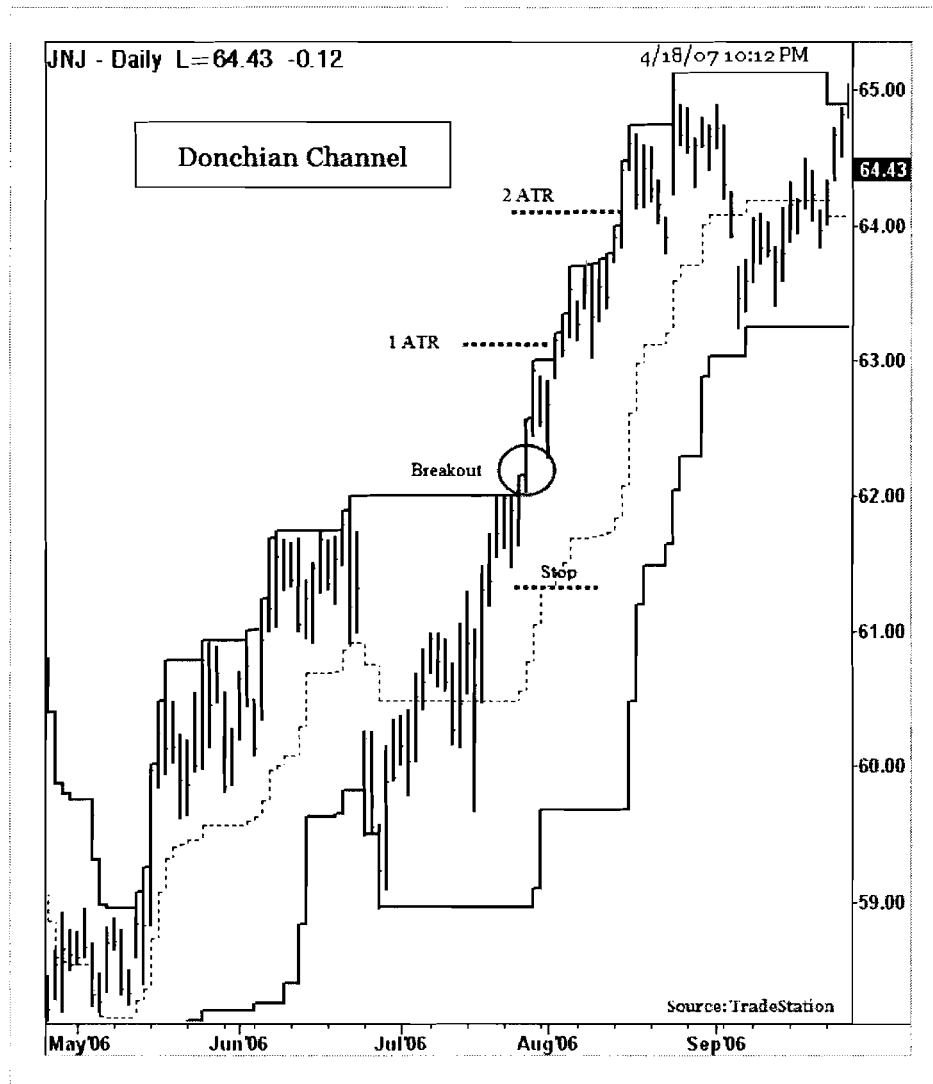


Trading Donchian Channel

The example above illustrates a Donchian Channel trading system from the daily Russell Emini (ER2) chart. In mid May 2006, ER2 triggered a “short” trade as it closed below the 4-week Donchian price channels at 755 levels. The Average True Range (ATR) in mid May was 12 points.

1. Enter a “short” trade below the low of the breakdown bar at 754.
2. Place a “stop” order at mid channel above the trade entry bar.
3. Set targets about one to two ATR levels from the breakdown levels.

Trading Donchian Channel



Trading Donchian Channel

The example above illustrates a Donchian channel from the Johnson and Johnson (JNJ) daily chart. A 4-week Donchian channel is plotted on JNJ's daily chart. In late July 2006, a long signal was triggered when JNJ closed above the 4-week upper channel at \$62. The Average True Range (10 period) for JNJ was 0.85 cents.

1. A "long" trade was entered above the breakout bar's high.
2. A "stop" order was placed at the mid Donchian channel line and used as a trail stop.
3. Targets were either set at fixed ATR lengths or used the center Donchian line as the trailing stop. The target was achieved around \$64 using a center line as the trailing stop.

7.3. Broadening Pattern (MegaPhone)

Broadening Top/Bottoms

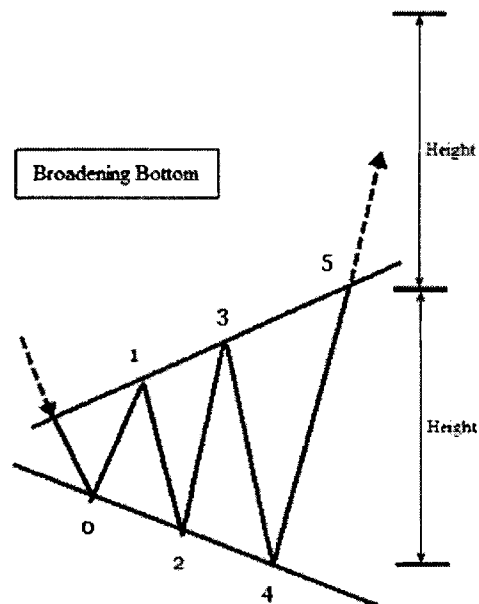
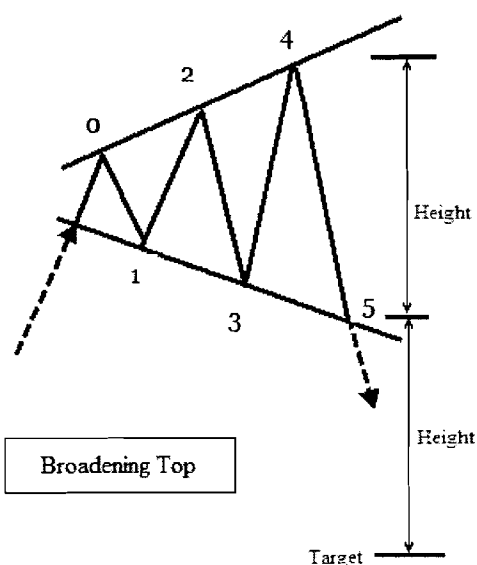
“Broadening Tops and Bottoms” are rare and difficult to detect, but they are very reliable patterns. They are also known as “Symmetric Inverted Triangles” or “MegaPhones.” This pattern consists of broadening each swing reversal with larger than the previous swing. The “Broadening top” consists of five swings points and usually the last swing reverses the trend direction that existed prior to the formed pattern. In a “Broadening top” formation, the first pivot, or turn, must be down, and in a “Broadening bottom” formation, the first pivot, or turn, must be up.

One of the critical points in “Broadening” tops and bottoms pattern is the mid line. In many cases, the rallies/declines stop at the mid line to retest the prior trend line. Usually this is the last swing before a clear breakout/breakdown occurs.

Trade: A trade is taken in the direction of the breakout/breakdown from of the pattern. When a price bar closes outside the pattern in the direction of the breakout/breakdown, and that price bar is followed by a close above the high, or a close below low of the breakout/breakdown price bar, a trade is signaled.

Target: The price objective of the trade is the height of the entire pattern added/subtracted from the breakout/breakdown levels.

Stop: The midpoint in the “Broadening” pattern is the critical point and trades should be protected with a “stop” at this level.



Trading Broadening Swing

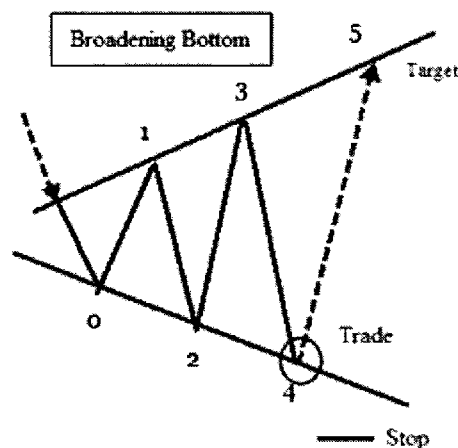
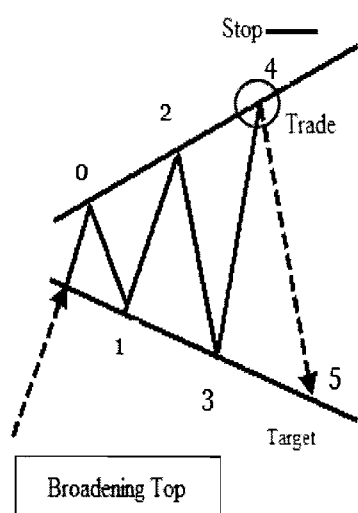
An important opportunity the “Broadening” patterns offer is trading the last swing. The number of swings in “Broadening” patterns is five. If the trend formation in the pattern is to the upside, then trading the fifth swing would be downside and can be traded from the top trendline. Similarly, if the trend direction before the pattern is to the downside, then the fifth swing will be to the upside. Keep in mind that “broadening” patterns are not infallible. However, they have a high success rate depending on where and when they form in a trend.

Trade: In “Broadening tops,” the potential trade is to go short at the beginning of the fifth swing. When the prices reach the top of the trend line, enter a short position when prices start declining and closed below the previous low. Similarly, in “broadening bottoms,” a long trade is possible at the lower trend line. Enter a long trade when prices start making higher highs above the lower trend line and when the price closes above the previous high.

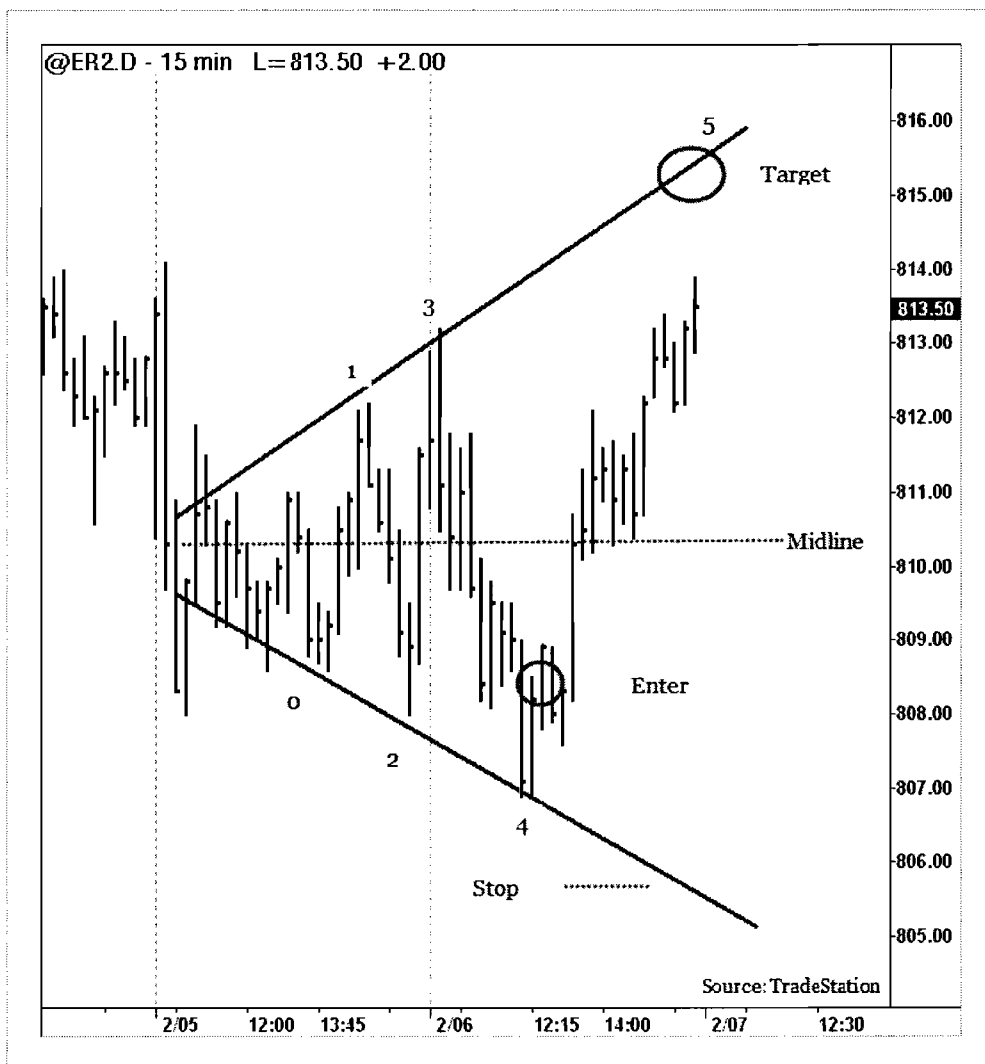
Target: A typical target for the last swing in a “Broadening” pattern would be the trend line on the opposite side of the pattern. Historically, this is the widest part of the pattern and is a very profitable setup.

Stop:

“Broadening” patterns are not infallible. They do fail and have significance as to how they fail. The pattern may be indicating a breakout/breakdown rather than a reversal of the trend. A “stop” order is placed if prices reverse and trade outside the trend lines.



Trading Broadening Swing Pattern

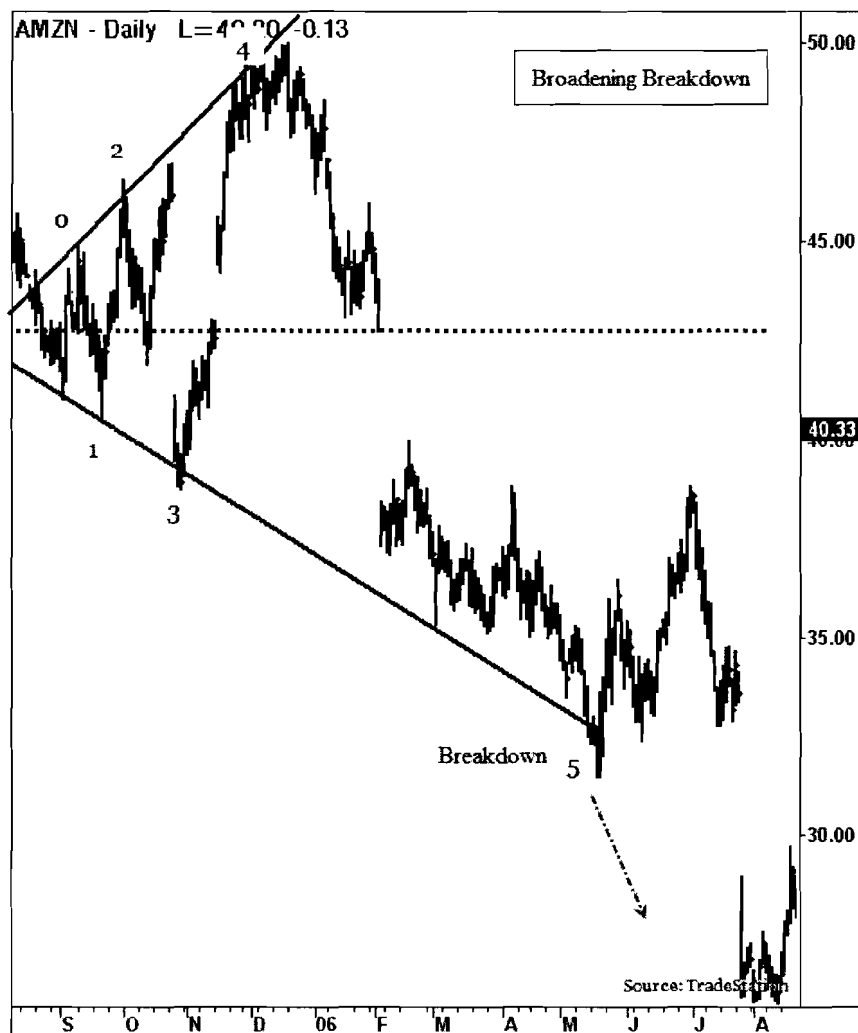


Trading Broadening Swing Pattern

The example above illustrates a “Broadening swing” trade from the Russell Emini (ER2) 15 minute chart. The ER2 chart shows two significant swings in a broadening channel as it made higher high swings and lower low swings. After two complete swings, ER2 presented an opportunity to trade the last swing from 4 to 5. When prices reach the bottom trend line at marker 4, wait for a reversal bar (close above previous high) before entering the trade.

1. Enter a “long” trade on the bar above the previous bars’ high.
2. Place a “stop” order below the trend line low.
3. Take partial profits at the mid channel line.
4. Target is set at the upper trend line.

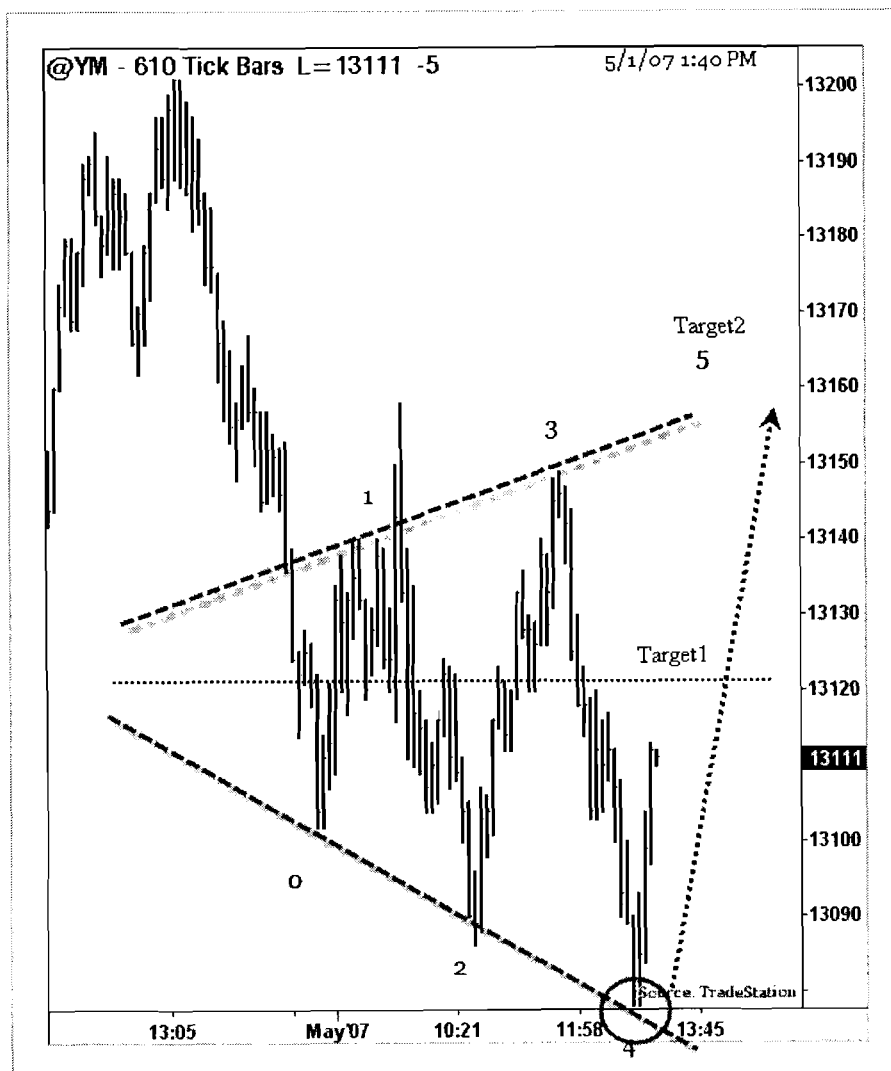
Trading Broadening Breakouts



Trading Broadening Breakdown

The example above shows a “Broadening” pattern from the AMZN daily chart. In September 2005, AMZN developed a series of higher highs and lower lows to form a “Broadening” pattern. The last swing of AMZN started in December, 2005 at \$50 and reached the other end of the pattern at \$33 (by May 2006). AMZN closed outside the “Broadening” channel in May signaling a potential reversal. Most Broadening patterns result in 5th swing continuation after breakout/breakdown. AMZN temporarily reversed its 5th swing from May to June 2005. In late June, AMZN traded below the lower trendline for a continuation of Broadening pattern.

Trading Broadening Bottom



Trading Broadening Bottom

The example above shows a Broadening bottom pattern in intra-day trading from the Dow Emini futures (YM) 610 tick chart. On May 06, 2007 Dow futures sold off and closed at 13100 levels. On May 07, 2007, YM attempted for a brief rally but continued its prior trend and closed at 13090 (2). YM attempted another rally to close above prior swing high (1) and sold off again to close below prior swing low (2) at 13080 level (4). The intra-day swings gave a “Broadening bottom” trading opportunity for the 5th swing.

1. Enter a “long” trade above the previous bars’ high at 13092.
2. Place a “stop” order below the low of the prior swing at 13078 (4).
3. Target the 5th swing above the prior swing’s high (3) at 13150. A partial position may be exited at the mid channel level (Target 1).

7.4. Linear Regression Channel

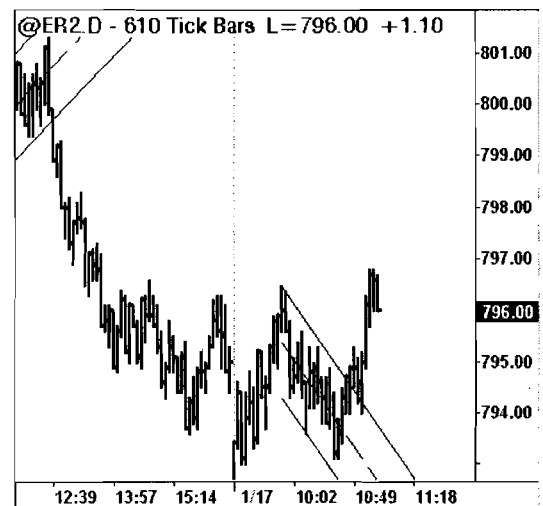
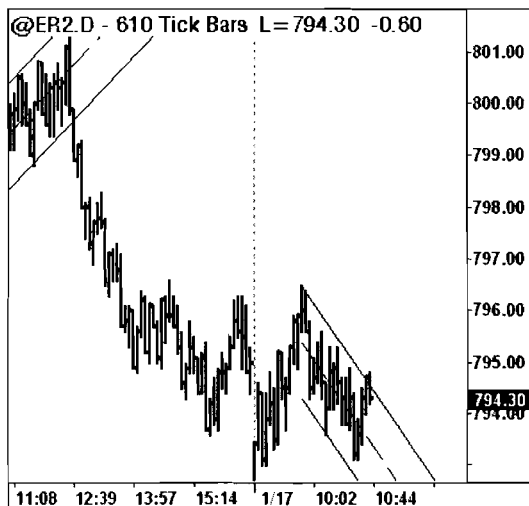
Linear Regression Channel

A "Linear Regression Channel" (LRC) is created by drawing an equal Standard Deviation (SD) distance from the "Linear Regression" based trend line. A "Linear regression" trend line shows equilibrium prices, where as "Linear regression" channels show the deviation of prices from the equilibrium or center line. A "Linear regression" channel is plotted on the price chart using the least squares method. In a LRC, the bottom channel indicates support and the top trend line indicates resistance. Prices trade within the LRC, and when prices exceed the upper or lower trend line, it signals a potential reversal. If the prices continue to close outside the LRC for about half of LRC length bars, then it may be signaling a trend change and a potential formation of a new LRC. Many traders use LRCs with the price-action to find key entry/exit trading opportunities.

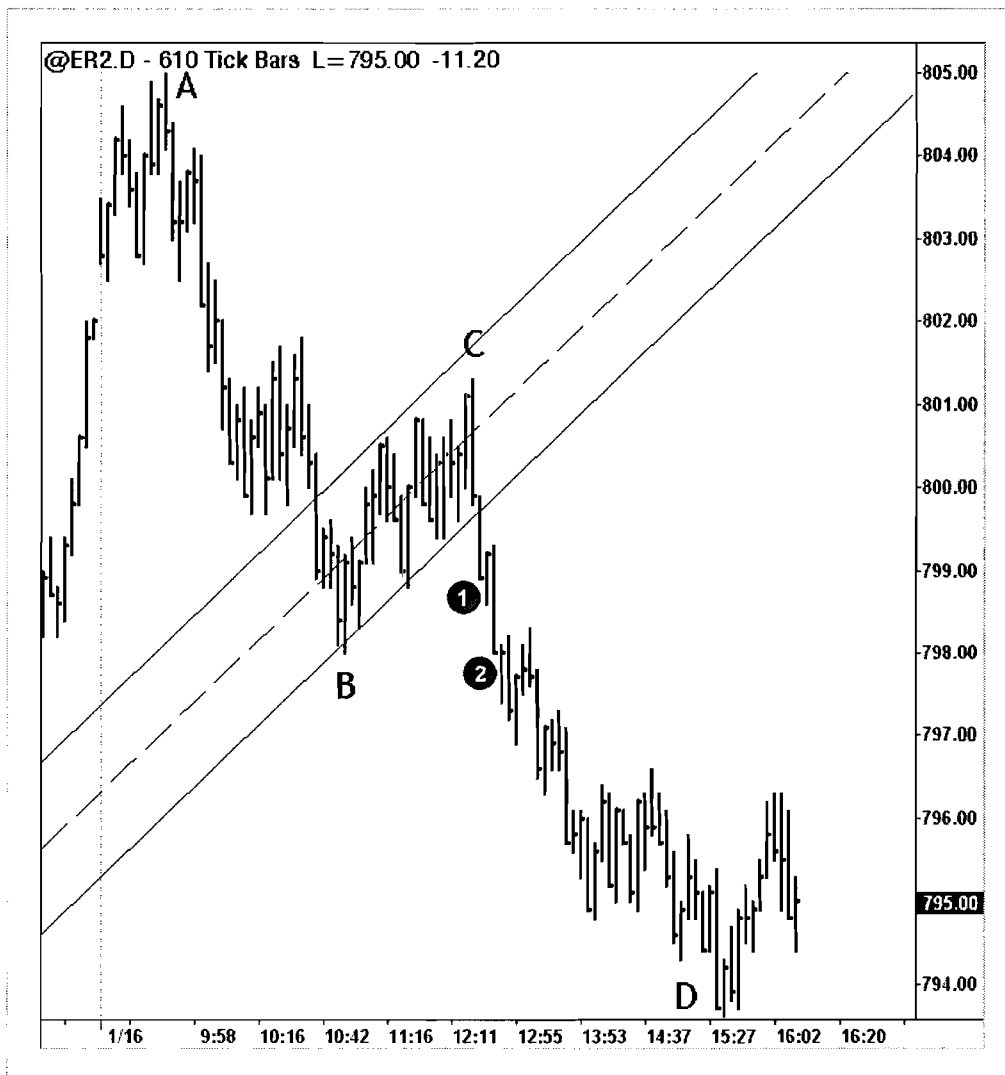
Trade: Wait for a LRC to form for at least 12-15 bars. Price closing outside the LRC suggests potential breakout/breakdowns. Traders enter above the high of the breakout bar for a "long" trade and below the low of the breakdown bar for a "short" trade.

Target: The range prior to the LRC formation would be the target from the breakout or breakdown level.

Stop: Place a "stop" order 1 tick above the high of the LRC for "short" trades and 1 tick below the low of LRC for "long" trades.



Trading LRC

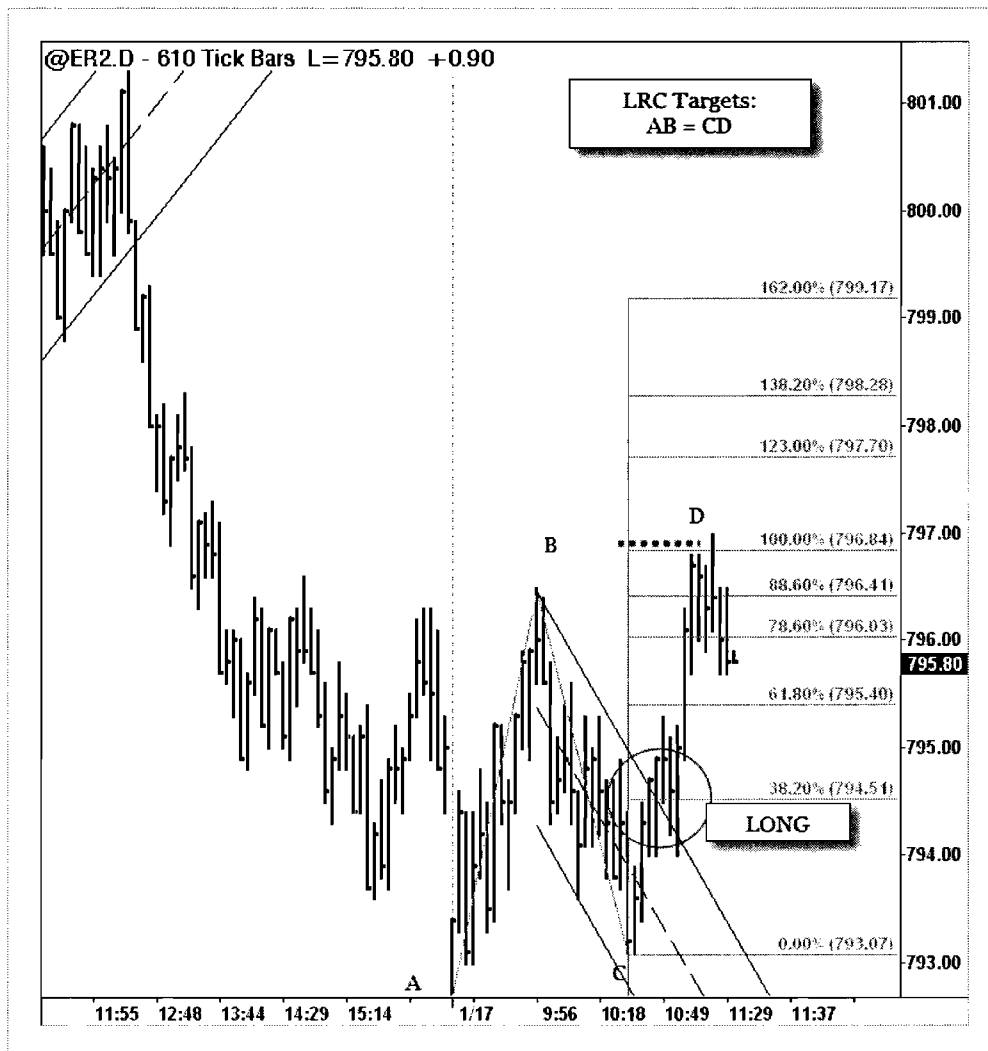


Trading Linear Regression Channel Breakdowns

The chart above shows a LRC pattern from the Russell Emini (ER2) 610 tick chart. On Jan. 16, 2007, ER2 sold off during the morning trade and closed near 798. At lunch hour, ER2 formed a LRC from 798 to 801 as prices tried to recover and traded higher. Around 12 pm, ER2 continued its prior down trend and closed below the LRC.

1. Enter a "short" trade below the low of the breakdown bar at 798.5.
2. Place a "stop" order to protect the short at 801.2.
3. The previous range prior to the LRC formation is set as the target range. The prior range is from 805 to 798. Subtract this range from level C at 801. The target is set at 794.

Trading LRC



Trading Linear Regression Channel Breakout

The example above shows a LRC breakout from the ER2 610 tick chart. On January 17, 2007, ER2 formed a LRC at about 11 am, where prices closed outside the LRC signaling a long trade at 795.

1. Enter a "long" trade at one tick above the "high" of breakout bar at 795.
2. Place a "stop" order below the low of LRC at 793.
3. Set the target (previous range, AB=CD) at 797.

7.5. Andrew's Pitchfork

Andrew's Pitchfork Pattern

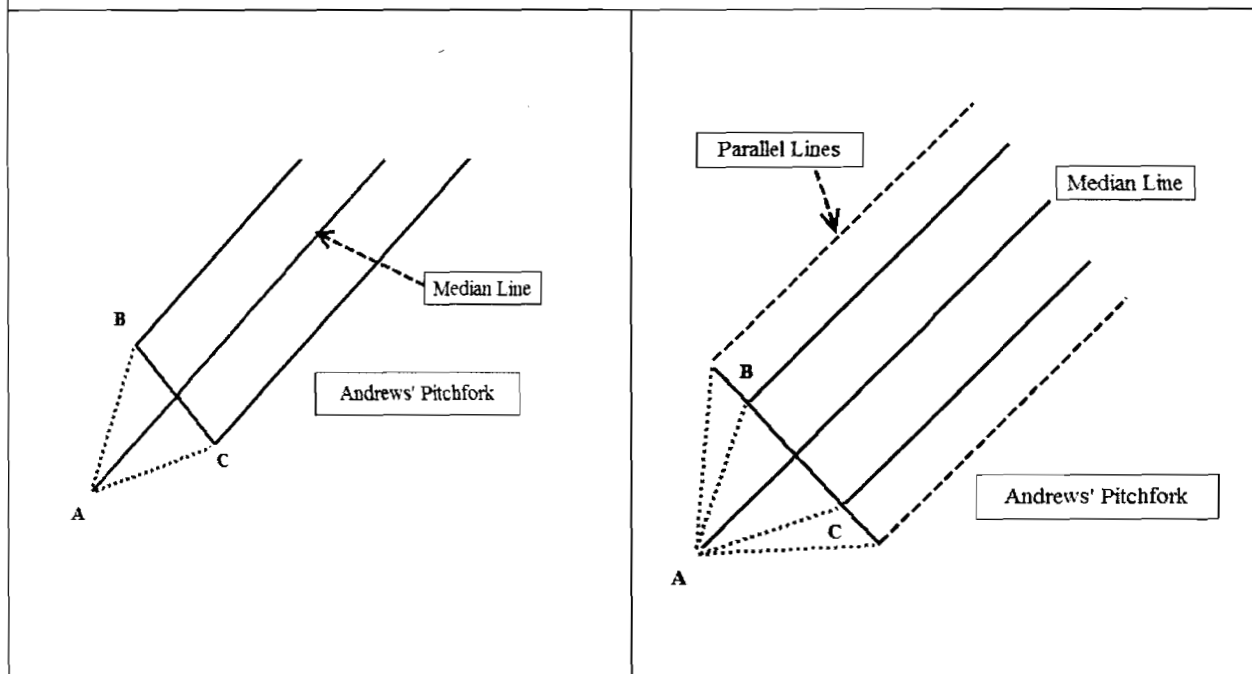
The concept of the "Pitchfork" pattern and its powerful trading method were first developed by Dr. Alan Andrews. "Pitchfork" and median line concepts work in all time-frame charts and in all instruments. Other "Pitchfork" theories were developed by Roger Babson and Timothy Morge.

Andrews' "Pitchfork" consists of three "swing" pivot points, A, B, C and three parallel lines. Construction of the "Pitchfork" is relatively simple. Draw a line connecting B and C points. From A pivot point, draw a line connecting A to the mid point of B-C. The middle line-from A is called the "median Line." Draw parallel lines from B and C along the median line to form a "Pitchfork."

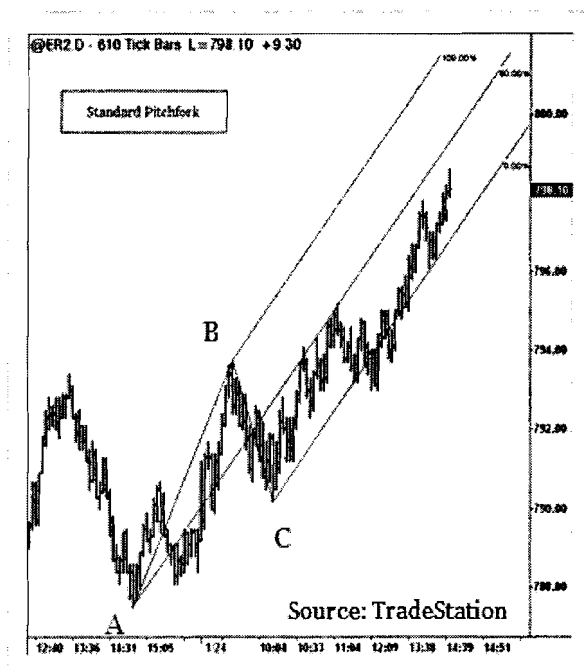
Trade: Dr. Andrews suggested that about 80% of the time, when trend is intact, prices gravitate towards the median line. Hence, when prices reach to the top trend line, take profits or short the market. Also, when prices reach the bottom trend line, enter a long position or close a short position.

Target: The targets for "Pitchfork" are usually at the median line. If the price does not stop at the median line, it will tend to move to the lower trend line (for shorts) or move to the upper trend line in case of a "long" trade.

Stop: Pitchfork upper and lower trend lines act as the "stop" levels for trades. For "short" trades, place a "stop" order above the upper "pitchfork" line. Similarly, for "long" trades, place a "stop" order below the lower pitchfork line.



Pitchfork Types



Here are a few variations of how to calculate a "Pitchfork."

Standard:

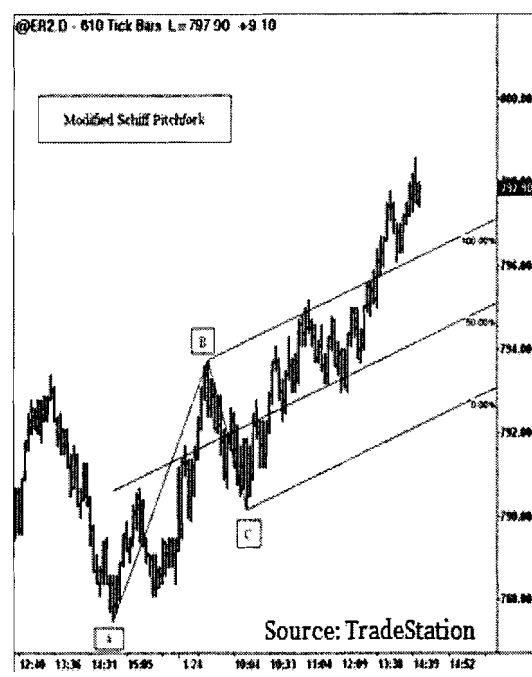
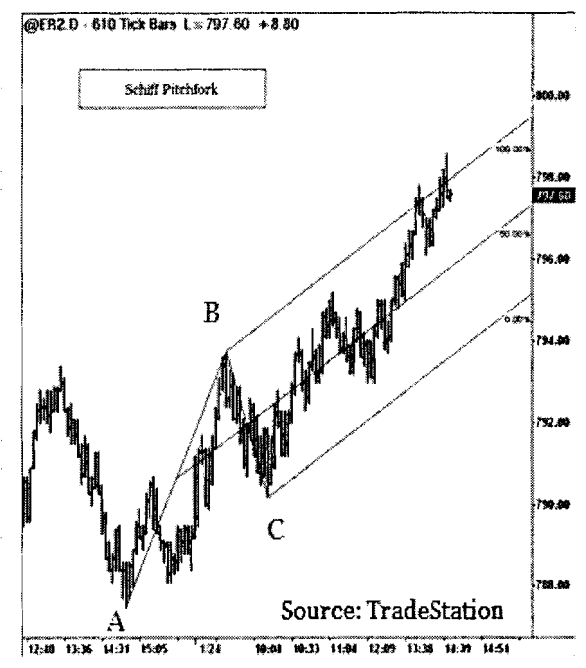
The median line is extended from A pivot point parallel to B and C pivot lines.

Schiff Method:

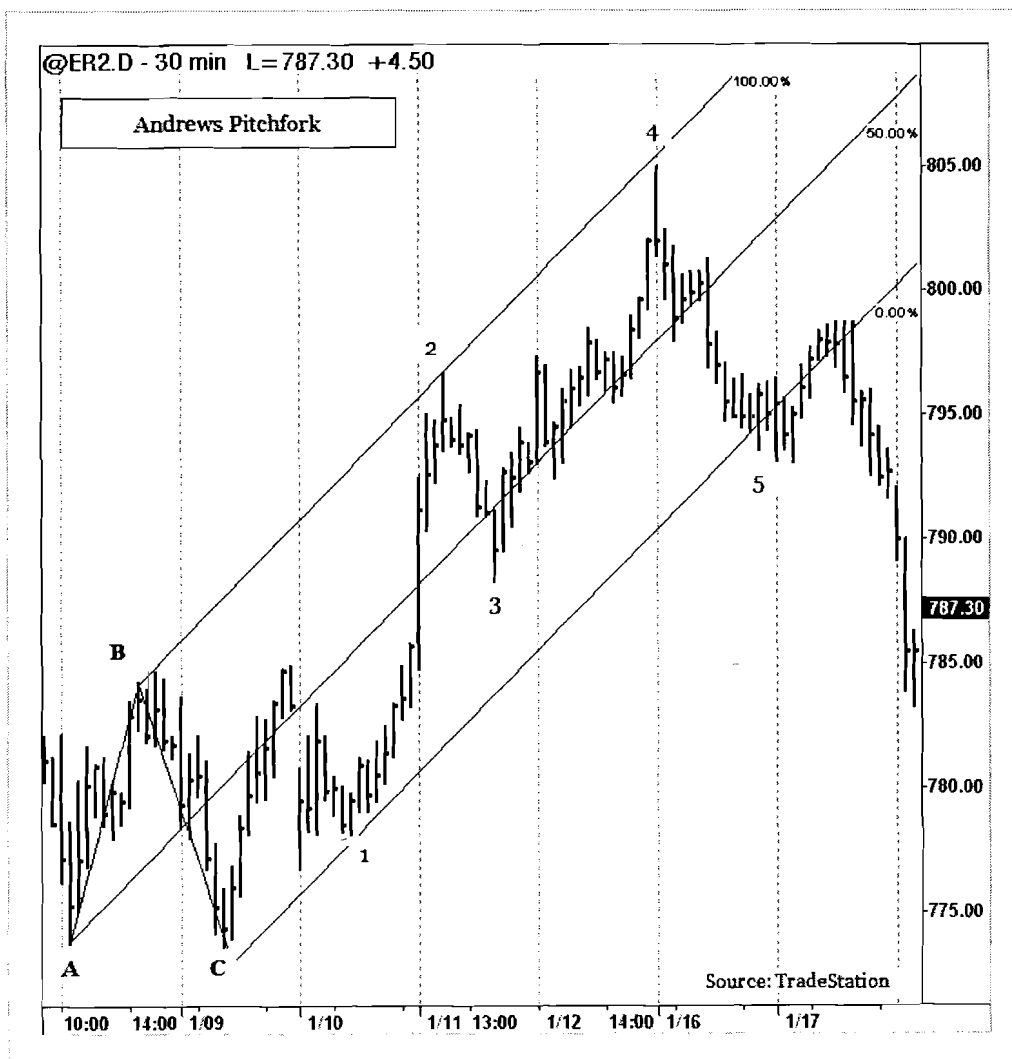
The mid point of A and B is used to draw the median line.

Modified Schiff:

A parallel line is drawn from the midpoint of A and B to B and C Pivot lines.



Trading Pitchfork



Trading Pitchfork Pattern

The example above displays a “Pitchfork” formation from the Russell Emini (ER2) 30 minute chart. The basic premise of “Pitchfork” trading is that prices trade from support to resistance in a channel format. In the example above, the “swing” pivots, A, B and C are identified and a “Pitchfork” is plotted to show the support and resistance areas. On January 10, ER2 reached the lower trend line support at 779. A “long” trade is triggered at 780 at point 1. A “stop” order was placed below the trend line. The first target is the median line at 788. ER2 rallied through the target level and closed at 796 (upper trend line). Next a “short” trade was triggered at point 4 at 805. A stop loss is placed at 807. The target was set at median line (at 799). This “short” trade continued to trade below the lower trend line (at point 5).

Chapter 8: Bands

8.1. Bollinger Bands

Bollinger Bands

“Bollinger Bands” were invented by John Bollinger, and are designed to identify price volatility using “standard deviation” around a simple moving average. As the volatility changes, these bands constrict and expand to give potential trade opportunities. “Bollinger bands” do not produce any Buy/Sell signals; however they help traders with other analysis techniques and indicators. The direction and width of “Bollinger bands” (%B) provide good trading ideas.

One of the techniques in trading “Bollinger bands” is when a new high/low price trade outside a band and subsequent highs/lows are falling inside the band. This occurrence may signal a price or trend reversal.

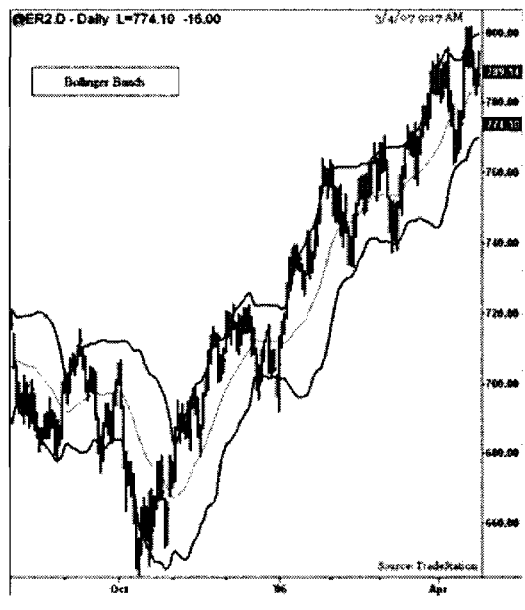
Another important technique of “Bollinger bands” is to compare the Bollinger bandwidth (%B) at key price levels. These bands constrict and provide an early warning signal before a big price change.

Traders use multiple “Bollinger bands” on multiple time-frames along with other momentum indicators. “Bollinger bands” dynamically change and provide a constant feedback to the momentum regarding the price and the direction of the trends. “Bollinger bands” are also used with other indicators to find trade opportunities.

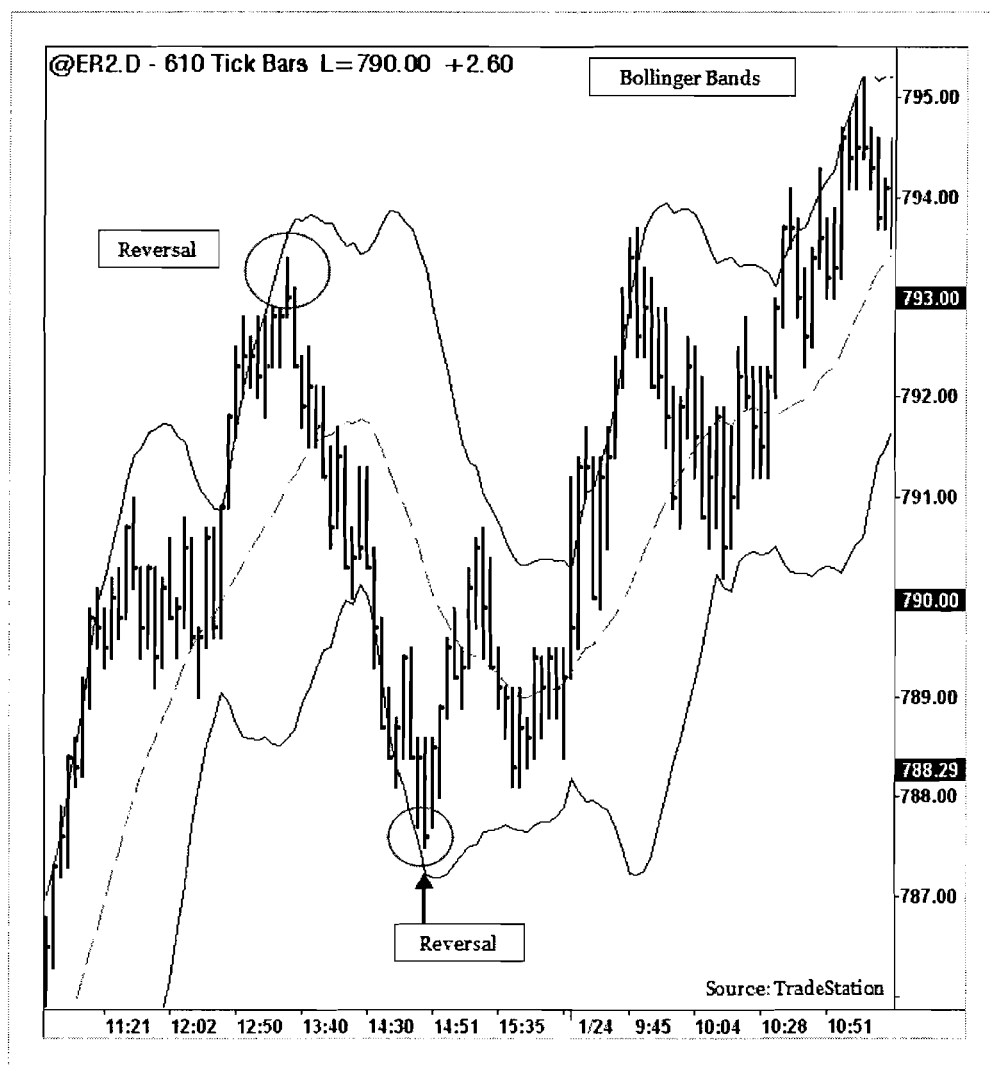
Bollinger Bands Construction

Bollinger Bands have three band like structures around a key simple moving average.

- Central: SMA (20 Periods)
- Upper : SMA + 2 Standard Deviations
- Lower : SMA – 2 Standard Deviations



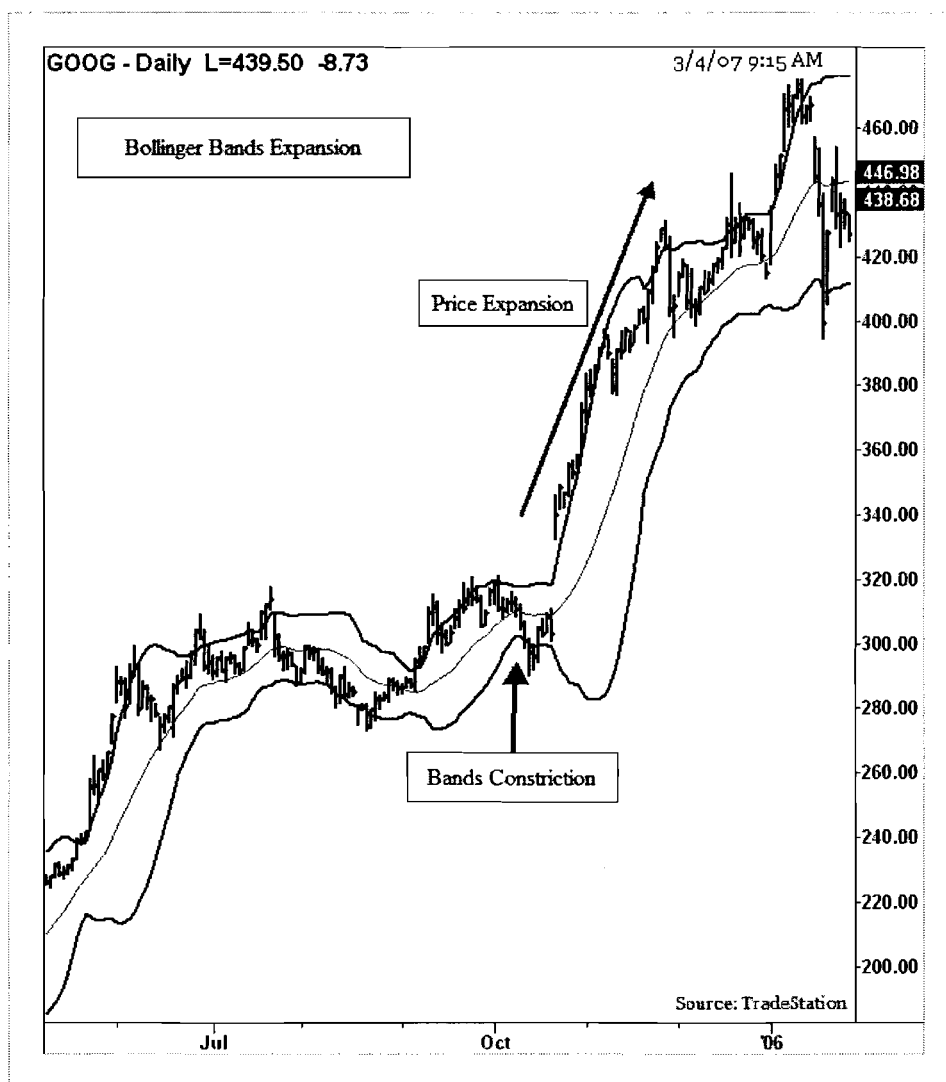
Trading Bollinger Bands



Trading Bollinger Bands

The example above illustrates a “Bollinger bands” trade setup from the Russell 2000 chart. On January 23, 2007, around 12.45 pm, the “Bollinger bands” expanded and made a new 20-bar high and pierced the bands outside of its price range. On subsequent highs, the price closed inside the bands to signal a potential trend reversal. At about 1.30 pm, the price reversed to test the lower bands. About 2.30 pm, the price made a 20-bar low and traded outside the bands. On subsequent trading, the price traded inside the bands to suggest a potential reversal to the upside. “Bollinger bands” trade setups can be traded along with other key technical indicator entry/exit signals.

Trading Bollinger Bands Squeeze



Trading Bollinger Bands

“Bollinger Bands” provide excellent feedback as their shape changes along with the price changes. They provide an early warning about “volatility” contraction and expansion. When prices trade in a consolidation range, the bands constrict (squeeze) and provide an early signal about an impending sharp price change. Price expansion makes the bands expand outward and provide a volatile shape and suggest that prices may trade outside the bands until the “volatility” returns to the norm within the bands.

The example above shows GOOG price chart with “Bollinger Bands.” In mid October 2005, the Bollinger bands contracted to suggest an impending move. After bands expansion, prices reached new highs. Trades are taken in the direction of the breakout from bands contraction.

8.2. Keltner Bands

Keltner Bands

"Keltner Bands" were first revealed from the book, *"How To Make Money in Commodities"* by Chester Keltner. "Keltner Bands" look similar to "Bollinger bands" and are computed using a single moving average with a fixed width envelope using Average True Range (ATR). The center line is the 10-period average of "high," "low" and "close" prices. The outer bands are constructed using a 10-day ATR. For a "lower" band, a 10-period ATR is subtracted from the 10-period Pivot MA and for "upper" band, A 10-day ATR is added to the 10 period Pivot MA. Trading signals are generated when prices reach outside the bands as prices may be out of the normal trading ranges and potentially may retrace back to the normal prices.

Linda Bradford Raschke, later developed various theories and added modifications for Keltner Bands using different exponential moving averages and Average True Ranges.

"Keltner Channel" trading is a trend based system method and only suitable for trending markets and fail in congestion or side-ways markets. "Keltner Bands" can be effectively used along with other indicators and patterns.

Another way to use "Keltner Bands" is to trade in the direction of the trend. When prices close above the upper band, it may signal strength and prices may continue to rise. Similarly, when prices close below the lower band, it may signal weakness and prices may continue to fall. When prices retrace back into the bands, the prior trend may signal a reversal.

Overall, "Keltner Bands" are like "moving average envelopes" or "Bollinger bands." The trading principles are similar except they are computed differently.

Keltner Bands Calculations

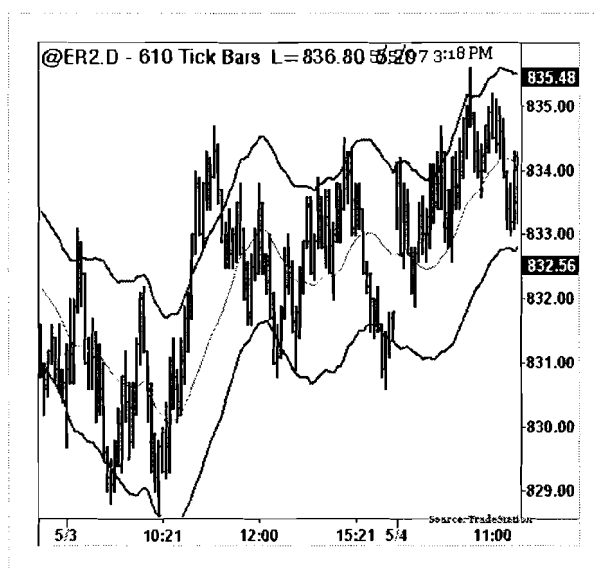
Pivot: $(H+L+C)/3$

ATR = Average(TrueRange, 10);

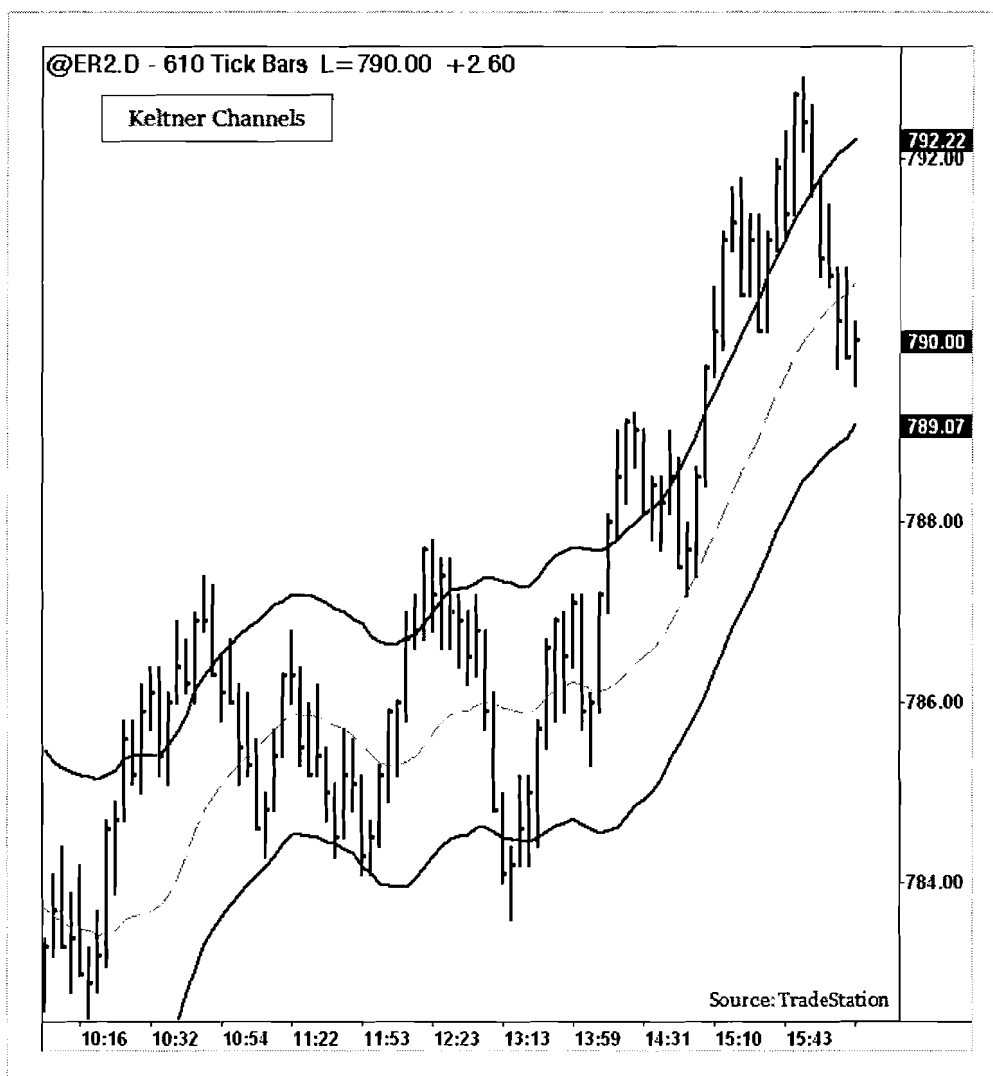
Middle Band: Pivot MA = MA(Pivot, 10);

Upper Band : Pivot MA + ATR (10);

Lower Band : Pivot MA - ATR(10);



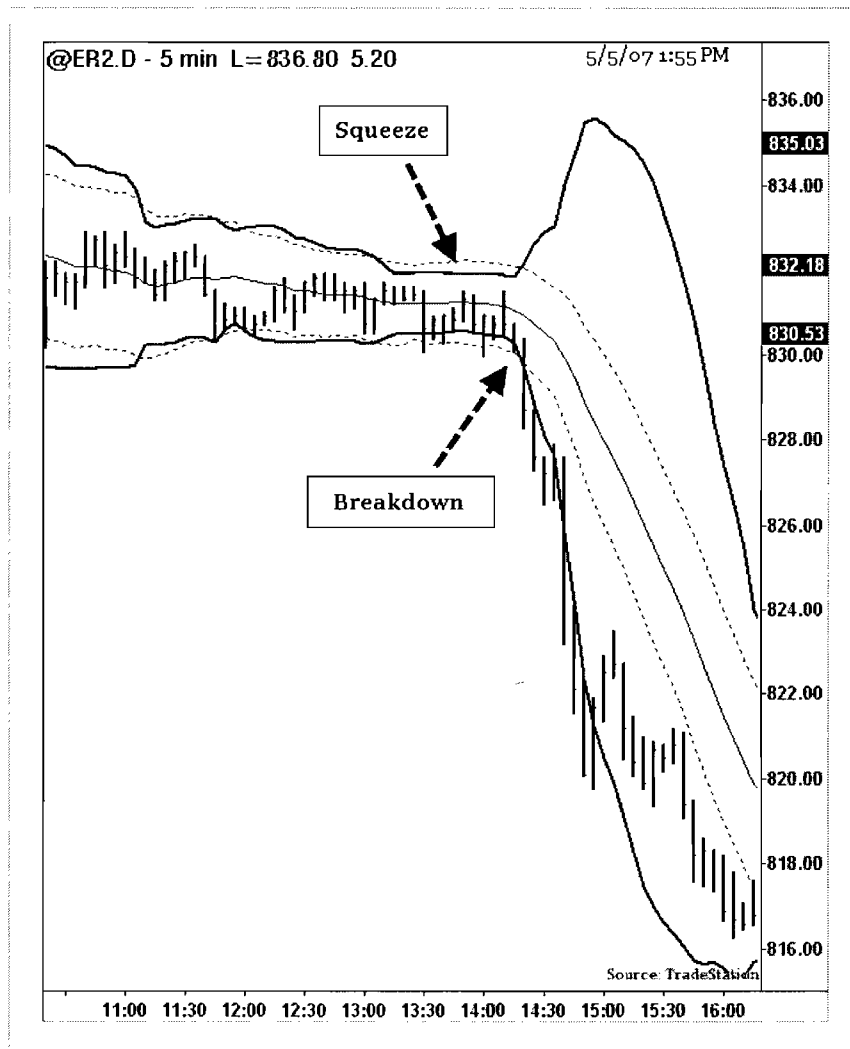
Trading Keltner Bands



Trading Keltner Bands

The example above shows “Keltner Bands” trading system from the Russell 2000 Emini chart. The middle band is represented by A 10 period Moving Average of Keltner average price $(H+L+C)/3$. The upper band is computed by adding 10-period average of ATR to the middle band. The lower band is computed by subtracting a 10-period average of ATR from the middle band. “Keltner Bands” Trading System includes buying when the markets penetrate the upper band and selling when the markets penetrate the lower band. A “stop” order is placed at the middle band to protect the trade. “Keltner Bands” are very effective in trending markets and should be avoided in side-ways or congestion zones.

Trading Keltner & Bollinger Squeeze



Trading Keltner Bands with Bollinger Bands Squeeze

The example above illustrates “Keltner Bands” and “Bollinger bands” trade setup from the Russell 2000 Emini chart. A 21 period Bollinger bands with a 2-Standard Deviation indicator and a 10 period Keltner channel with an average of a 10 period ATR is plotted. This trade setup shows volatility contraction within a trading channel as Bollinger bands squeeze inside the Keltner channel. On April 30, 2007, at about 1:30 pm, Bollinger bands squeezed inside the Keltner Bands to suggest a contraction of volatility and potential expansion move. When Bollinger bands break out of the Keltner Bands to the downside, a “short” trade is entered at the 830 level. By the end of the day, Russell 2000 closed near 816; 14 points from the entry.

8.3. Fibonacci Bands

Fibonacci Bands

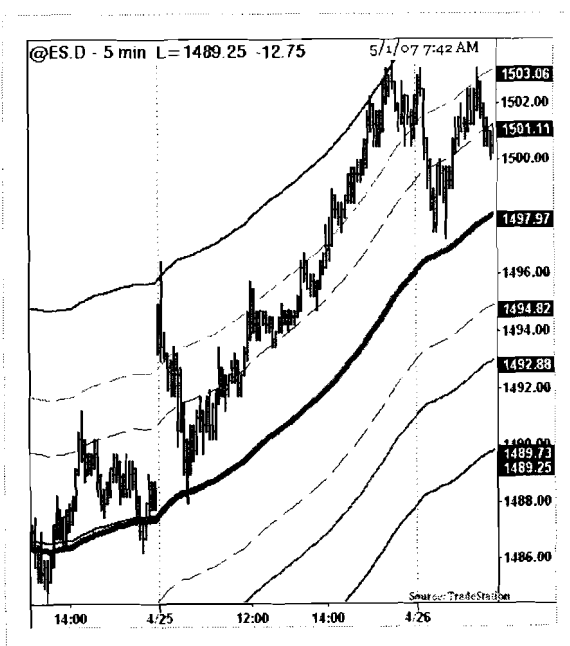
Fibonacci Bands are derived from Fibonacci ratios expansion from a fixed moving average. These bands help traders find key areas of support and resistance. "Fibonacci bands" are computed by adding a Fibonacci ratio distance (Up and Down) from a "key moving" average (21, 34, 89 periods). An 8 period average of "True range" is computed. The multiples of Fibonacci ratios of this range are added to the fixed moving average to compute Fibonacci Bands (Fib. Bands formulas are provided below).

One of the best ways to find trend reversals is to watch the price action near the extreme bands (both lower and higher). Markets tend to reverse when prices trade outside of the band for a few bars and again trade inside the bands. After reversals, markets also tend to trade from one extreme band to the other end (opposite) of the extreme bands.

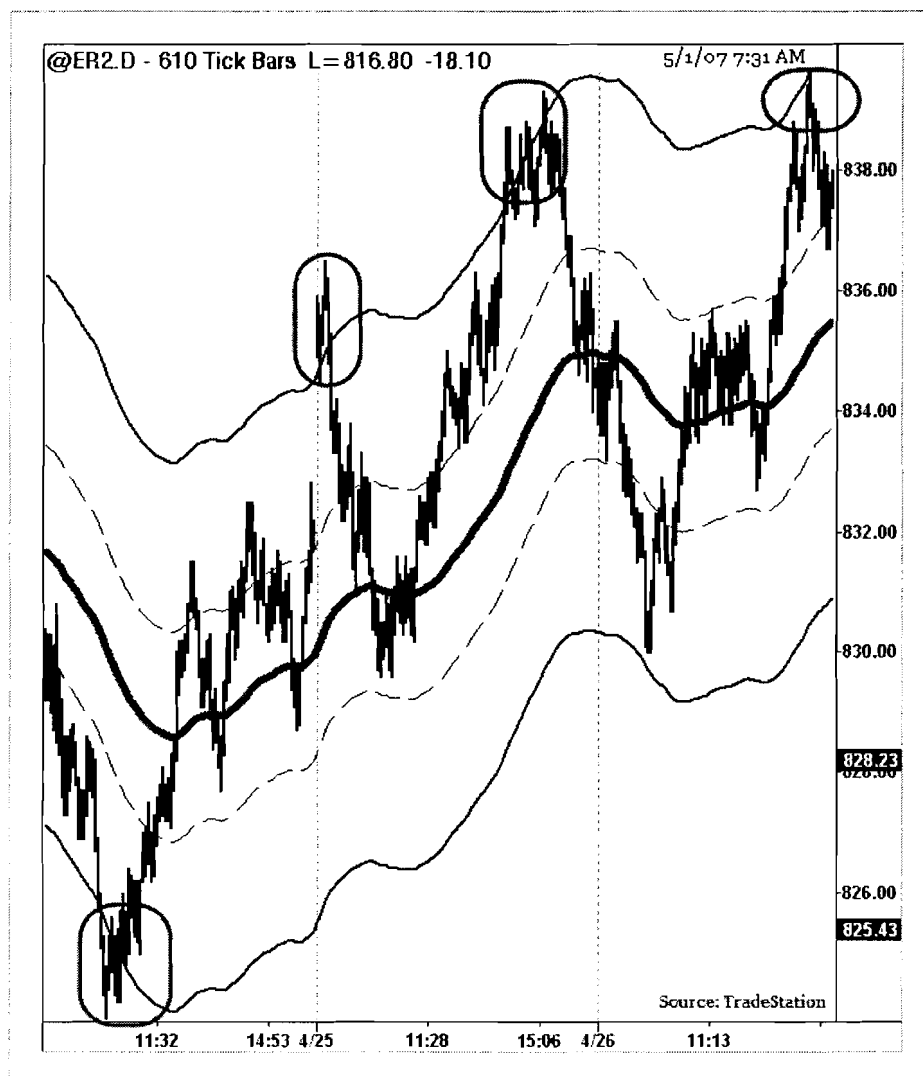
In my view, the best results can be obtained from using multiple time-frames for the same instrument. Fibonacci Bands indicator may be used along with the other indicators. When shorter time-frames signals are aligned with a larger time-frame trade signals, the Fibonacci Bands signals and the market may be in sync. Trades are usually taken in the direction of a larger time-frame. The "short-term" band support/resistance areas are used for "entry" and "exit" setups. During market rallies, utilize exhaustion or trend based market indicators along with Fibonacci Bands. During congestion zones use non-momentum based indicators for better results.

Computing Fibonacci Bands

- $MA = \text{Exp. Moving Avg. (Close, 34)}$;
- $TR = \text{Exp. Average(TrueRange, 8)}$;
- $\text{UpperBand3} = MA + 4.23 * TR$;
- $\text{UpperBand2} = MA + 2.62 * TR$;
- $\text{UpperBand1} = MA + 1.62 * TR$;
- Moving Average (MA);
- $\text{LowerBand1} = MA - 1.62 * TR$;
- $\text{LowerBand2} = MA - 2.62 * TR$;
- $\text{LowerBand3} = MA - 4.23 * TR$;



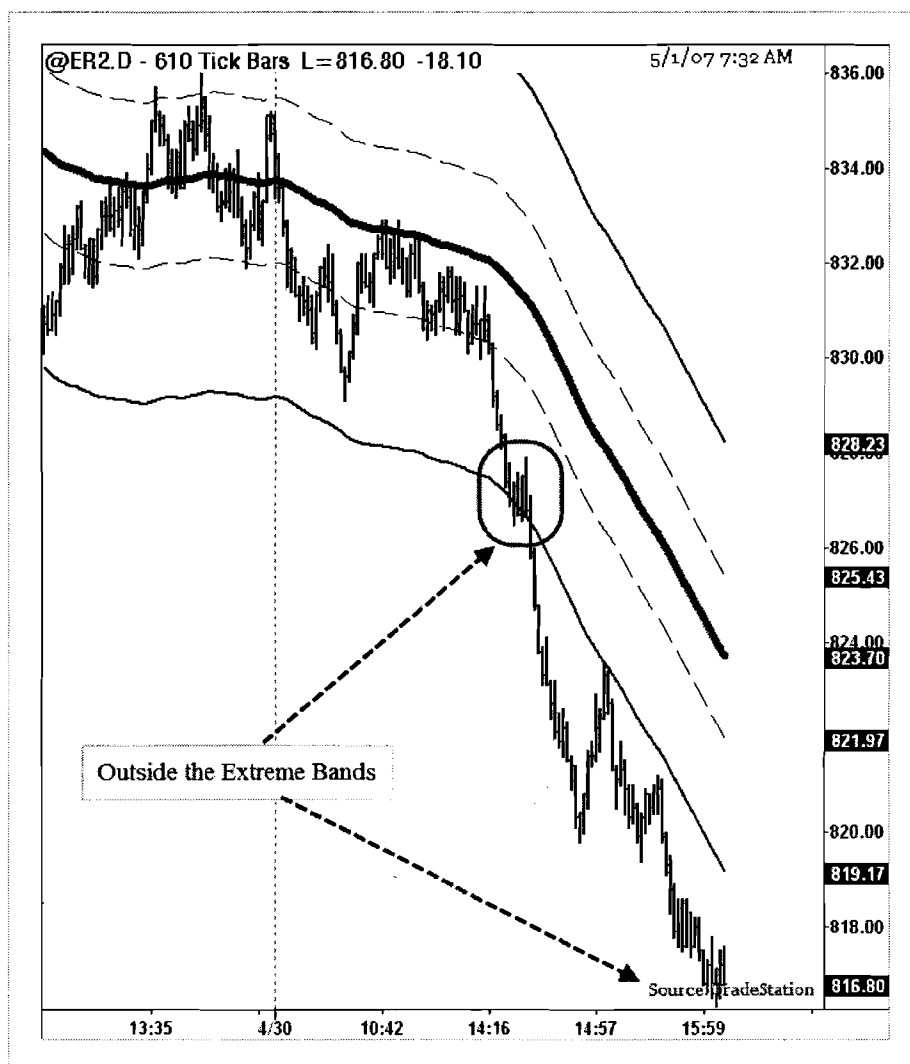
Trading Fibonacci Bands



Trading Fibonacci Bands

The example above shows Fibonacci Bands plotted from the Russell 2000 Emini 610 tick chart. The middle line shows the 89 period moving average and 1.62 and 4.62 Fibonacci Bands (Upper and Lower). Trades are only initiated in the direction of market reversals. On April 24, 2007, the markets sold-off in the morning session and traded in a steep price decline. At about 11.30 am, the price closed outside the Fibonacci's extreme lower bands and signaled a potential reversal. When prices are traded inside the band, enter a "long" trade 1 tick above the high of the previous bar. Place a "stop" order below the low of the recent "swing low" outside the Fibonacci Band. Targets can be set using the moving average (89 periods) and extreme band range.

Trading Fibonacci Bands



Trading Fibonacci Bands

The example above shows Fibonacci Bands from the Russell 2000 Emini 610 tick chart. On April 30, Russell 2000 (ER2) experienced weaker markets and traded below the moving average (Center Line). At 2:30 pm, the price closed outside the Fibonacci Bands to signal a potential reversal. Subsequent bars have not produced any trades as the price did not close above the high of the break-in bars for a reversal. Price also quickly traded outside the extreme Fibonacci Bands to signal further weakness. Upside trade reversals did not occur until the price re-enters the band. Traders can use Fibonacci Bands to effectively find key entry and exits based on the trend reversals occurring near these extreme bands.